

23EAC212

Electronic Systems Design and Automation

S4 EAC

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EMC

- Electromagnetic compatibility(EMC), concerns the unintentional generation, propagation and reception of electromagnetic energy.
- “ The capability of electrical and electronic systems, equipment, and devices to operate in their intended electromagnetic environment within a defined margin of safety, and at design levels or performance, without suffering or causing unacceptable degradation as a result of electromagnetic interference” (ANSI C64.14-1992).
- Every electronic system must meet EMC standards, evidenced, for example, by the mandatory CE mark in the European Union or the EMC compliance mandated by the Federal Communications Commission (FCC) in the United States.
- EMC is often the hidden culprit behind undesired circuit functionalities and unwanted signals.
- Mostly due to the unintentional coupling of circuits that are either part of, only partially part of, or completely outside the system of interest.

EMI

- Electromagnetic interference (EMI) is the lack of EMC
- EMI is the process by which disruptive electromagnetic energy is transmitted from one electronic device to another via radiated or conducted paths (or both).
- Mostly for RF signals in 10 kHz to 100 GHz, commonly used for communication
- But EMI can occur in the frequency range commonly identified as "anything greater than DC to daylight.
- RF EMI can come as a byproduct of an equipment's operation
- RF is transmitted through two basic modes: radiated and conducted emissions
- Radiated emissions: The component of RF energy that is transmitted through a medium as an electromagnetic field (Mainly through free space)
- Conducted emissions: The component of RF energy that is transmitted through a medium as a propagating wave, generally through a wire or interconnect cables.
- The conducted emission through power lines is called Line Conducted Interference (LCI).

Immunity, susceptibility, and suppression

- Immunity. A relative measure of a device or a system's ability to withstand EMI exposure while maintaining a predefined performance level.
- Radiated immunity. A product's relative ability to withstand electromagnetic energy that arrives via free-space propagation. A 1W transmitter will have electric fields of 5V/m at 1 m distance. That is within the range of 1-10 V/m that can lead to failure of the equipment due to interference.
- Designing equipment such that it does not corrupt EM environment and prevent failure of nearby devices or cause interference to communication is a legal requirement.
- Conducted immunity: A product's relative ability to withstand electromagnetic energy that penetrates it through external cables, power cords, I/O interconnects, or chassis. EMI may also couple to a chassis if interconnects are improperly implemented.
- Susceptibility: A relative measure of a device, or a system's propensity to be disrupted or damaged by EMI exposure to an incident field or signal. It is the lack of immunity.
- Suppression: The process of reducing or eliminating RF energy that exists without relying on a secondary method, such as a metal housing or chassis. Suppression may include shielding and filtering as well.

ESD

- Electrostatic discharge (ESD). A transfer of electric charge between bodies of different electrostatic potential in proximity to each other or through direct contact. This definition is observed as a high-voltage pulse that may cause damage or loss of functionality to susceptible devices. Although lightning differs in magnitude as a high-voltage pulse, the term ESD is generally applied to events of lesser amperage and more specifically to events triggered by human beings.
- Direct ESD cause catastrophic failures
- Radiated ESD events cause interference

EMC and EMI

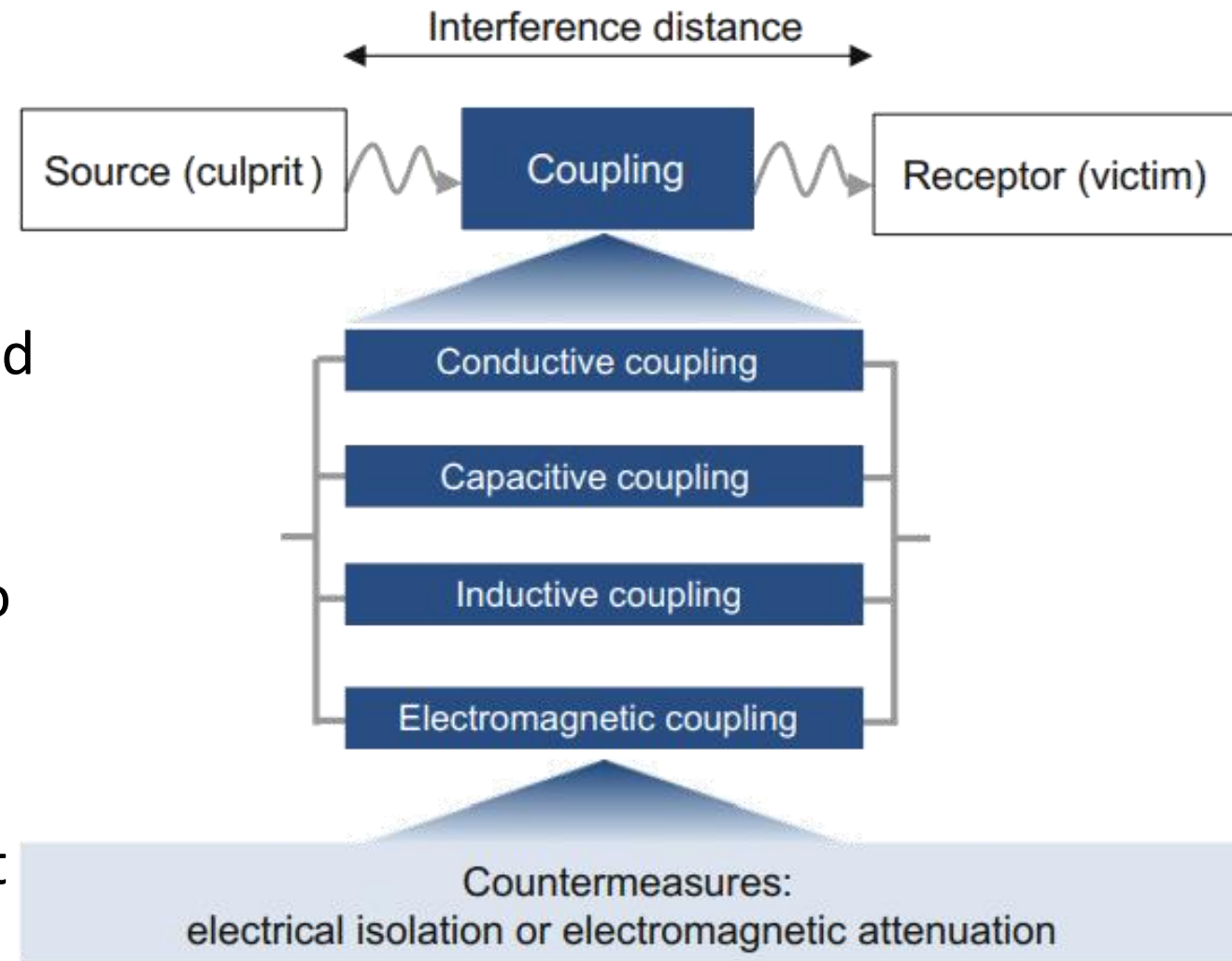
- EMC guidelines ensures that equipment is developed to function properly in its electromagnetic surrounding, and that it does not interfere with other equipment.
- Two conditions must be met for compliance with these requirements.
 - First, a system acting as an emitter should not emit unapproved disturbances (interference), and
 - second, a system acting as a receiver should not be susceptible to external disturbances.
- The second aspect is a system's immunity to outside/external disturbances.
- Selective EMC measures should be put in place to maintain approved emission and immunity levels so that electronic systems can operate properly alongside one another.
- Any system can transmit or receive unintentional electromagnetic energy and can be an "emitter" or "receiver"

Coupling

- Disturbances, emitted by a source of disturbance (“culprit”), travel via a coupling path to the receiver, a susceptible device or receptor (“victim”).
- While disturbance sources and susceptible devices can easily be characterized by measuring their interference and immunity, insight into the physical fundamentals of electrical engineering is needed for identifying intermediate coupling mechanisms
- knowledge of the coupling mechanisms is needed for selecting suitable countermeasures, such as shielding for sources as well as receptors and thus for compliance with statutory emission and immunity levels.
- **Electrically small or large**: depends on how the system dimensions L compare with the wavelength λ
- **Electrically small $L \ll \lambda$** : the wave that occurs can be ignored.

Coupling mechanisms for electrically small structures

- The coupling of electrically small structures can be of **three types**:
- **conductive coupling** (galvanic/common impedance coupling): where there is a conductive connection between the source and the receptor
- **capacitive coupling**: the coupling is caused by a varying electric field between the two adjacent conductors
- **inductive coupling** (magnetic coupling): induced voltages in one conductor due to a varying magnetic field (due to varying currents) in another parallel conductor.
- These are conductor-bound.
- Can be modeled with a simple equivalent circuit comprising discrete components, like resistors, capacitors, and coupled inductances



Coupling in electrically long structures

- wavelength is on the same order of magnitude or smaller than the system dimensions: ex. The wavelength is only 30 cm at 1 GHz
- The propagation of these waves can be either bound to conductors, such as cables or wave guides, or can be transmitted wirelessly, i.e., radiated.
- This phenomenon is called electromagnetic coupling (also: radiative coupling).
- Inductive and capacitive coupling effects cannot be separated in this scenario.

Countermeasures-Answer three questions

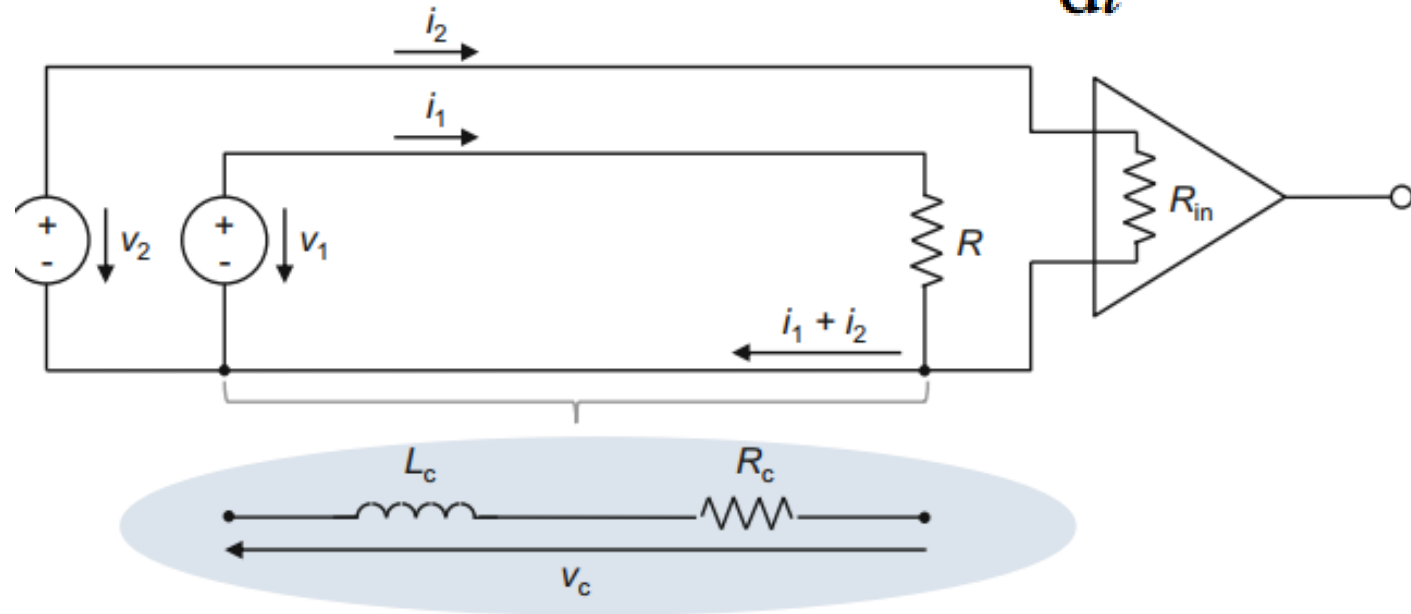
- How do the disturbances arise?: Identifying the disturbance sources and eliminating or minimizing their effects
- How do they impact the circuits? : The coupling path between the disturbance source and susceptible device, in particular, should be attenuated
- How do they penetrate a system?: Hardening the receptor, i.e., raising its immunity limits
- two active circuits are considered to illustrative examples for the four modes
 - Culprit: circuit #1 (v1) with higher power rating
 - Victim: circuit #2 (v2)

Conductive Coupling

- A current flows in a circuit (“culprit circuit”) through an impedance that is also part of another circuit (“victim circuit”), causing interference
- The current flowing in the culprit circuit produces a voltage drop across the common impedance, which superimposes itself on the signal in the victim circuit, and thus corrupts and possibly invalidates it.
- The common impedance is often a common reference conductor or the common ground for different circuits on a printed circuit board.
- Conductive coupling leads to compatibility issues, such as the 50 Hz mains hum from many loads on the same supply line.

Conductive Coupling-how it happens

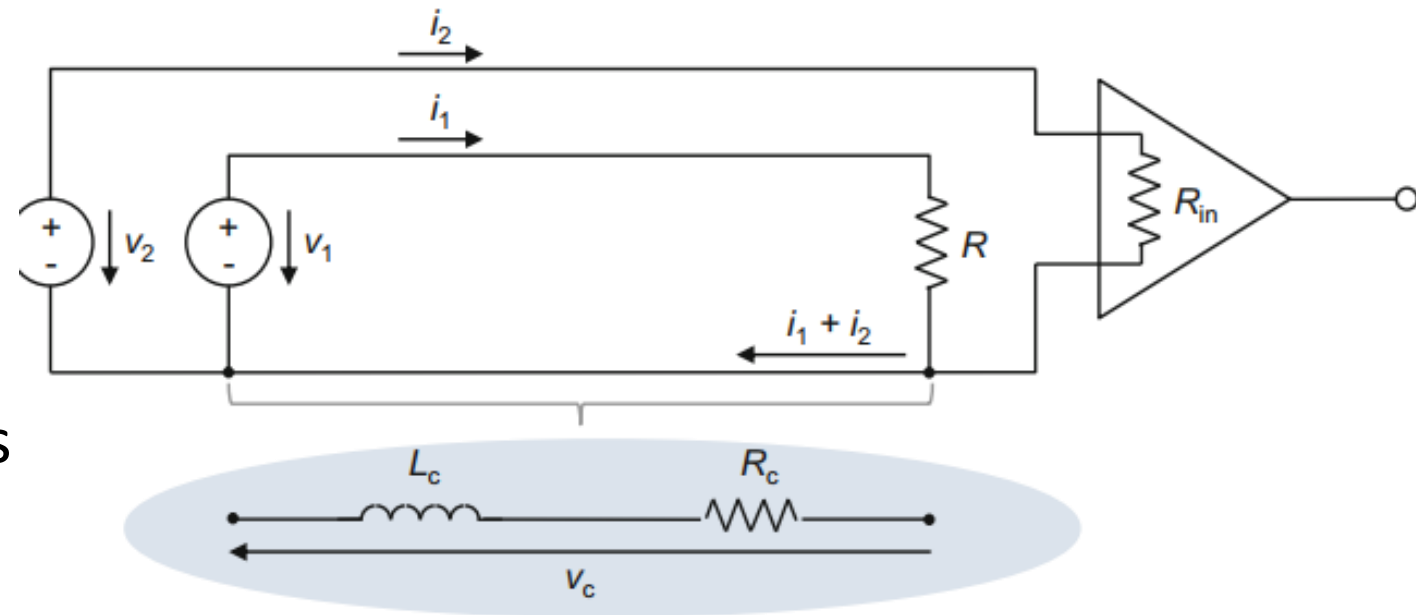
- Currents i_1 and i_2 flow through the two circuits with common impedance Z_c , line resistance R_c and line inductance L_c .
- The potential difference v_c across the coupling impedance Z_c is proportional to the sum of two currents $i_1 + i_2$.
- The proportion of the external voltage based on the victim circuit (v_2) is due only to i_1 .
- The voltage induced in the victim circuit is thus:
$$v_{c2} = R_c \cdot i_1 + L_c \cdot \frac{di_1}{dt}$$
- v_{c2} is the interference voltage induced in circuit #2 in V,
- di_1/dt is the rate of change of the current in circuit #1 in A/s,
- R_c is the resistance of the conductor common to both circuits in Ω , and
- L_c is the inductance of the conductor common to both circuits in H.



Conductive Coupling-how it happens

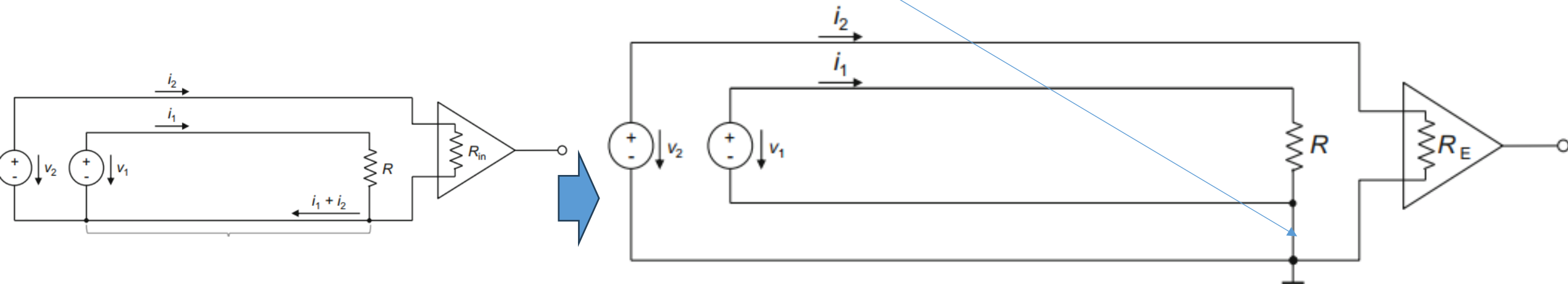
- The induced voltage v_{c2} is in series with the input voltage v_2 .
- Currents in electronic systems are generally not greater than a few amperes.
- One can often ignore the $R_c i_1$ component if the value of R_c is on the order of milliohms.
- Currents change rapidly in digital systems.
- Thus, relatively small currents through the inductance L_c produce a coupling effect if the currents are switched on or off rapidly.
- Coupling resistance R_c and coupling inductance L_c depend on frequency as well, due to the skin effect

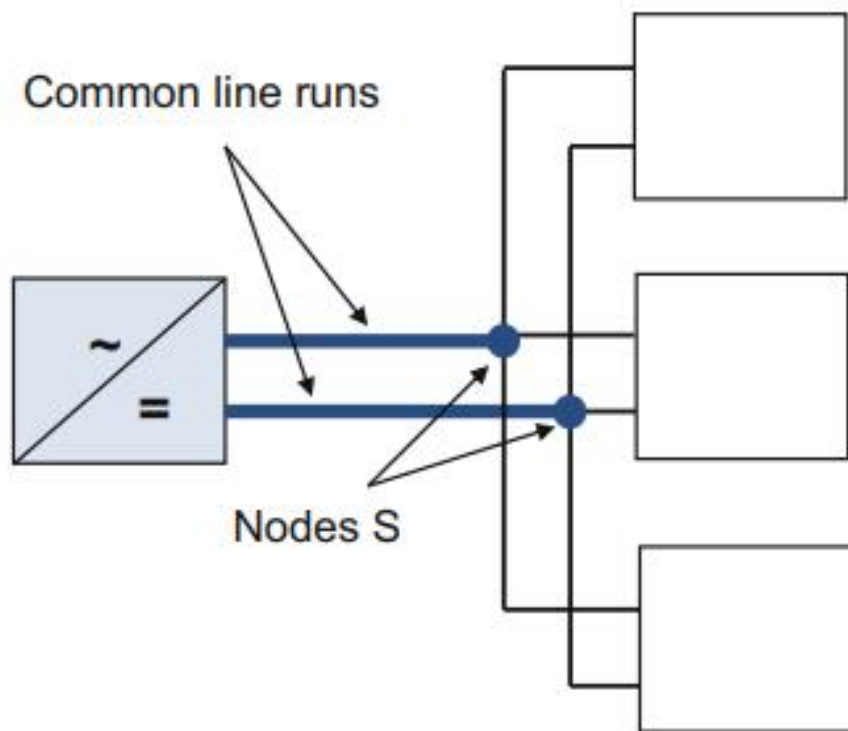
$$v_{c2} = R_c \cdot i_1 + L_c \cdot \frac{di_1}{dt}$$



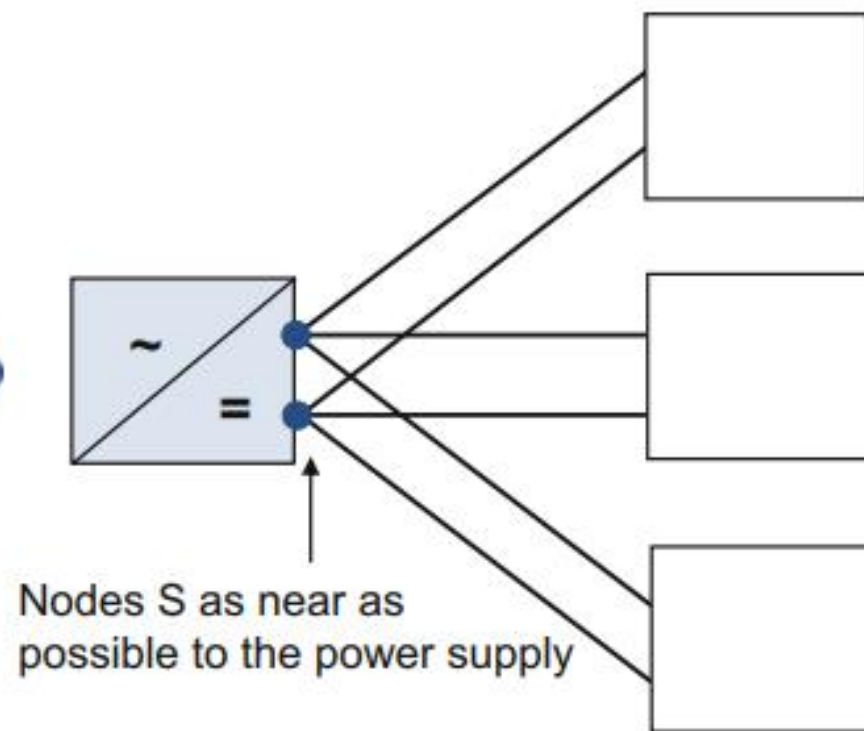
Conductive Coupling-how it can be mitigated

- Optocouplers, coupling transformers, etc., can be used to electrically isolate parts of circuits
- The use of separate ground and supply lines for loads drawing high currents (lamps, relays, power transistors, etc.) on the one hand, and logic circuits or analog amplifier circuits on the other, is encouraged.
- These separate reference and ground conductors should be interconnected only at one point to avoid common paths.
- A star-shaped connection is typically selected, for example, to power analog and digital units separately





bad



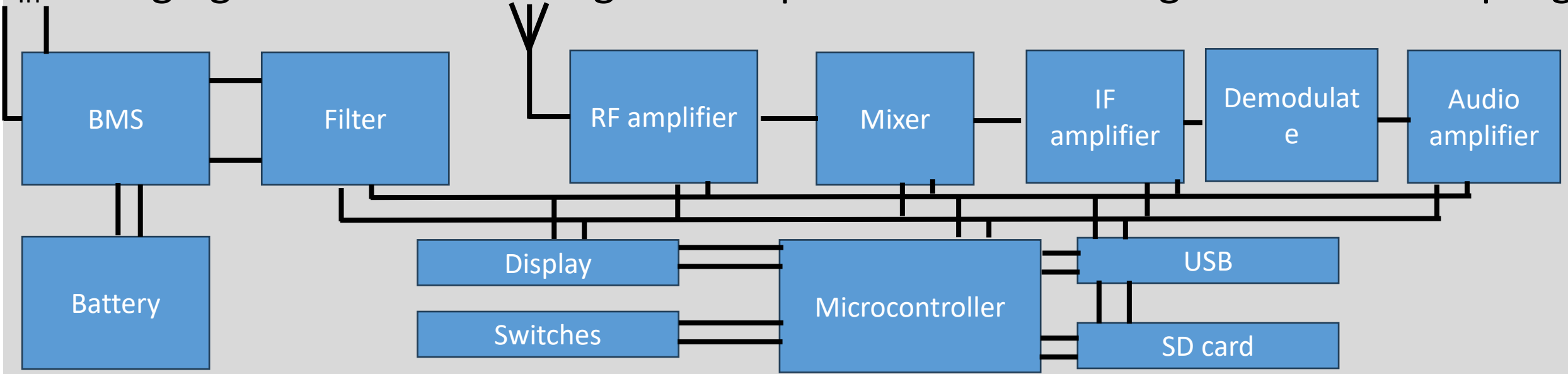
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...Conductive Coupling-how it can be mitigated

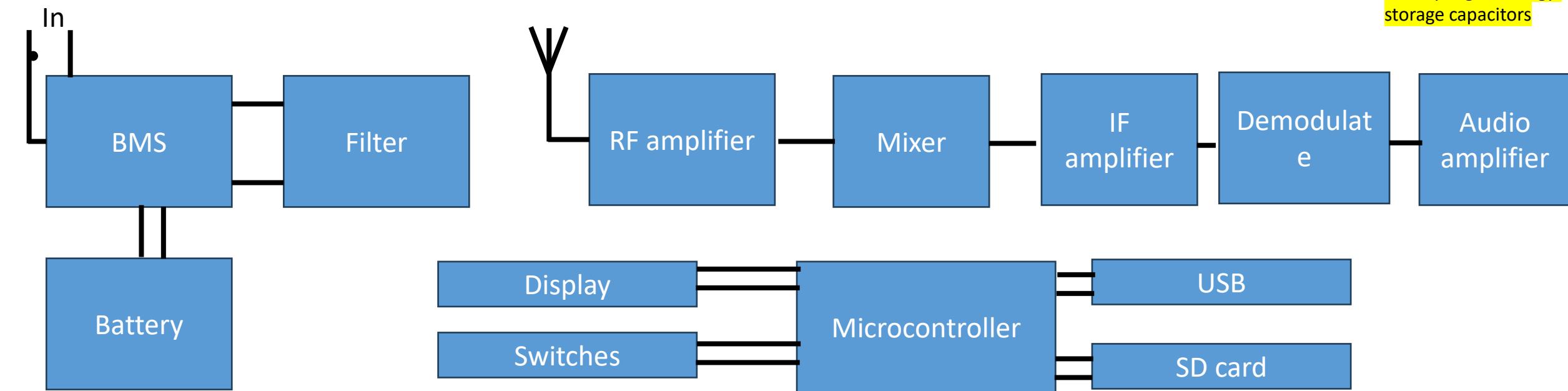
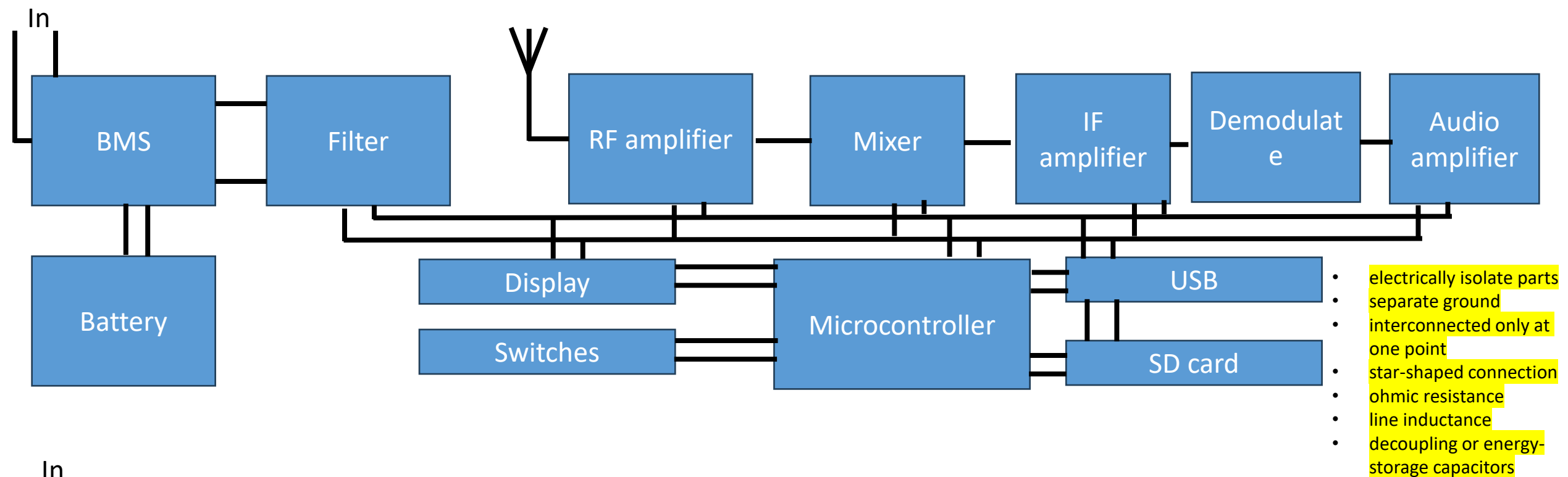
- The **ohmic resistance** can be limited by sufficiently large cross sections and low contact resistances at terminals and connectors.
- The **line inductance** should be kept to a minimum using
 - short lines in the common current path;
 - wide ribbon cables (inductance per unit length for parallel plate transmission line is $L_0 = \frac{\mu d}{w}$ for d thick dielectric and w wide plate-proof not required);
 - routing signal and return lines close together (use thin film as a dielectric) : This reduces the area between the supply and return path and hence reduces the inductance of a loop since inductance increases with the loop area; and
 - routing cables straight: reduces the length
- Current transients caused by switching in the circuit are hard to control.
- Voltage dips arising from short-term current spikes can be mitigated by placing capacitors, such as **decoupling or energy-storage capacitors**, at suitable locations in the circuits.
- These capacitors **are placed parallel to the power supply** for every chip or piece of circuitry, and they will act as charge or energy storage devices in times of high current or high power consumption

Q

- Relayout the below circuit laid out in blocks to reduce conductive coupling through ground. List the mitigation steps taken for reducing conductive coupling.



- electrically isolate parts
- separate ground
- interconnected only at one point
- star-shaped connection
- ohmic resistance
- line inductance
- decoupling or energy-storage capacitors



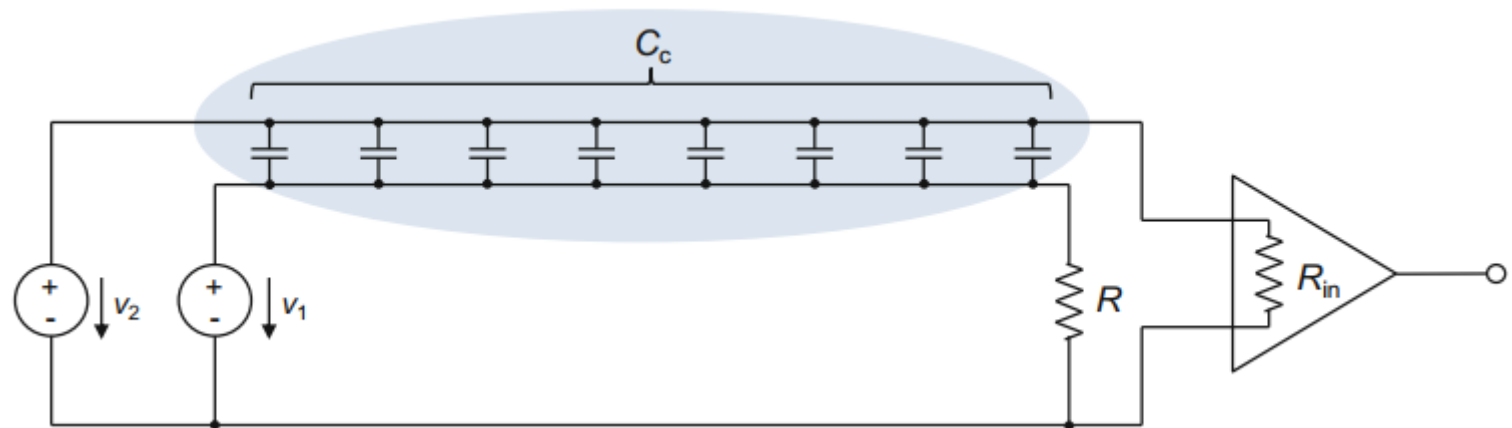
Capacitive coupling

- Capacitive coupling takes place between two circuits in which there is a potential difference, and thus an electric field, between the conductors in the circuits.
- If the voltage is time dependent, a displacement current will flow between the conductors and an additional line current of the same magnitude will flow in the conductors.
- The latter produces interference voltages in the impedances in the circuits, which are superimposed on the signal.
- Ex. Interference between power conductor of mains and the instrument conductor of a low voltage sensor due to proximity. Typically, mV of interference for 120 or 240 V supply near it.
- Capacitive coupling is very much a function of the distance between culprit and victim; it occurs only at close range.

- Capacitive currents flow from conductor to conductor via the capacitance C_c , which we have introduced in the circuit shown.
- A partial current flows to the ohmic input resistor R_{in} in an amplifier and produces an ohmic disturbance voltage drop v_{c2} :

$$v_{c2} \sim i_c \cdot R_{in} \sim C_c \cdot \frac{dv_1}{dt} \cdot R_{in}$$

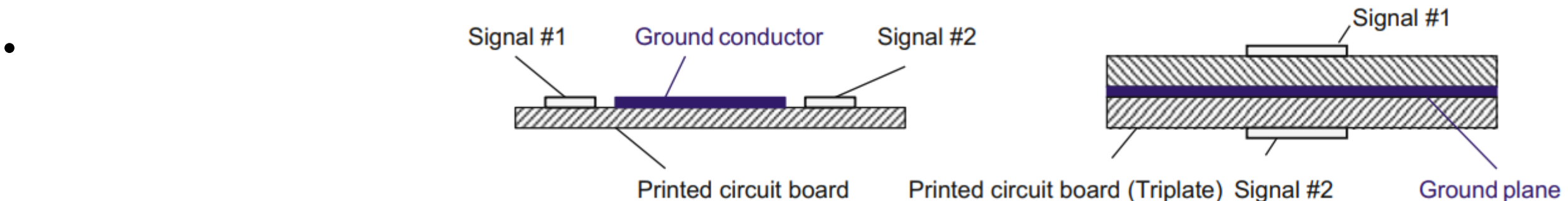
- v_{c2} : the interference voltage induced in circuit #2 at the conductor end in V
- dv_1/dt : the voltage change in circuit #1 in V/s
- C_c the line capacitance between the two circuits in F and
- R_{in} the terminating resistance at the end of conductor #2 or the amplifier input resistance in X.



- The interference voltage increases with the line length and the frequency of the disturbance source
- Hence, unshielded interconnects in LF and HF systems are exposed to large capacitive interference from voltage changes in other circuits.
- Switching DC circuits produces interference voltages in the same way.
- These voltages drop with smaller terminating resistors at the start and end of the conductor.
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Capacitive coupling- mitigation

- “Slow logic” will cause fewer disturbances than “fast logic” for the same signal level.
- Capacitive coupling from digital signals is mitigated by lower signal amplitudes and greater signal rise and fall times.
- Interconnects with grounded shielding provide complete interference decoupling.
- The extra capacitance, which is 70–100 pF/m for coaxial cables, is a load for the drive circuit and reduces the logic speed in digital systems.
- Culprit and victim can also be isolated by placing shielding conductors (ground wires) between individual signal wires or by inserting shielding planes (ground planes) between circuit layers

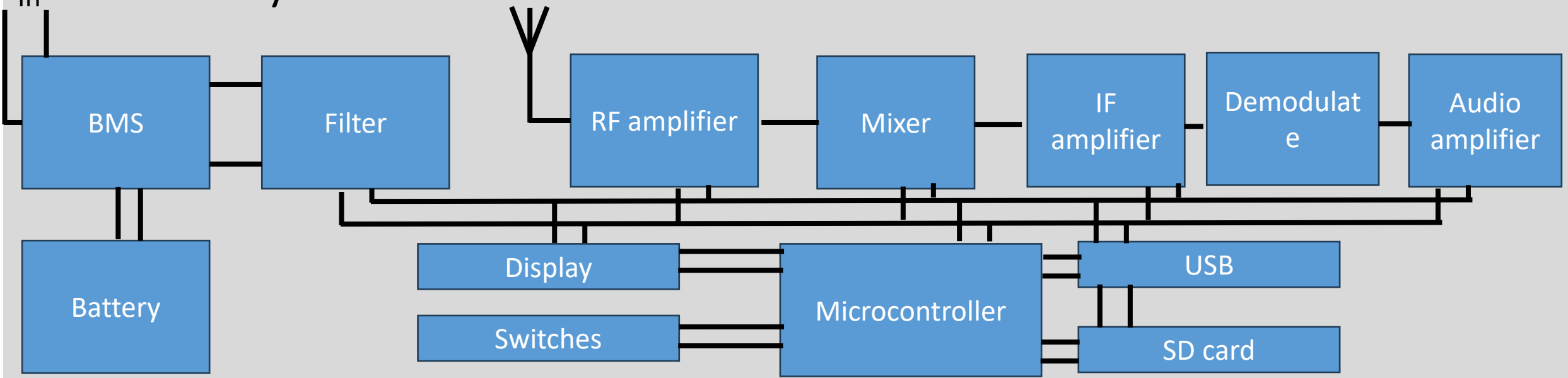


Capacitive coupling- mitigation

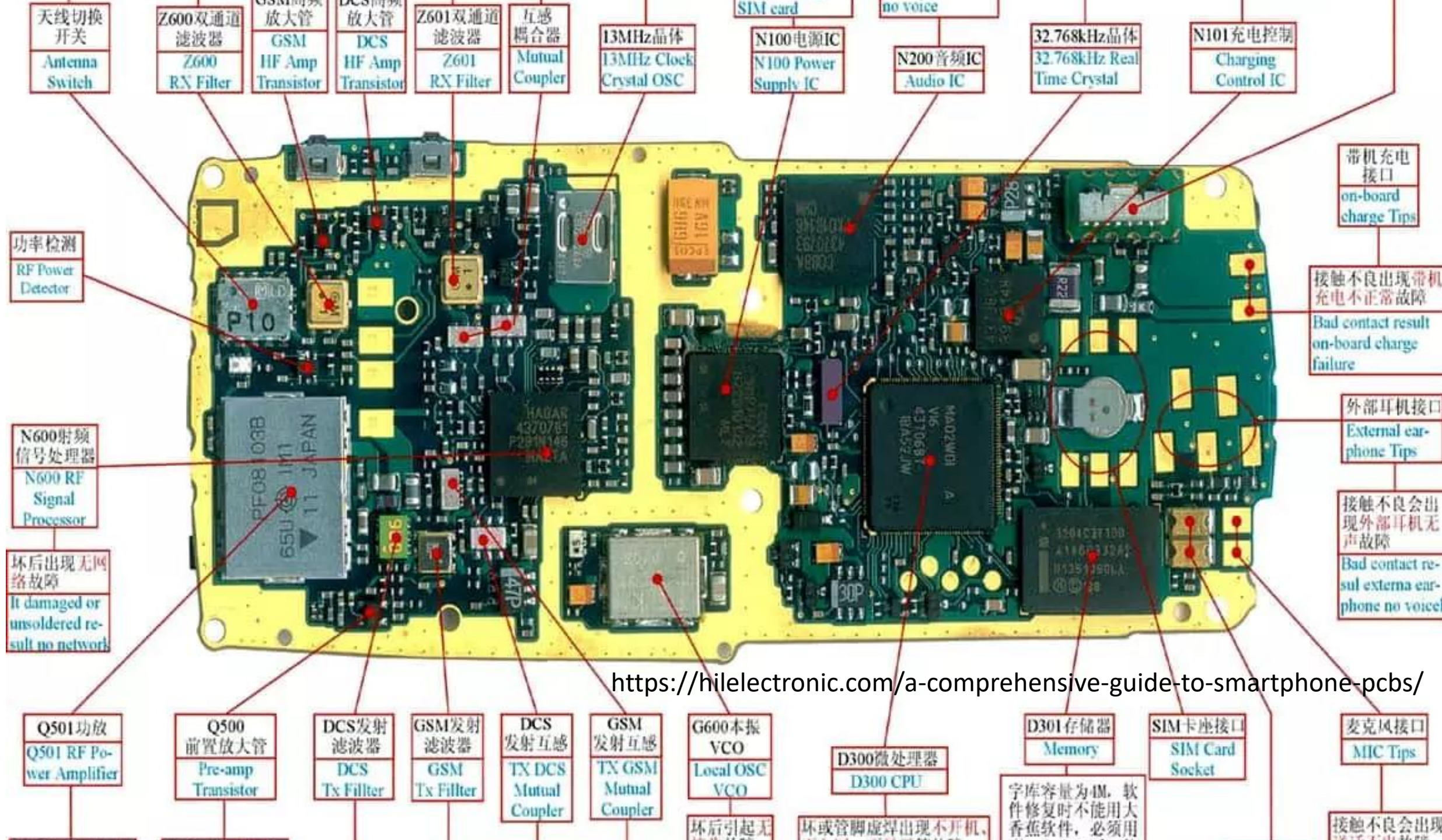
- Recommendations for reducing capacitive coupling are:
 - Maximize distances between parallel signal lines.
 - Decouple possible culprits and victims on PCB with shielding conductors.
 - Do not use cable harnesses/bundles that put signal-carrying conductors with those that carry power or other high-speed signals.
 - Use point-to-point wiring for signal lines. Avoid intermediate connections with branches or joints
 - Twist signal lines with additional ground lines.
 - Reduce the steepness and amplitude of voltage changes within the culprit circuit
 - the victim system should have minimal impedance.

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- Discuss the mitigation steps that can be taken for reducing capacitive coupling in the below system.



- Maximize distances
- shielding conductors
- Do not use cable harnesses/bundles
- point-to-point wiring for signal lines
- Twist signal lines with additional ground
- Reduce the steepness and amplitude
- should have minimal impedance



Unshielded twisted pairs, with each pair having a signal and a ground, are used in LAN

Cable bundling-Avoid bundling unshielded high-frequency signals with power lines.

<https://www.fiberoptics4sale.com/blogs/archive-posts/95046918-what-is-unshielded-twisted-pair-utp-cable>

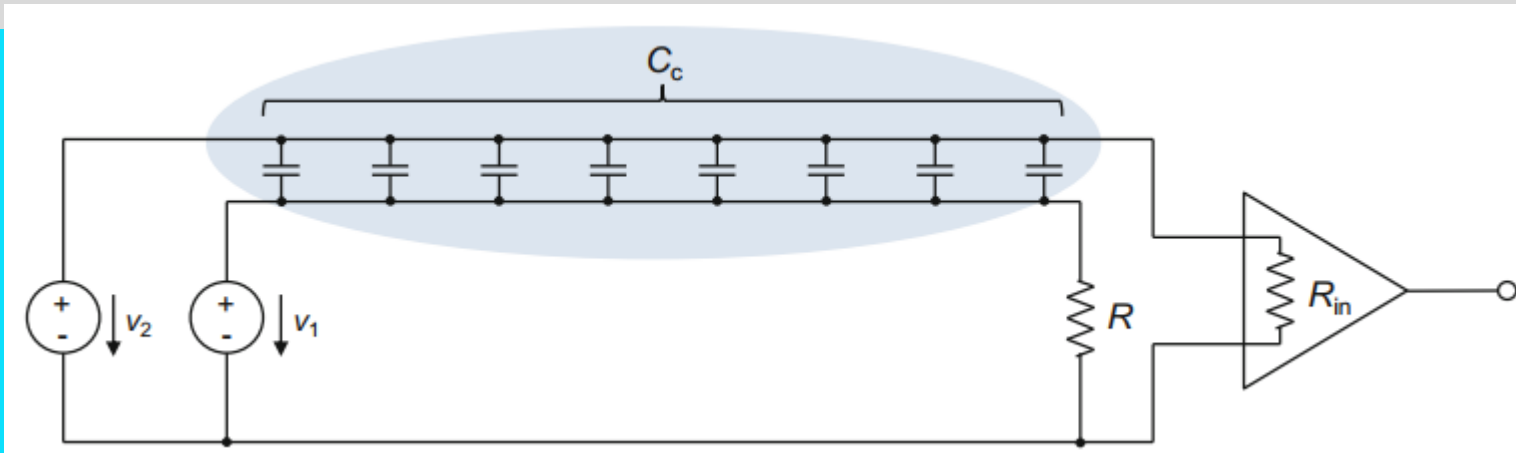
<https://resources.altium.com/p/types-wiring-harnesses-electronics>

Q

- Calculate the amount of capacitive coupling due to adjacent traces in circuit 2. Given that the capacitance per unit length is 0.2fF/cm, and the input impedance of the amplifier is 1M Ω . There is a voltage spike in circuit 1 due to a switching of v_1 . The ramp rate is 1.3 V/100ns. The lines are near over a length of 10 cm.

$$v_{c2} \sim i_c \cdot R_{in} \sim C_c \cdot \frac{dv_1}{dt} \cdot R_{in}$$

- Here capacitance is given and
- Hence no need to use parallel plate capacitance formula.
- $v_{c2} = 0.2 \times 10^{-15} \times 10 \times 0.013 \times 10^9 \times 10^9 = 26 \text{ mV}$
- Ans=26 mV



Inductive coupling

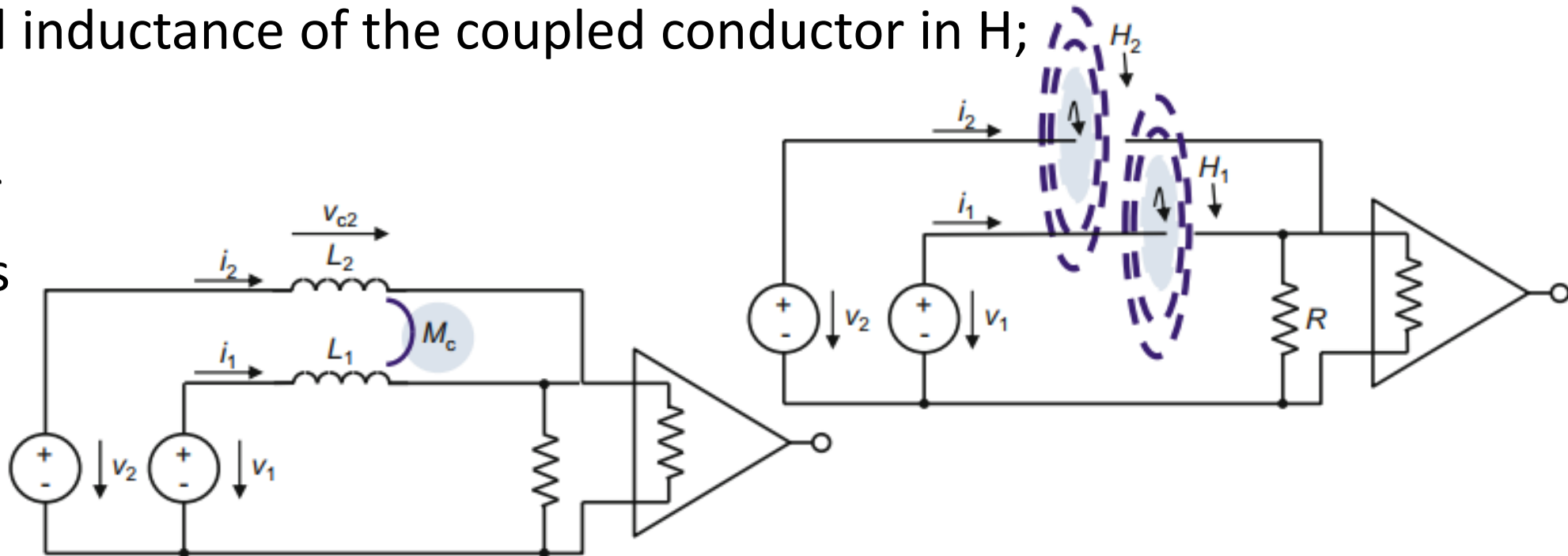
- Inductive coupling, also known as magnetic coupling, occurs between two current-carrying conductor loops.
- Current flowing in a conductor produces a magnetic field around the conductor.
- Inductive coupling occurs because a time-dependent current in the culprit circuit produces a time-dependent magnetic flux throughout the victim circuit, inducing an interference voltage on the signal in this circuit
- The effects of the culprit current on the victim circuit are represented in an equivalent circuit schematic either by a mutual inductance (coupling inductance) or by an induced source voltage
-

- The voltage induced in the victim circuit is:

$$v_{c2} = -M_c \cdot \frac{di_1}{dt}$$

$$M_c = k \cdot \sqrt{L_1 \cdot L_2}$$

- where v_{c2} is the interference voltage induced in circuit #2 in V;
- di_1/dt the change in current over time in circuit #1 in A/s;
- M_c is the mutual inductance of the coupled conductor in H;
- L_1, L_2 , the self-inductances of the conductors 1 and 2 in H
- k the coupling coefficient



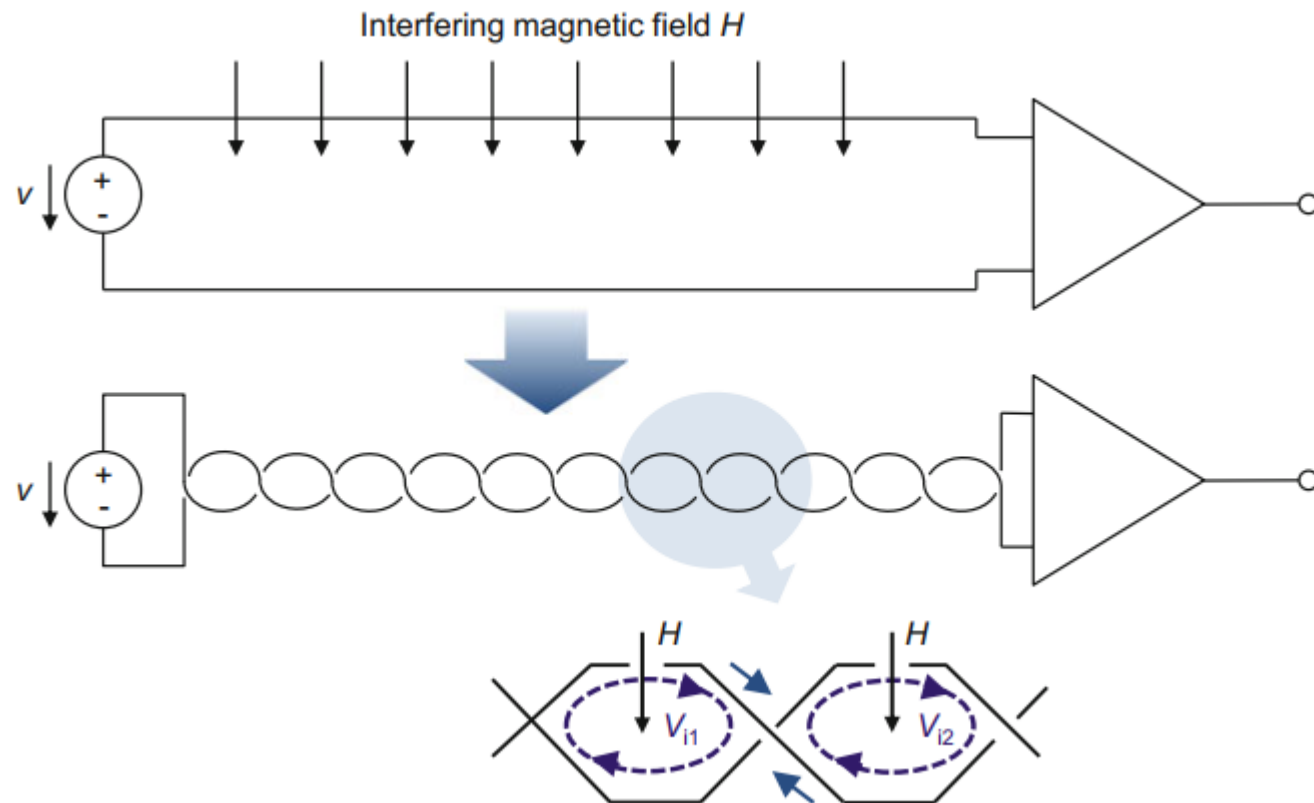
Inductive coupling- mitigation

$$v_{c2} = -M_c \cdot \frac{di_1}{dt} \quad M_c = k \cdot \sqrt{L_1 \cdot L_2}$$

- To reduce the inductive interference, the amplitude and steepness of the current changes in the culprit system should be minimized.
- The coupling coefficient k can be reduced by increasing the distance between the coupled circuits.
- Inductances in conductor loops are at their lowest when signal and return lines are closest together.- small loop area
- Hence, one should keep the surface area of pulsed conductor loops as small as possible.

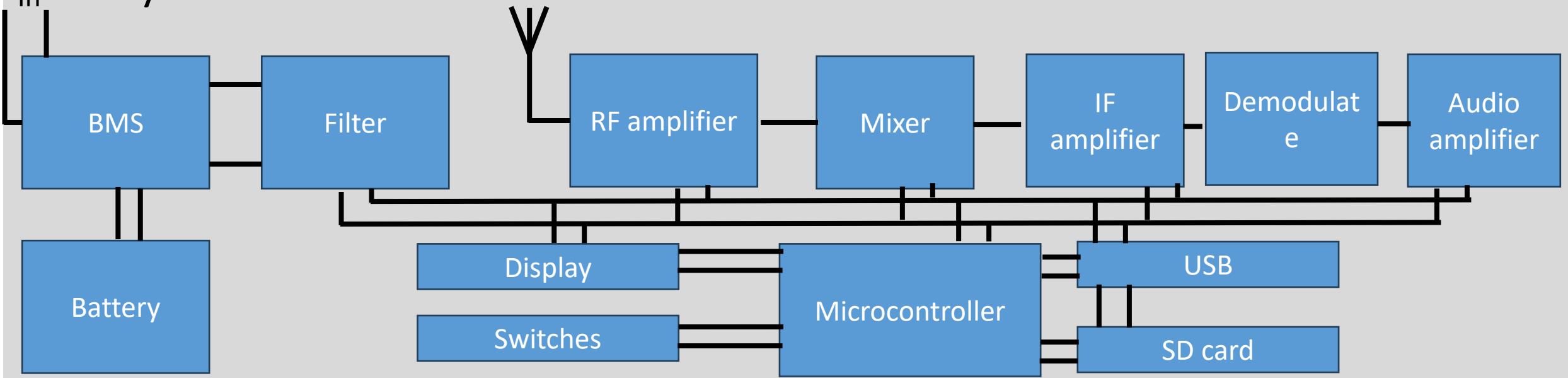
...inductive coupling- mitigation

- If the **conductor is twisted**, interference voltages from external magnetic fields are largely neutralized.
- This is especially true for location-independent interference fields, such as in the vicinity of large transformers or electrical equipment.
- Twisting the conductor also eliminates the EMI from own magnetic field on its surroundings.
- If the above measures are not sufficient, then use shielding to reduce interference further
- High-permeability magnetic materials can divert magnetic fields away from the victim



Q

- Discuss the mitigation steps that can be taken for reducing inductive coupling in the system below.



- Minimise the amplitude of the current changes
- Minimise the steepness of the current changes to reduce di/dt
- increasing the distance between conductors to reduce k
- signal and return lines are closest together.- small loop area
- conductor is twisted – cancel induced current, cancel own net magnetic field
- Shielding

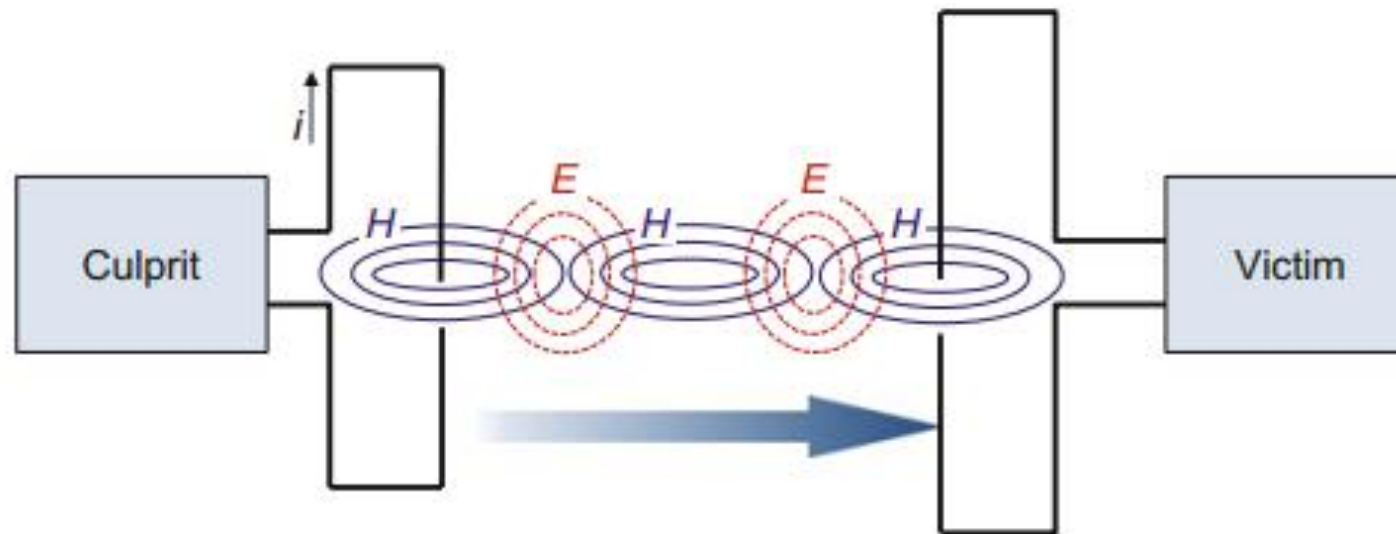
Electromagnetic Coupling

- When a **time-varying electric or magnetic field exists**, the other will automatically be present by virtue of Maxwell's equations
- Electromagnetic coupling, also known as radiative coupling, exists if the victim is situated in a field where the electric and magnetic fields occur simultaneously.
- The two can be **treated separately, when the frequency is small** such that length L is small ($\lambda \gg L$)
- For fast circuits ($\lambda \approx L$) the current and voltage are in the same order of magnitude as the physical dimensions of many electronic circuits
- This causes the current and voltage waves to propagate in the circuit
- The associated field is an electromagnetic **radiation that disengages** from the circuit and **propagates** through free space.
- Such a circuit or such a system radiates EMI ("culprit"), and receive ("victim") EMI

- Electromagnetic waves can be transmitted via **interconnecting conductors** as well as by **radiation**. In the former, the wave travels between two conductors.
- If the electric and magnetic fields emanating from this wire pair impinge on a second wire pair, electromagnetic interference will affect the second wire pair.
- The general case of coupling involves both **capacitive and inductive interference occurring simultaneously**
- In interference by radiation, on the other hand, an electromagnetic wave with electric field strength E and magnetic field strength H is the disturbance generated by a culprit.
- The fields E and H are orthogonal to each other.
- In the **proximity of the culprit**, i.e., in the near field at a distance $r \ll \lambda/2\pi$, **either the electric field E** (high voltages and low currents in the culprit) **or the magnetic field H** (high currents and low voltages in the culprit) **dominates**, depending on the type of culprit.

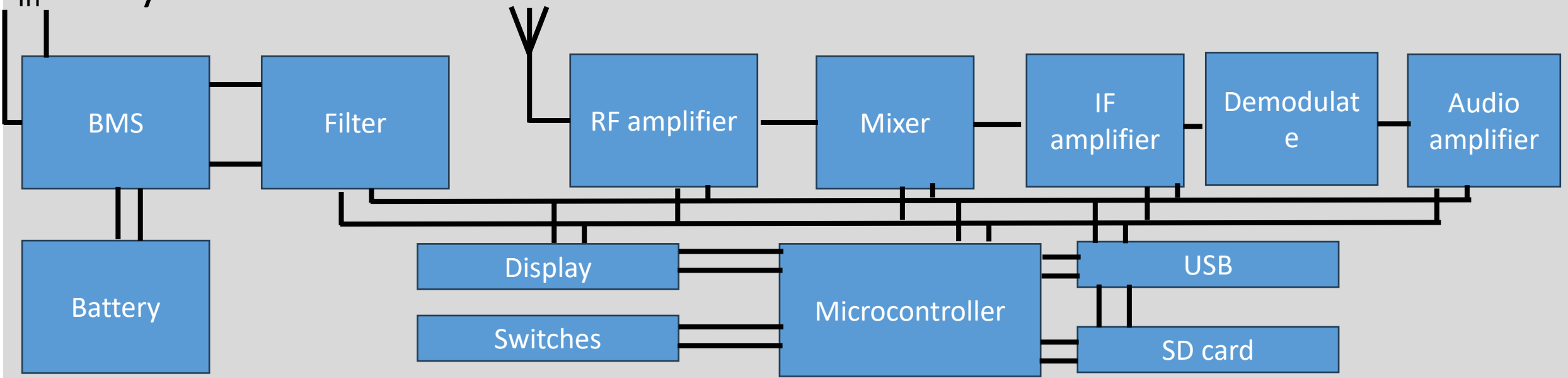
Electromagnetic coupling-mitigation

- **Shielding systems** deployed to suppress conductor- and radiation-based interference can shield electromagnetic fields
- Further countermeasures include techniques such as
 - **compact system construction**,
 - **symmetrical signal transport (transmission line)**, and
 - **filtering** of the power supply lines (decoupling filters).
- In addition, metal system enclosures should have **only one entry point** for power and signal lines to **avoid HF currents over the enclosure**.



Q

- Discuss the mitigation steps that can be taken for reducing inductive coupling in the system below.



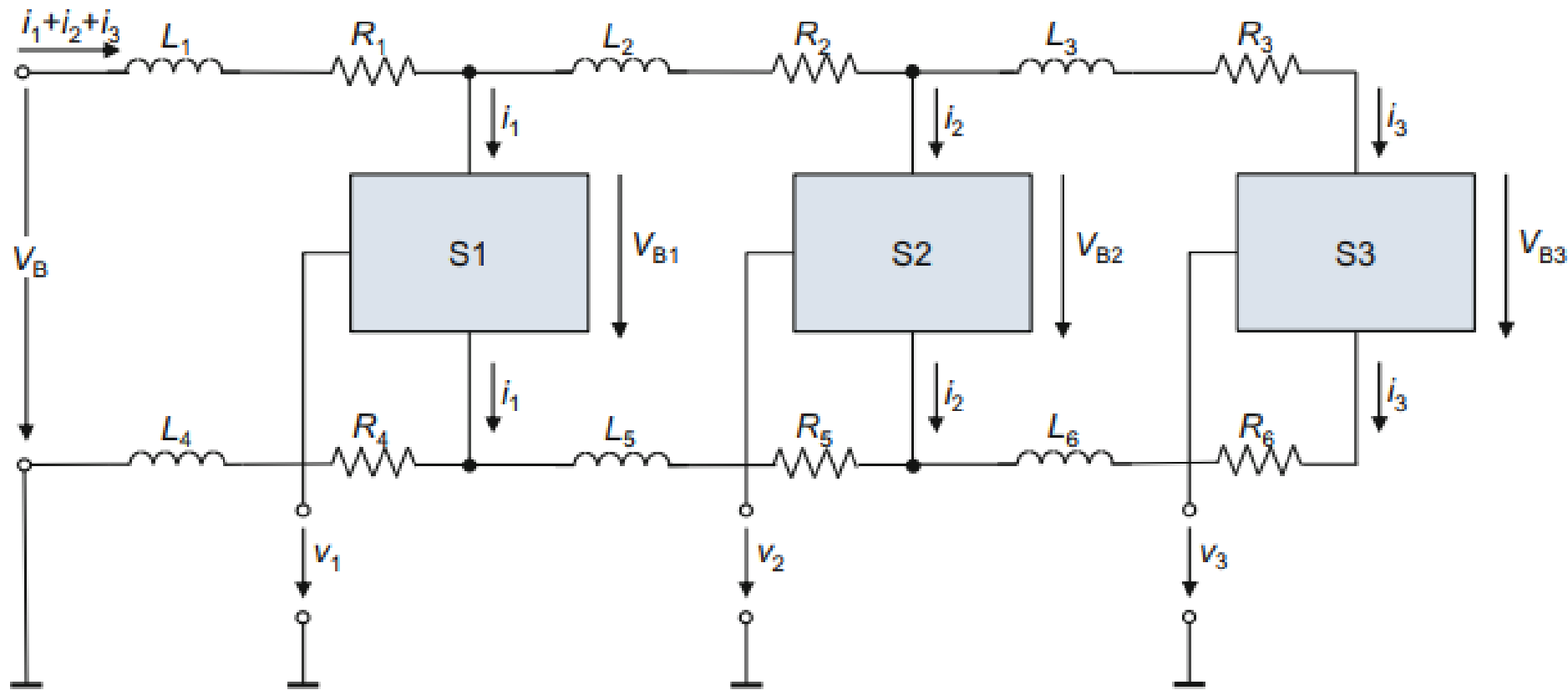
- Shielding systems
- compact system construction,
- symmetrical signal transport (transmission line),
- Filtering
- only one entry point

Grounding-Conductive coupling

- Input, output, and source voltages in electronic systems are referenced with respect to a common given 0 V potential point called ground.
- In reality, there is no such thing as ground in many electrical systems. Ex. a system operating on a battery often uses 0V as the negative terminal of the cell
- This low impedance conductor to a 0 V reference must be designed into the system right from the beginning.
- A good reference or grounding methodology avoids noise pickup and improves signal integrity by protecting against unwanted interference reception and radiated emissions

...Grounding-Conductive coupling

- three circuits S_1 – S_3 powered by the source voltage V_B via signal lines with line inductances L_1 – L_6
- The circuits (digital logic, amplifiers, power stages, etc.) are driven by control voltages v_1 – v_3 .



...Grounding-Conductive coupling

- If there is a current spike di_3/dt due to a change in the control voltage v_3 in S_3 , the supply voltage V_{B2} for S_2 is altered during the current spike by the value:

$$\Delta V_{B2} = (L_1 + L_2 + L_4 + L_5) \cdot \frac{di_3}{dt} + (R_1 + R_2 + R_4 + R_5) \cdot i.$$

- Furthermore, the base point of circuit S_2 is moved by the voltage:

$$\Delta V_{O2} = (L_4 + L_5) \cdot \frac{di_3}{dt} + (R_4 + R_5) \cdot i$$

- with respect to the reference potential 0 V (minus terminal of V_B).
- The effective potentials at inputs S_1 and S_2 thus change as well, as the **voltage drops at L_4 and $L_4 + L_5$ are added to input voltages v_1 and v_2** w. r. t. the ground potential.
- The combined input voltages may trigger unintended functions or outputs in S_1 and S_2 .
- This **unwanted voltage rise on the circuit ground is also known as ground bounce.**

...Grounding-Conductive coupling

- The change in supply voltage also typically causes malfunctions in the culprit circuit as well.
- The direction of the change in current (rising or falling edge) dictates whether the supply voltage is increased or reduced for the duration of the change in current.
- Current transients and high-frequency currents cause this effect due to inductance
- High DC currents can produce a significant voltage drop at the ohmic part of the conductor, particularly in analog circuits.
- The proposed countermeasures for conductive and inductive coupling should therefore be used to mitigate interference effects in the reference or ground conductor.

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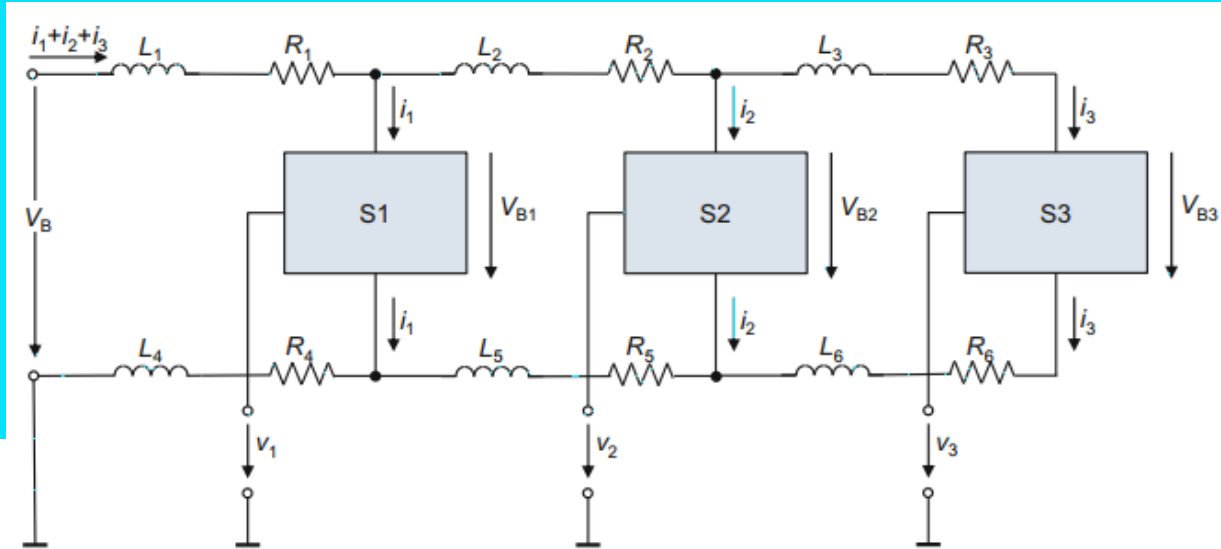
- The equivalent circuit diagram for part of a commercial printed circuit board with circuits S_1 – S_3 is given below. The length of the traces between the connector (plug) and S_1 and between S_1 – S_3 is $l = 7$ cm, respectively. The trace inductance is approximately $L' = 10$ nH cm⁻¹. The resistance is $R' = 85$ mΩ m⁻¹ = 0.85 mΩ cm⁻¹ at 20 °C. If a current of 20 mA flows through circuit S_3 for 5 ns with a current slope of 4 mA/ns, find the (a) inductive and (b) ohmic voltage drops produced at the reference conductor (connector to S_3).

$$\Delta v_{O3(L)} = (l_1 + l_2 + l_3) \cdot L' \cdot \frac{di}{dt}$$

$$\Delta v_{O3(R)} = (l_1 + l_2 + l_3) \cdot R' \cdot i$$

$$\begin{aligned} \Delta v_{O3(L)} &= (7 \text{ cm} + 7 \text{ cm} + 7 \text{ cm}) \cdot \frac{10 \cdot 10^{-9} \text{ H}}{\text{cm}} \cdot \frac{20 \cdot 10^{-3} \text{ A}}{5 \cdot 10^{-9} \text{ s}} \\ &= 0.84 \text{ V.} \end{aligned}$$

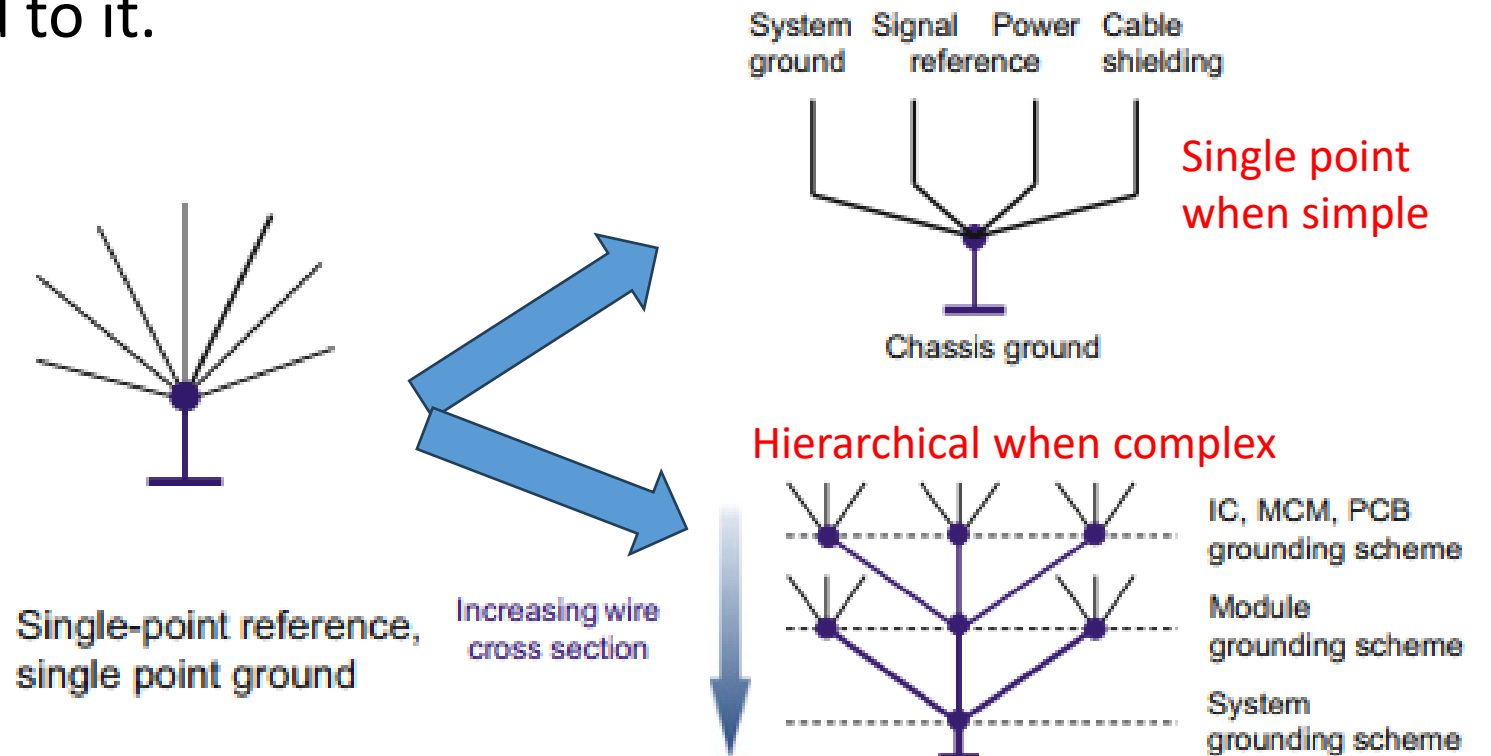
$$\begin{aligned} \Delta v_{O3(R)} &= (7 \text{ cm} + 7 \text{ cm} + 7 \text{ cm}) \cdot 0.85 \cdot 10^{-3} \frac{\Omega}{\text{cm}} \cdot 20 \cdot 10^{-3} \text{ A} \\ &= 0.36 \text{ mV} \end{aligned}$$



An interference of this magnitude can generally be ignored in digital circuits, but can be significant in analog circuits.

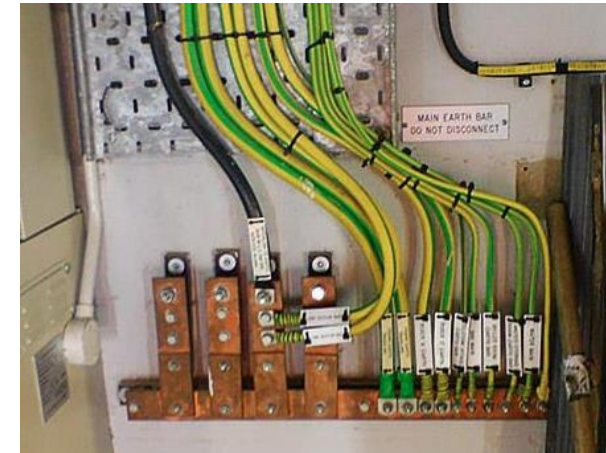
Reference/Grounding Systems Schemes

- Two recommended grounding schemes
 - a star-shaped single-point reference or ground
 - reference or ground plane/multipoint ground
- Star/Single point grounding: the individual ground categories in a system all meet at a single point in an equipotential bonding busbar, called the chassis or system ground
- A single-point ground terminal ensures that there are no potential differences between individual ground lines strapped to it.
- A single-point scheme becomes impractical in a large system as it requires a massive amount of wiring.
- A hierarchical grounding scheme is recommended for large electronic systems,
 - comprised of individual equipotential circuit planes

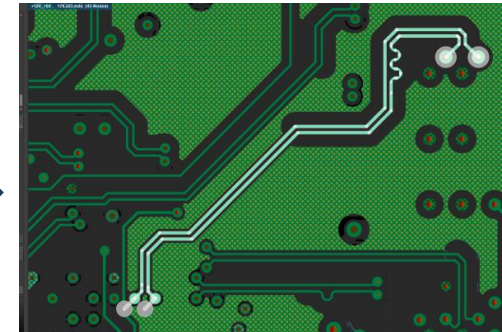


Single point/hierarchical- Main ground types

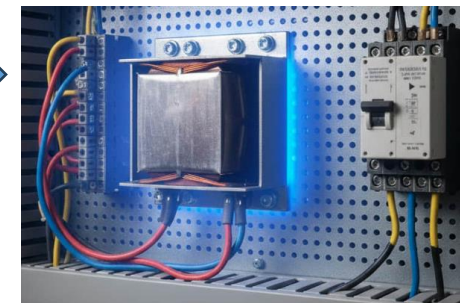
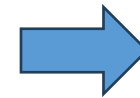
- System ground (also: safety ground or chassis ground): to prevent an electric shock should a fault condition occur. It comprises all electrically conductive parts of modules and systems; the system ground for systems in protection class I is connected to the protective earth PE, usually identified by a green-yellow color.



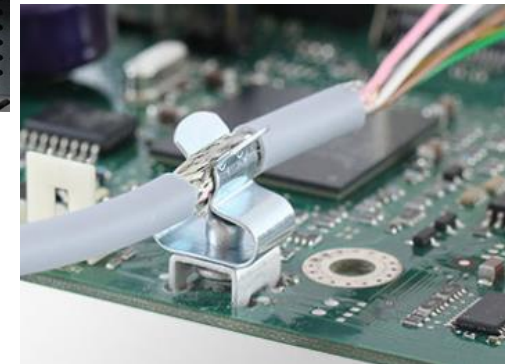
- Signal reference: all signal return conductors that provide a return path for intended signal flows.



- Power reference: all power supply return conductors, also known as the neutral conductor N, are usually identified by a blue color.



- Cable shielding ground: all shields. The shield reduces electrical noise and interference with other devices.



https://electrical-engineering-portal.com/earthing-protection-system#google_vignette

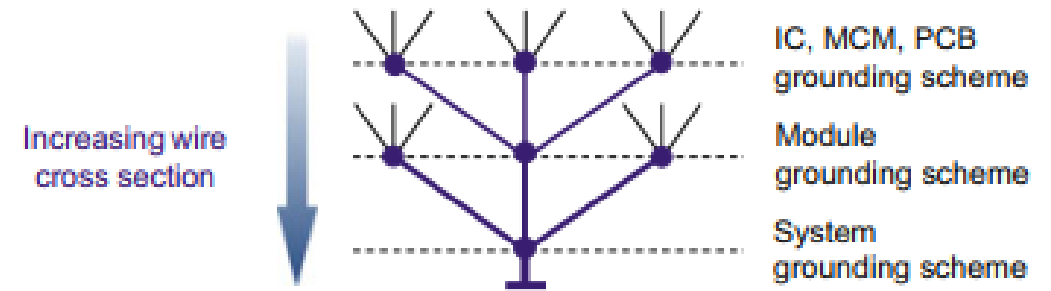
<https://electronics.stackexchange.com/questions/650101/controlled-impedance-signals-and-non-adjacent-reference-planes>

<https://www.weishoelec.com/Blog/what-is-a-control-power-transformer/>

<https://www.icotek.com/en/products/emc-cable-shield-clamps/pcb>

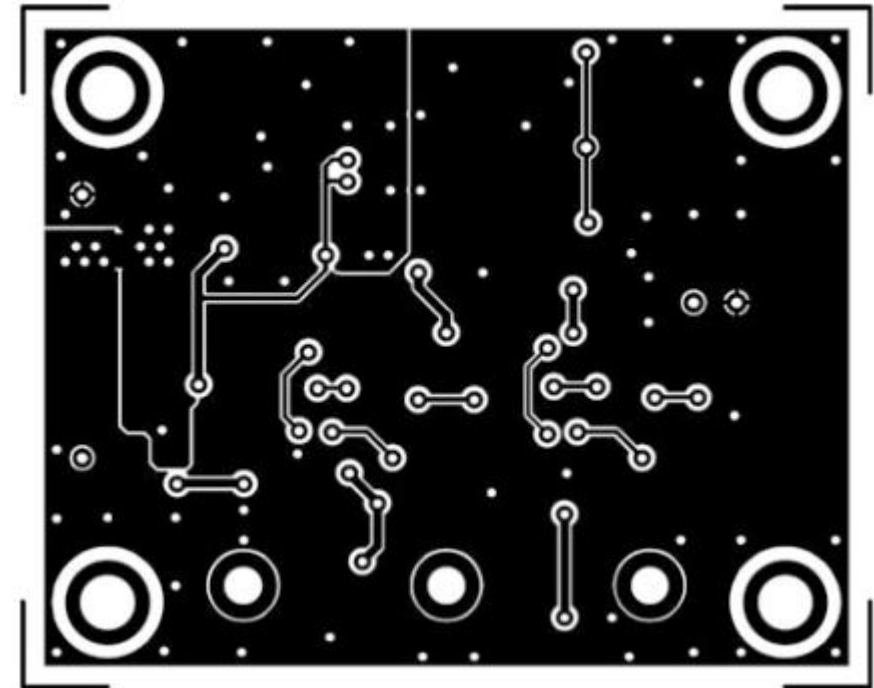
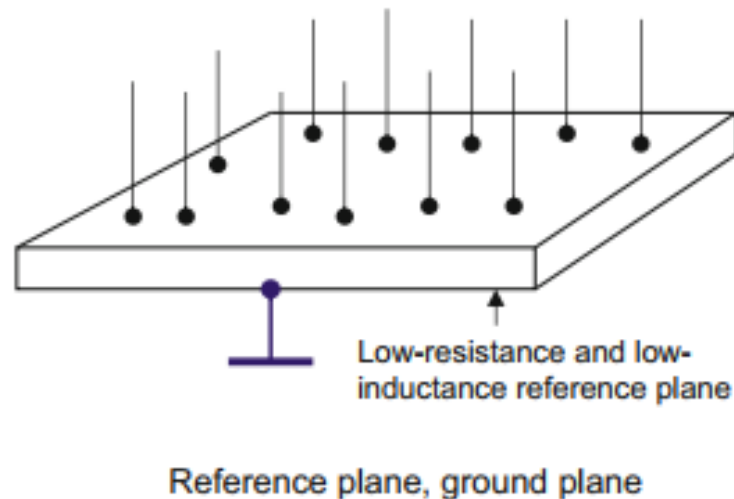
Single-point with Hierarchical grounding

- The impedance of the ground lines should be gradually reduced from plane to plane in the direction of the system ground point to minimize any potential differences that may arise between the planes.
- This can be facilitated by increasing the cross-sections of the ground lines in the direction of the ground terminal.



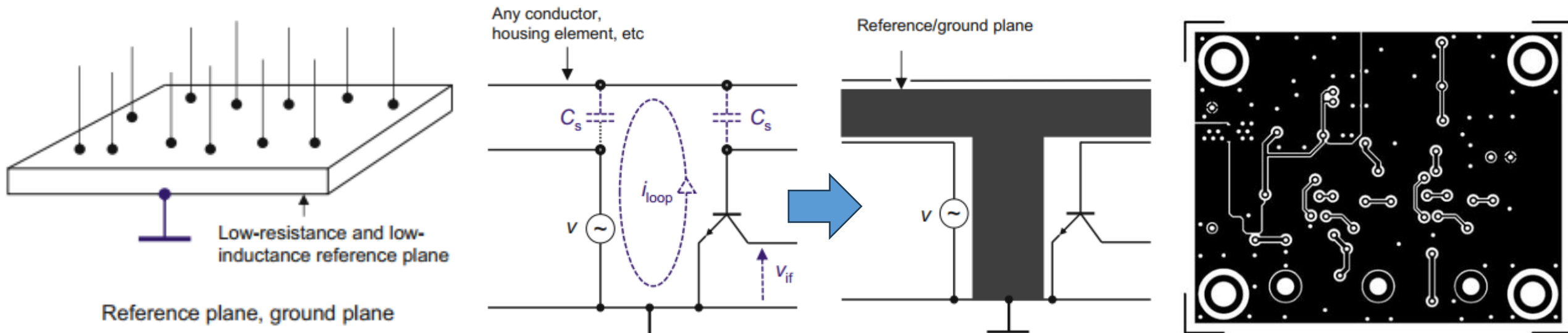
Ground plane scheme

- A reference plane provides a low impedance ground path available throughout the board
- It is especially beneficial at frequencies above 100 kHz, as then a circuit's ground potential is not only determined by the main ground point but also by other points in a circuit due to the effects of stray capacitances
- These stray capacitances can also cause unwanted loops with loop currents, which in effect neutralize the single-point grounding.
- It is only by implementing a reference plane as per Fig. 6.13 (on the right) that these stray capacitances can be fully isolated and a uniform reference potential achieved. Figure 6.14 shows an example of a reference plane on PCB.



Ground plane scheme

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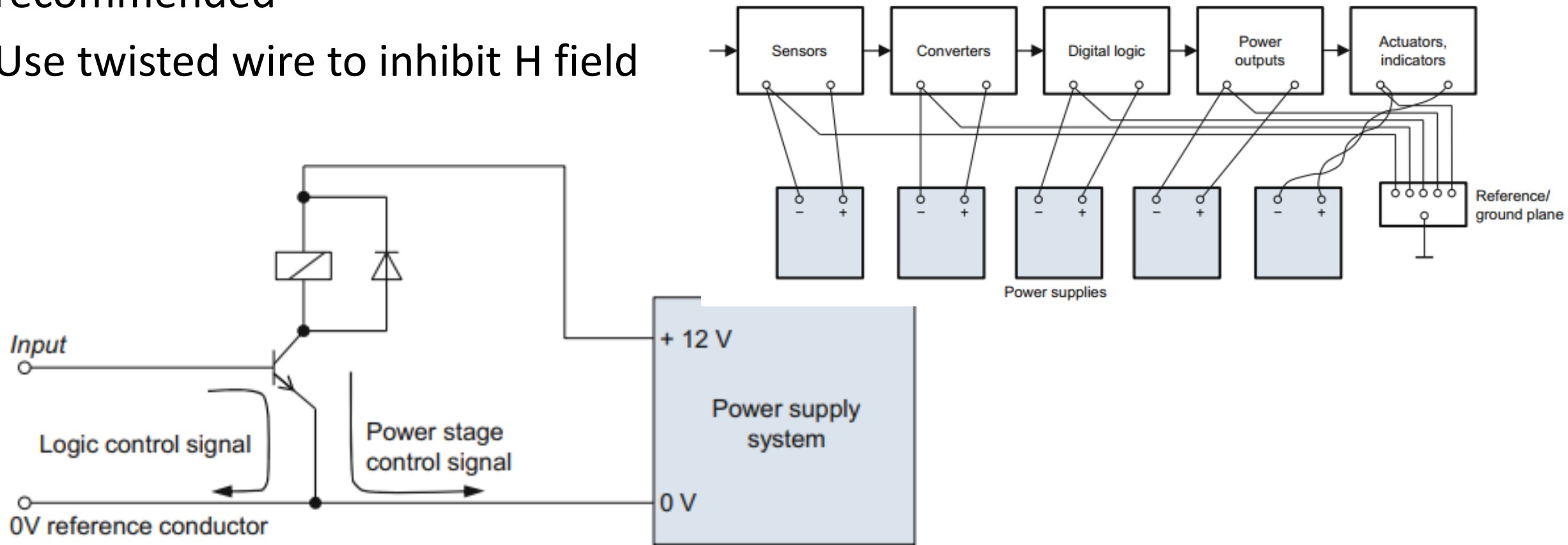


Reducing contact resistance

- Low-impedance contacts with minimum contact resistance can be achieved by using
 - planar contacts,
 - avoiding oxide and coating layers by fitting locking washers,
 - assuring contact force stability using spring washers, and
 - preventing corrosion with suitable protective coatings.
- Electrochemical corrosion is another issue to be avoided by deploying connected partners in the electrochemical galvanic series with a low potential difference
 - ($\Delta v \leq 0.25$ V).

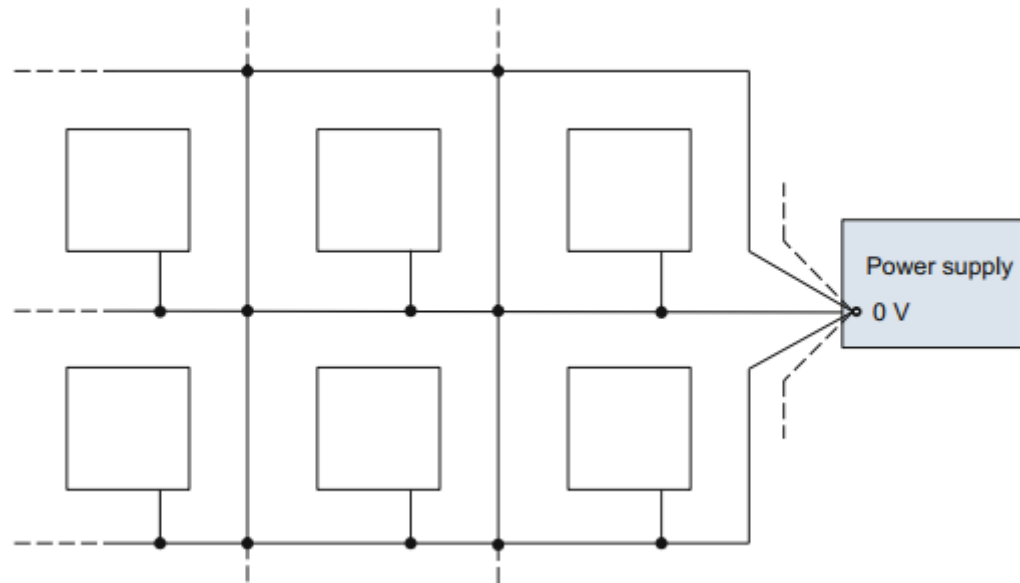
Return conductors for signals - Digital

- When relays or contactors are controlled by digital outputs, the return signal from the load should not flow through the logic 0 V reference conductor –Use separate return lines
- In large systems, a dedicated power supply for each functional unit is recommended
- Use twisted wire to inhibit H field



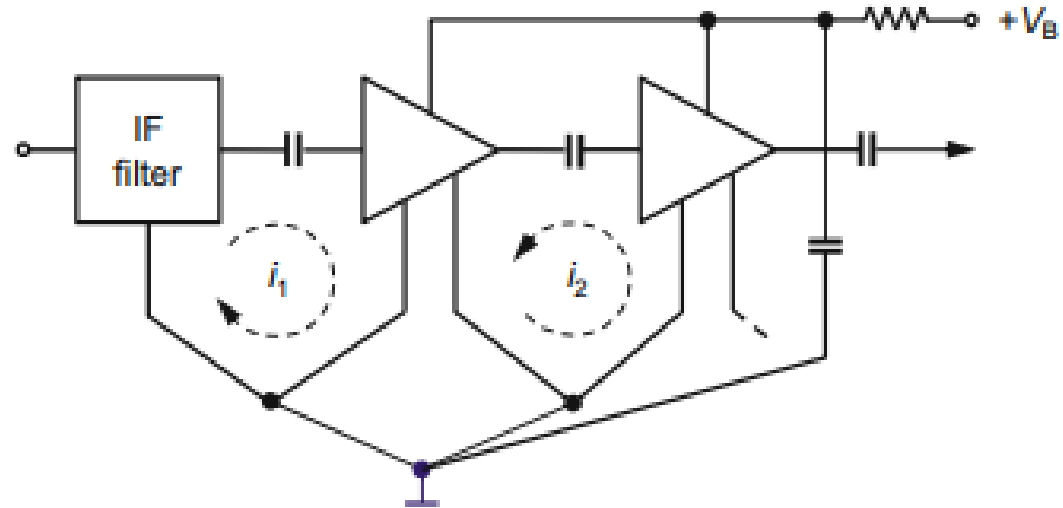
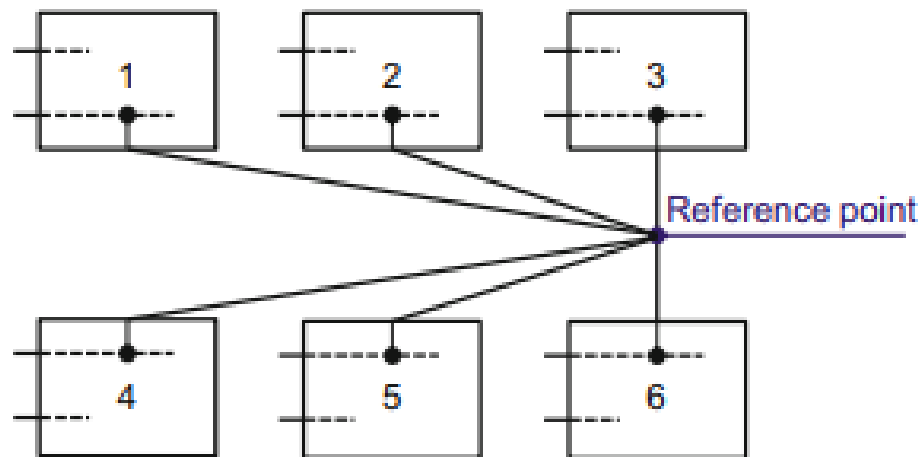
Grounding

- Inductance should be kept to a minimum in the digital logic due to the presence of transients
- a meshed grounding scheme (and possibly a meshed power supply arrangement) and ribbon cables can help



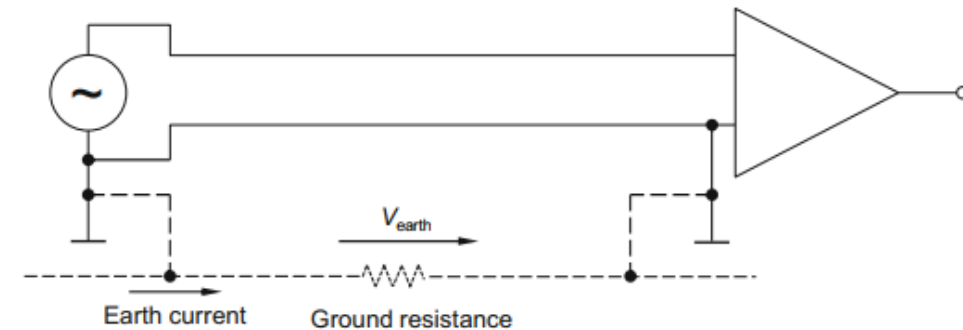
Return conductors for signals - Analog

- Analog input signals can be very weak
- Single point and hierarchical grounding can be used to reduce conductive coupling
- RF grounding shielding are critical
- Shields are treated as separate conductors and are also connected with the chassis or system reference.
- Shields must not carry any system currents.

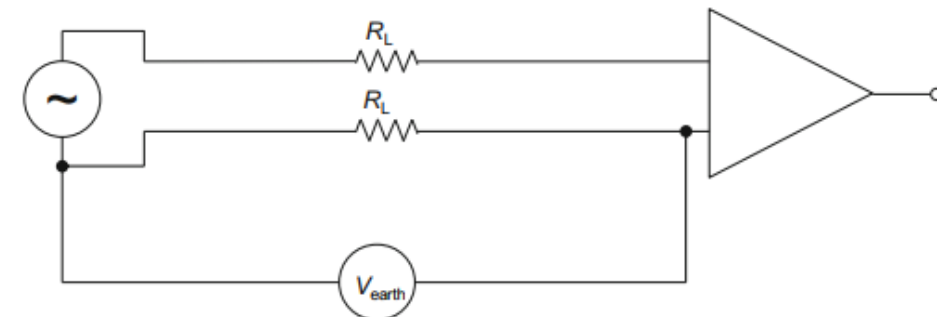


Ground loops

- If power systems are grounded in many places, potential differences will arise between the different earth grounds.
- These “earth currents”, which mostly originate from AC grid currents, are superimposed as mains hum on system signals in the event of a ground loop.
- The earth’s resistance is parallel to the line resistance R_L of the return signal conductor
- The same coupling issues arise if the chassis or rack is used as a return conductor for the power supply.
- If the reference potentials are unequal, a disturbance that is the same amplitude as the difference in reference potential is produced.

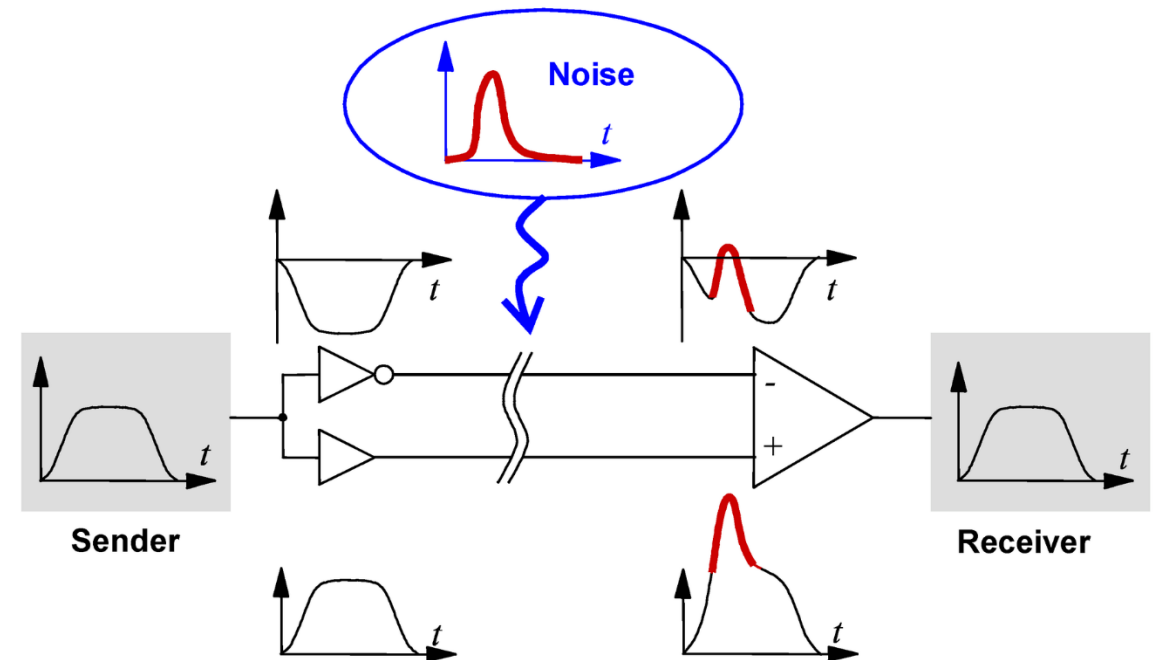


Equivalent circuit



Differential transmission

- Differential signal transmission (also: differential signaling) with two wires, which are not connected to the systems' ground conductors or are connected to a single point only, is a solution for these issues.
- The receiving circuit responds to the electrical difference between the two signals, rather than the difference between a single wire and ground (single-ended signaling).
- Symmetrical signal transmission ensures that the impact of disturbances affects both conductors (signals) identically.
- As the signal is transmitted with opposite polarity, the noise is subtracted away at the differential receiver. Variations in the reference potentials (grounds) of different systems are eliminated the same way.



Shielding from Fields-Shielding factor

- shielding is one of the key countermeasures to ensure EMC of systems
- An electrically conductive shield has the following key functions:
 - It blocks external electromagnetic fields from entering the system or attenuates them to such an extent that they cannot degrade system functionality (system immunity is thus assured),
 - It prevents internal electromagnetic fields from leaking out of the system or it attenuates them to such an extent that the outside is not disturbed (an approved system emission level is thus assured).
- Shields attenuate the electric and/or magnetic field intensity.
- Consider an external electromagnetic field impinging upon a system that is protected by a shield.
- The ratio of the remaining field intensity H_{in} or E_{in} behind the shield (“inside”) to the field intensity H_{out} or E_{out} of the external disturbance field (“outside”) is often referred to as **shielding factor S**.

...Shielding factor

- A dimensionless quantity between 0 and 1
- perfect shielding corresponds to a shielding factor of 0, while no shielding corresponds to a shielding factor of 1.
- The shielding factor for the magnetic field is given by
- The shielding factor for the Electric field is given by

$$S_H = \frac{|H_{in}|}{|H_{out}|}$$

$$S_E = \frac{|E_{in}|}{|E_{out}|}$$

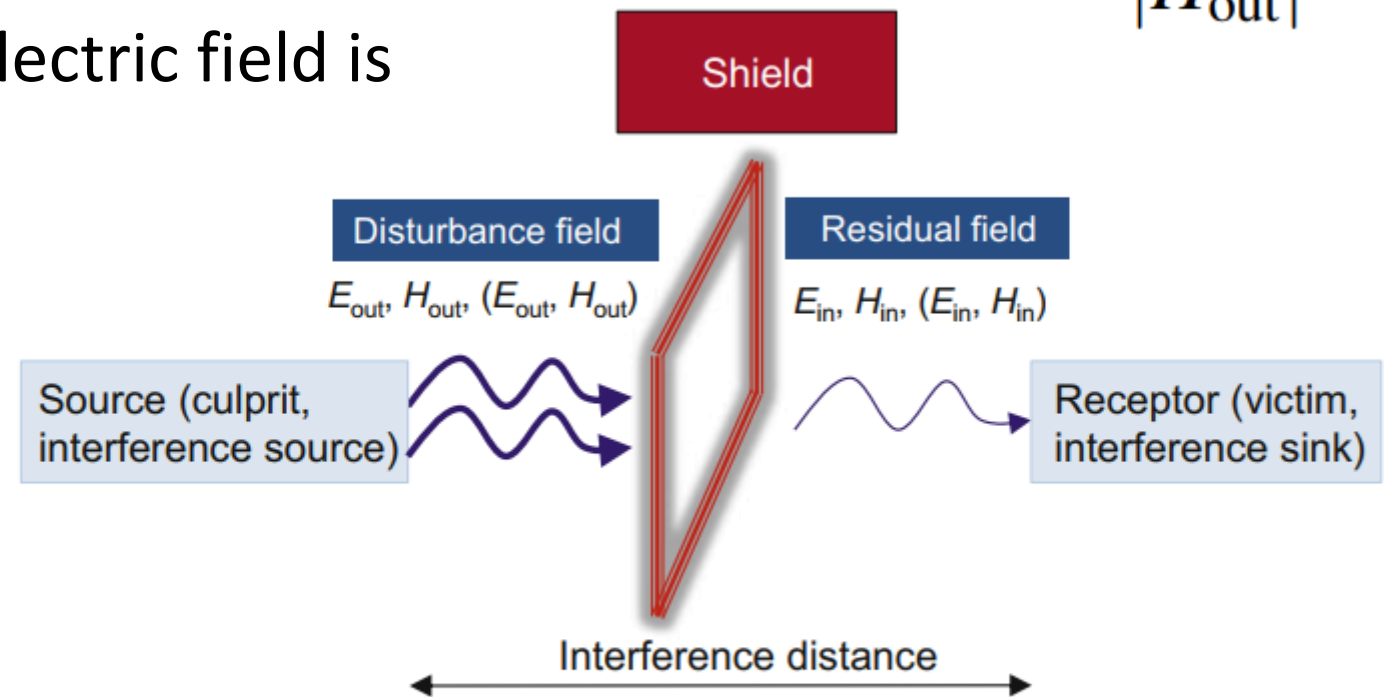


Fig. 6.20 Attenuating the disturbance by shielding (E_{out}, E_{in} external, internal electric field, H_{out}, H_{in} external, internal magnetic field, $(E_{out}, H_{out}), (E_{in}, H_{in})$ external, internal electromagnetic field)

Shielding effectiveness (SE)

- A more widely used metric, and dimensionless
- It is defined as “the ratio of the signal received (from a transmitter) without the shield, to the signal received inside the shield”
- It represents the insertion loss when the shield is placed between the transmitting antenna and the receiving antenna
- Often expressed in logarithmic form in decibels (dB) for magnetic and electric fields as:

$$SE_H = 20 \cdot \log_{10} \frac{|H_{\text{out}}|}{|H_{\text{in}}|} \quad \text{and} \quad SE_E = 20 \cdot \log_{10} \frac{|E_{\text{out}}|}{|E_{\text{in}}|}$$

- Shielding factors for very good shields approach zero, and the shielding effectiveness should be at least 80 dB for very good shields.
- **Ex.** At 80 dB shielding effectiveness, the ratio of the disturbing (outside) to the residual (inside) field is 10,000 to 1, which means 99.99% of the disturbance is blocked

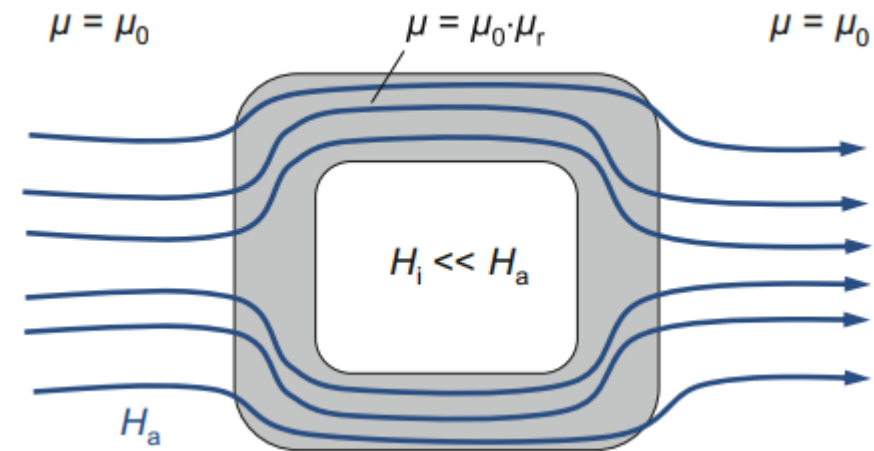
- The job for the engineer is to design a shielding with the most **effective shielding structure** and **with suitable shielding materials**.
- Shielding is required for systems of various shape, size and “type” and subsystems like filters, coils, transformers
- For far-field (large distance) electromagnetic coupling:
 - Shielding is bi-directional, i.e., it both protects the system from external interference and prevents the system from radiating outward.
- For near-field conditions (close to source): The same shield is used, but the design approach must consider field type, since the near-field may be:
 - E-field dominated (capacitive coupling) → easier to shield (conductive enclosure)
 - H-field dominated (inductive coupling) → harder to shield (requires loop minimization, thicker or magnetic materials)

Source types

- **Static fields:** constant over time and permanently distributed in free space.
 - all terms involving differentiation with respect to time vanish.
 - **electrostatic field:** charges, parallel-plate, fully-charged capacitor
 - **magnetostatic field:** from constant current, from a permanent magnet and Earth's field
- **Dynamic fields:** variable fields that change over time and they can also change both spatially and over time. They are complex fields, and Maxwell's equations in their entirety must be satisfied
- **Quasi-static fields:** class of dynamic fields for which certain features can be analyzed as though the fields were static.
 - Depending on the dominant static field is magnetic or electric, quasi-static fields are called **magnetoquasistatic** fields or **electroquasistatic** fields.
- **Electromagnetic (wave) fields:** made up of interdependent electric and magnetic fields, varying with time (dynamic fields). Size such that $\lambda \approx L$. Propagates over time in free space in the far field

Shielding Magnetostatic Fields

- Ferromagnetic materials with relative permeability $\mu_r \gg 1$ are used to shield magnetostatic fields.
- Permeability $\mu = \mu_0 \mu_r = 4\pi \times 10^{-7} \frac{Vs}{Am} \mu_r$
- magnetic reluctance of materials with high relative permeability μ_r is much lower or the magnetic conductivity is higher than materials in the surroundings, such as air or dielectric PCB material
- The magnetic flux follows the path of least reluctance “pulling” the flux into a highly permeable body, such as a shield made of a magnetic material.
- Magnetic field lines are always continuous and, in contrast to electric field lines, they do not terminate at the shield, but go through it.
- Thus, the disturbing field can be diverted around the object to be shielded through a magnetically conductive material (bypass effect).



Ferromagnetic materials for shielding magnetic fields

- Ferromagnetic materials have their own magnetic moments that are uniformly aligned in regions inside the material, called magnetic domains.
- Under the influence of an external magnetic field, these randomly oriented domains rotate in the direction of the field so that the permeability increases.
- If all the magnetic domains are oriented in the field direction, the magnetic flux is saturated. This field orientation, in the case of magnetically soft materials, is negated if the external field is removed. (Magnetically soft materials are easily magnetized and demagnetized.)
- Magnetically soft materials with high relative permeability μ_r are therefore used as magnetostatic shields

Table 6.1 Approximate relative permeability μ_r for different magnetically soft materials indicating their field distorting ability

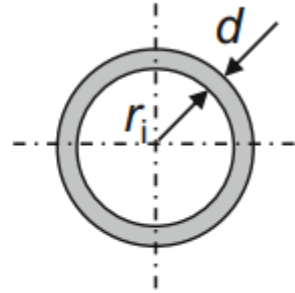
Material	Supermalloy	Mu-metal	Permalloy	Nickel–iron	Carbon steel	Nickel
μ_r	100,000	25,000	4500	1000	200	100

Shielding structures

- A shield must be able to allow magnetic field lines to pass through it without resistance and without saturating the material.
- Saturation occurs when the material's magnetization reaches its limit.
- Beyond this point its effective permeability collapses (it behaves almost like air), and additional flux leaks through the shield.
- On a B–H curve, this is the flat region where increasing H no longer increases B.
- A sphere—with its homogeneous shield material—is the most effective shape for a magnetostatic shield.
- Multiple shields made of a number of thin layers are a good choice for more demanding applications, as the shielding factor multiplies or the shielding effectiveness in dB adds up.
- An outer layer with high saturation field (ex. Silicon steel / low-carbon steel-lower permeability but higher saturation field $\sim 1.5\text{--}2\text{ T}$) and inner layer with higher μ_r (ex. Mu-metal- very high permeability, but low saturation field $\sim 0.7\text{--}0.8\text{ T}$) can give best shielding without saturation.

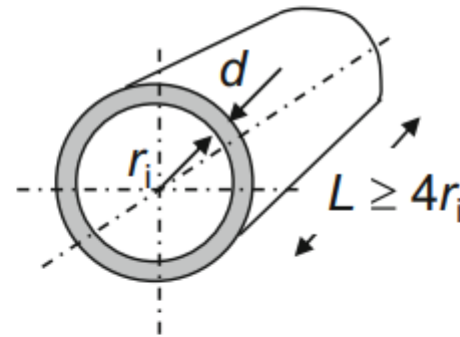
The shield effectiveness of a sphere, a cylinder, and a cube

Sphere



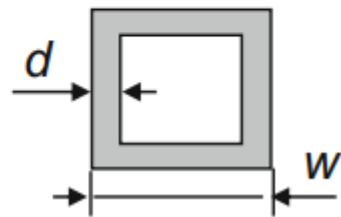
$$SE_H = 20 \cdot \log_{10} \left(1 + \frac{2}{3} \mu_r \cdot \frac{d}{r_i} \right)$$

Cylinder



$$SE_H = 20 \cdot \log_{10} \left(1 + \frac{1}{2} \mu_r \cdot \frac{d}{r_i} \right)$$

Cube (approximation)



$$SE_H \approx 20 \cdot \log_{10} \left(1 + \frac{4}{5} \mu_r \cdot \frac{d}{w} \right)$$

Magnetic shielding-avoid break in field path

- Bonded or press-fit connections should be used to join shield elements in order to avoid or reduce air gaps.
- Only (tungsten inert gas) TIG welding with ferromagnetic materials should be used for bonded joints, as a soldered joint would be equivalent to a magnetic air gap because solder is non-magnetic.
- Distances between welds in spot-welded joints and between screws in shields screwed together should be as small as possible to minimize air gaps.
- Magnetic shielding is used for shielding for frequencies in the 50 Hz–20 kHz range also.
- Layers of electrically-conductive metals like aluminum or copper are necessary for higher frequencies.

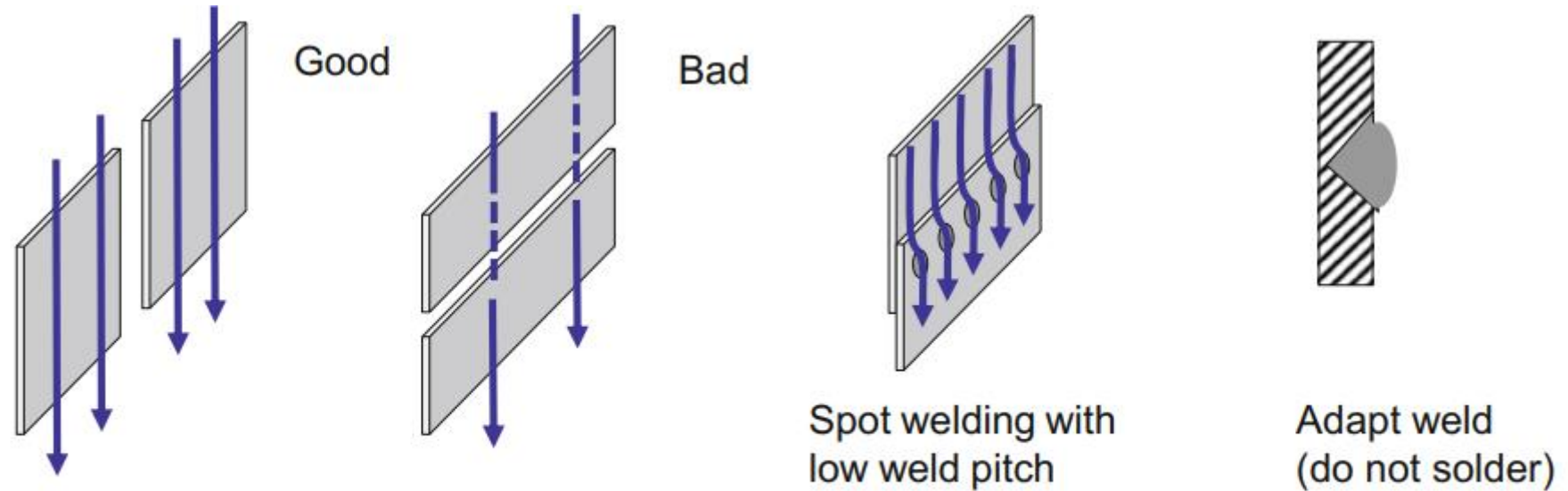
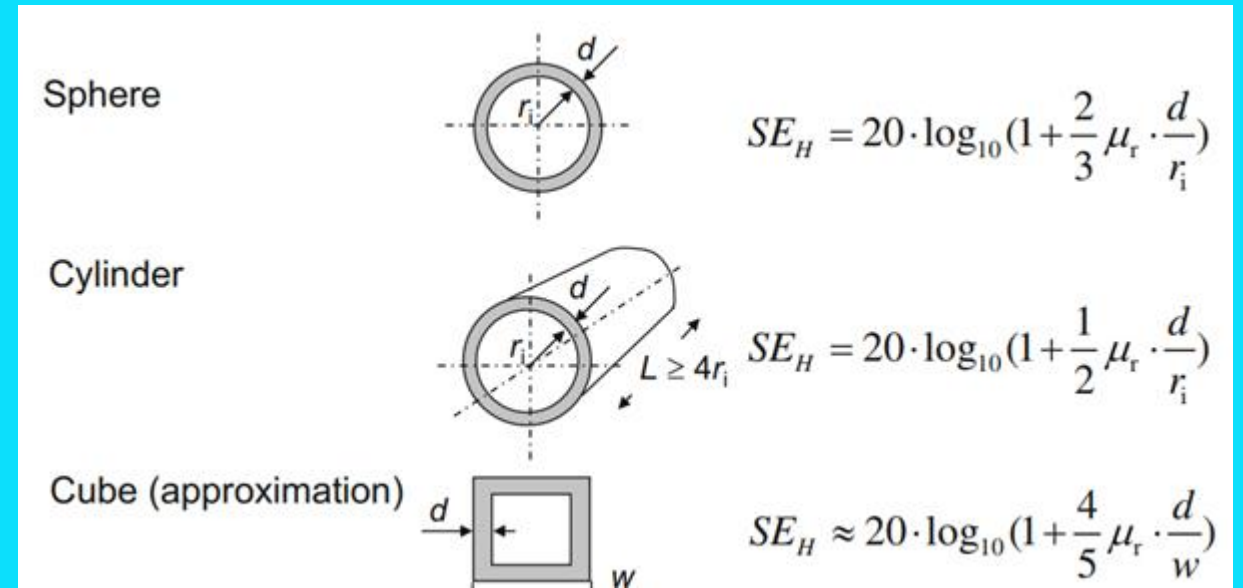


Fig. 6.24 Air gaps in the shield material should be avoided or they should be aligned with the disturbance to maintain a homogeneous field distribution (i.e., continuous field lines)

Q

- An iron sphere has $\mu_r = 200$, inner diameter $r_i = 50$ mm and thickness of 1 mm. (a) Find the Shielding effectiveness and shielding factor. (b) Compare with that for 10 mm thick sphere. Is the obtained shielding sufficient? (c) Compare for 1 mm cylindrical one with same radius and square shields with 100 mm side. (d) Compute the overall SE for two 1 mm shields one with inner radius of 52 mm and the second with 50 mm inner radius.

- (a) $SE_H = 20 \log_{10} \left(1 + \frac{2}{3} \mu_r \frac{d}{r_i} \right)$
- $SE_H = 11.2$ dB for 1 mm shield thickness $S = 10^{-SE_H/20} = .281 \Rightarrow 72\%$ reduction
- (b) $SE_H = 28.8$ dB for 10 mm shield thickness $S = 10^{-SE_H/20} = .04 \Rightarrow 96\%$ reduction
- $SE_H < 80$ dB which is insufficient



Q

- An iron sphere has $\mu_r = 200$, inner diameter $r_i = 50$ mm and thickness of 1 mm. (a) Find the Shielding effectiveness and shielding factor. (b) Compare with that for 10 mm thick sphere. Is the obtained shielding sufficient? (c) Compare for 1 mm cylindrical one with same radius and square shields with 100 mm side. (d) Compute the overall SE for two 1 mm shields one with inner radius of 52 mm and the second with 50 mm inner radius.

- (c) $SE_{Hcube} = 20 \log_{10} \left(1 + \frac{4}{5} \mu_r \frac{d}{w} \right) = 8.3 \text{ dB}$

- $SE_{Hcyl} = 20 \log_{10} \left(1 + \frac{1}{2} \mu_r \frac{d}{r_i} \right) = 9.5 \text{ dB}$

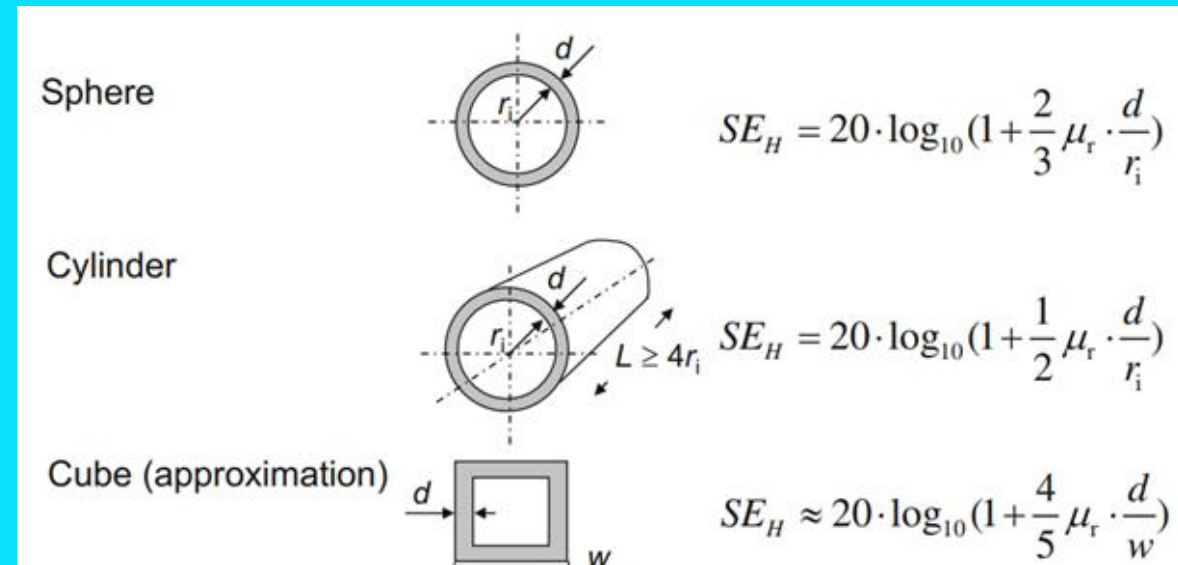
- (d) $SE_{H1} = 20 \log_{10} \left(1 + \frac{2}{3} \mu_r \frac{1}{52} \right) = 11 \text{ dB}$

- $SE_{H2} = 20 \log_{10} \left(1 + \frac{2}{3} \mu_r \frac{1}{50} \right) = 11.2 \text{ dB}$

- $SE_{H\text{overall}} = SE_{H1} + SE_{H2} = 22.2 \text{ dB}$

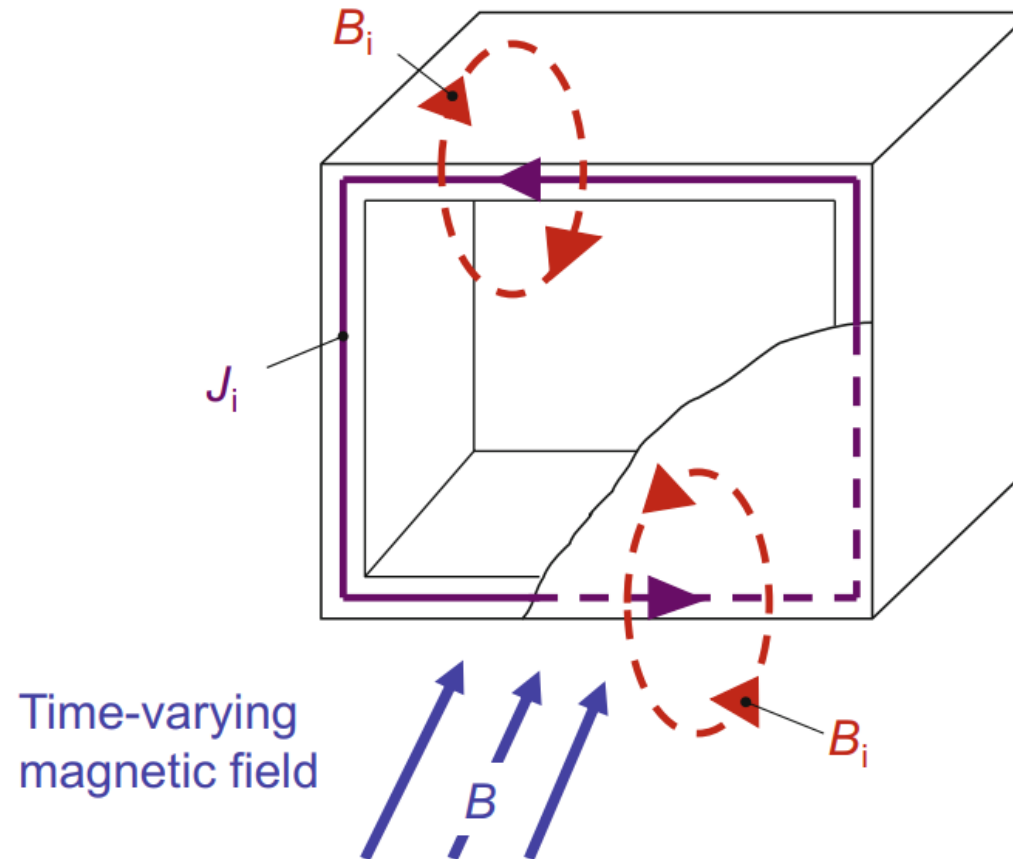
- $SE_{H2\text{mm}} = 20 \log_{10} \left(1 + \frac{2}{3} \mu_r \frac{2}{50} \right) = 16 \text{ dB}$

- Multiple single layer is better than one thick one



Shielding Magnetoquasistatic Fields

- A magnetoquasistatic field has a **dominant slowly alternating magnetic field**.
- Shielding should handle coupled electrical and magnetic processes:
 - **Law of induction**-Changing magnetic flux leads to induced EMF
 - **Ampère's circuital law**-Current or changing electric flux causes magnetic field.



Two mechanisms in Magnetoquasistatic shielding

Mode 1: Eddy current

- Change in magnetic flux density B with time $\rightarrow$ electrical eddy current field E_i in conductive shield
- E_i drives a current density $J_i \rightarrow$ magnetic flux density B_i that envelopes J_i
- B_i inside the shield is in the opposite direction of the primary flux B
- The disturbance field is thus attenuated inside the shielding enclosure

Mode 2: Skin effect

- Reduction in time-varying current and field from the outside to the inside of a metallic conductor is skin effect
- Skin effect is determined by frequency and the electrical material constants of the conductor (**permeability and conductivity**).

Skin depth

- The disturbance field can only penetrate to the **skin effect**.
- The **skin depth** d is a characterizes this depth of penetration.
- **Skin depth** is the distance from the surface at which the disturbance field is attenuated by a factor of “1/e” (that is, approximately 37% of its value at the surface; Euler’s number $e \approx 2.718$).
- The skin depth d is defined for good conductors as

$$\delta = \frac{1}{\sqrt{\pi f \mu \sigma}}$$

- frequency f , permeability $\mu = \mu_0 \mu_r = 4\pi \times 10^{-7} \frac{Vs}{Am} \mu_r$, and electrical conductivity σ of the shield material

Magnetoquasistatic Shielding material

$$\delta = \frac{1}{\sqrt{\pi f \mu \sigma}}$$

- Low skin depth δ and thus high shielding effect is achieved with
 - a high frequency disturbance field,
 - a highly permeable shield material, and
 - a shield material with high electrical conductivity
- Ferromagnetic conductors are needed for low frequencies since skin depths will be large for small μ and good conductivity σ alone may not give small δ
- The skin depth reduces with frequency so shielding due to high conductivity will be sufficient and non-magnetic shield will work

Table 6.2 Skin depth δ for different flat conductors and frequencies to illustrate the skin effect

Material	δ in mm				
	50 Hz	800 Hz	1 MHz	100 MHz	10 GHz
Copper	9.6	2.4	0.067	$6.7 \cdot 10^{-3}$	$6.7 \cdot 10^{-4}$
Aluminum	13.3	3.3	0.94	$9.4 \cdot 10^{-3}$	$9.4 \cdot 10^{-4}$
Iron ($\mu_r = 300$)	1.5	0.38	0.011	$1.1 \cdot 10^{-3}$	$1.1 \cdot 10^{-4}$
Mu-metal ($\mu_r = 25,000$)	0.333	0.084	$2.36 \cdot 10^{-3}$	$2.36 \cdot 10^{-3}$	$2.36 \cdot 10^{-5}$

Shielding effectiveness - Magnetoquasistatic

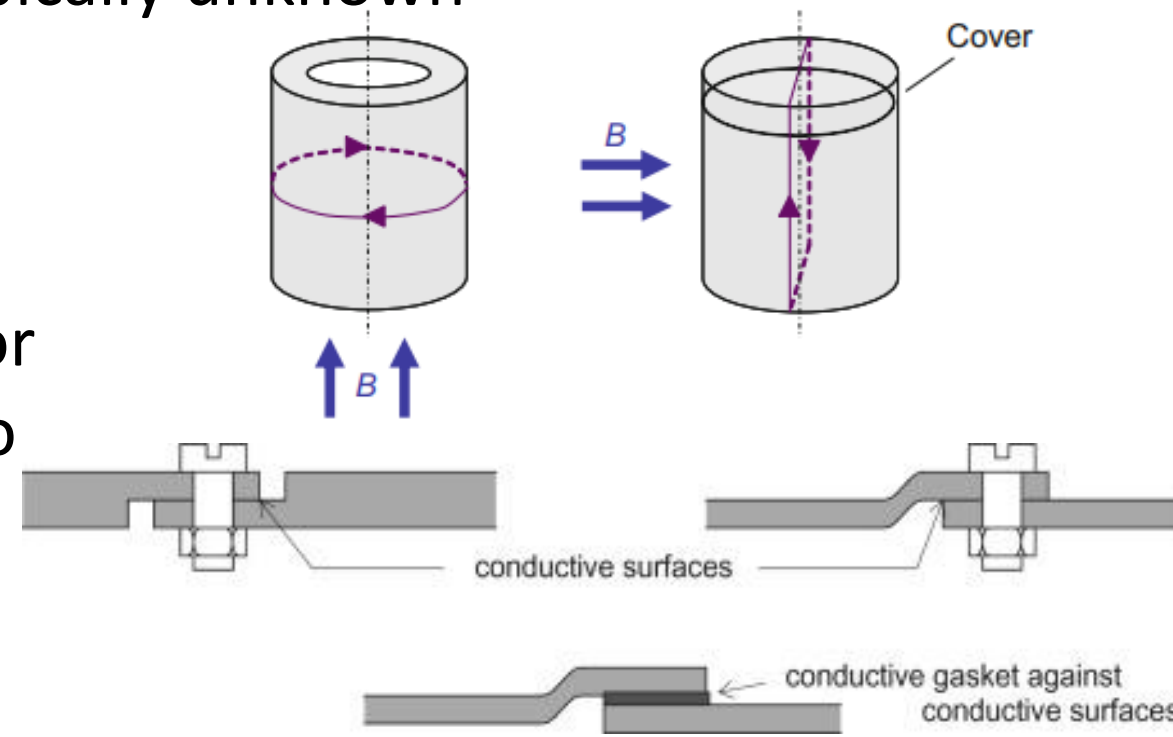
- The shielding effectiveness SE_H is calculated as:

$$SE_H = 20 \cdot \log_{10} \left(e^{d/\delta} \right) \approx 8.7 \frac{d}{\delta}$$

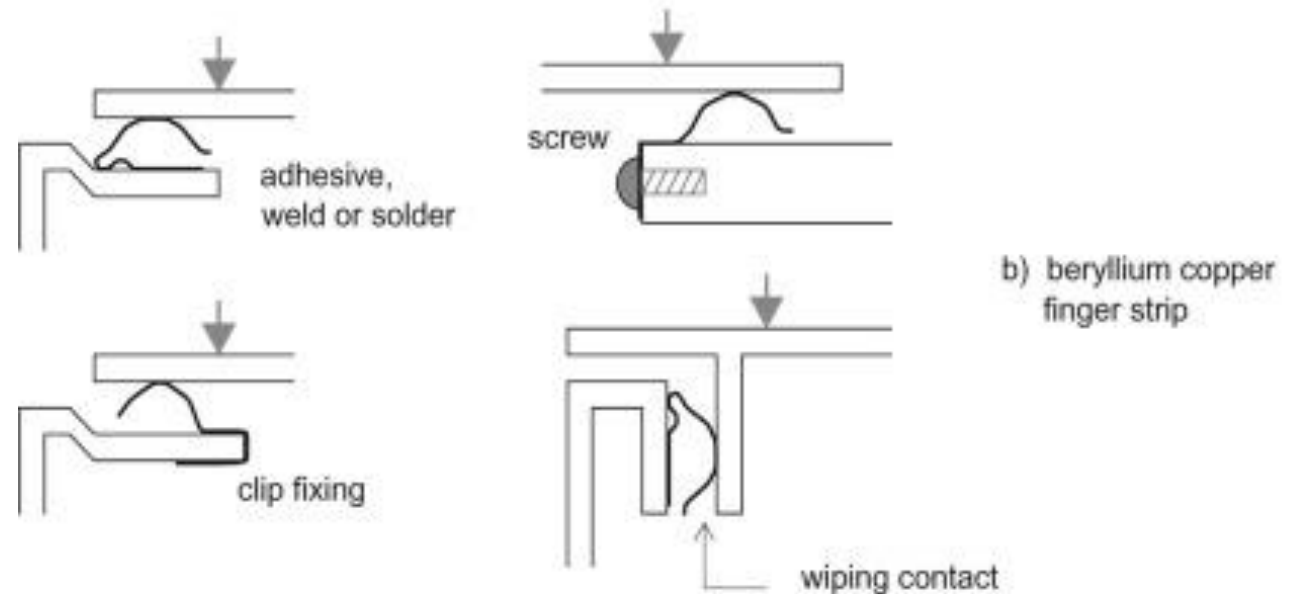
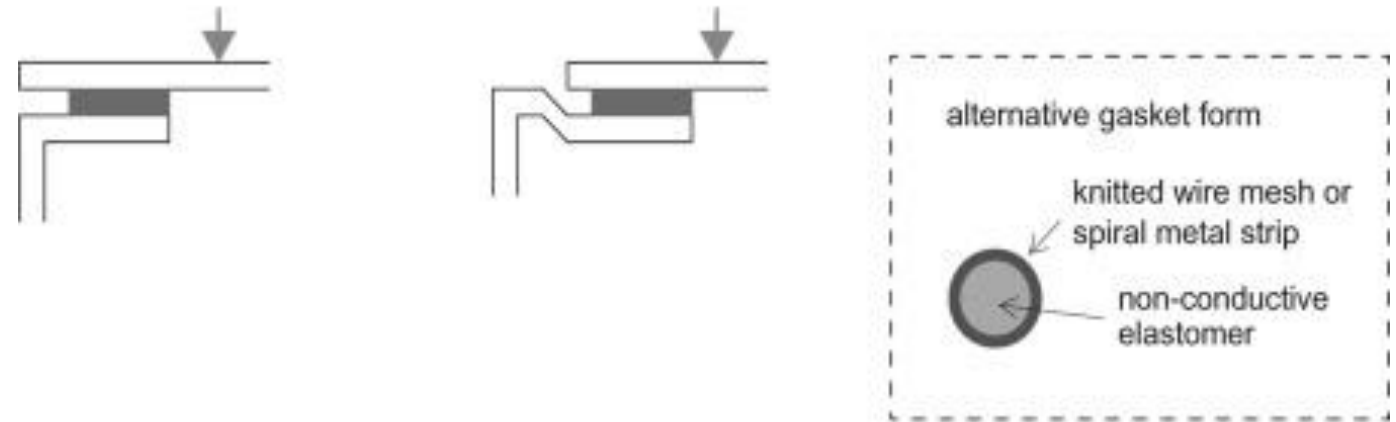
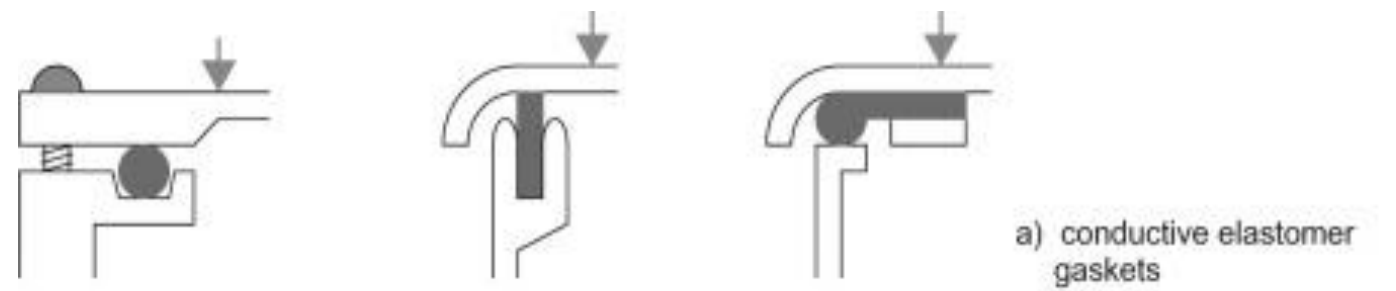
- with shield thickness d and skin depth $\delta = \frac{1}{\sqrt{\pi f \mu \sigma}}$
- A shielding thickness of $d \geq 5\delta$ is sufficient to provide attenuation to <1% giving a shielding effectiveness of $SE_H=43.4$ dB.
- Hence a shield thickness of $d \geq 5\delta$ can be taken as a rule

Shielding design - Magnetoquasistatic

- Shielding enclosures must be designed to allow **eddy currents** to flow freely
- Impedance of the shield should be reduced so that Eddy currents can flow freely.
- **Joints and seams should be normal to the direction of the disturbance field.**
- Metallic joints and seams should be smooth when they run in the same direction as the disturbance field.
- The direction of the disturbance field is typically unknown
- Hence the **resistances of joints and seams in the shielding enclosure should be kept minimum**
- Ex. a removable cover might be needed for the enclosure and the joint of the cover to the enclosure should have low resistance
- **Conductive contacts or seals** are recommended for such a scenario.



EMI shielding conductive gaskets

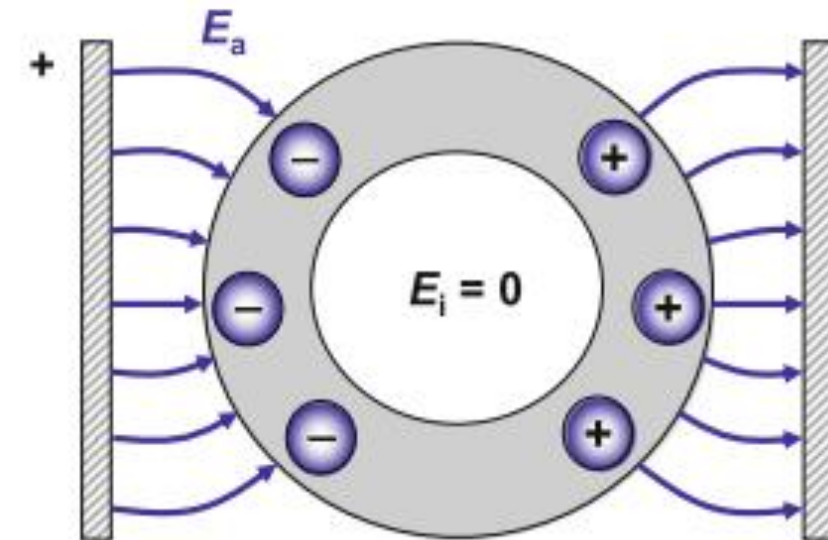


Q

- Calculate the shielding effectiveness for a shielding against a Magnetoquasistatic field when the shield thickness is 2δ . What is the skin depth of copper with a conductivity of 5.96×10^7 S/m at 10 MHz?
- $SE_H = 8.7 \frac{2\delta}{\delta} = 17.4$ dB
- Copper is non-magnetic giving $\mu_r = 1$
- $\delta = \frac{1}{\sqrt{\pi f \mu \sigma}} = \frac{1}{\sqrt{\pi \times 10^7 \times 4\pi \times 10^{-7} \times 5.96 \times 10^7}} = 20 \mu m$
- The above shield is then a $40 \mu m$ thick layer.

Shielding Electrostatic Fields

- The static charge is a source of an electric field
- In response to an external electric field, charge on an electric field shield redistributes itself
- As illustrated in induced charges on the outer surface of a conductive object will shield against electric fields as there is no surface charge along the inner surface.
- Thus, an electrically conductive object in an electrostatic field E_{out} is an ideal electrical shield since there is no field inside the body (electric field strength $E_{\text{in}} = 0$).
- This is also known as a Faraday shield.

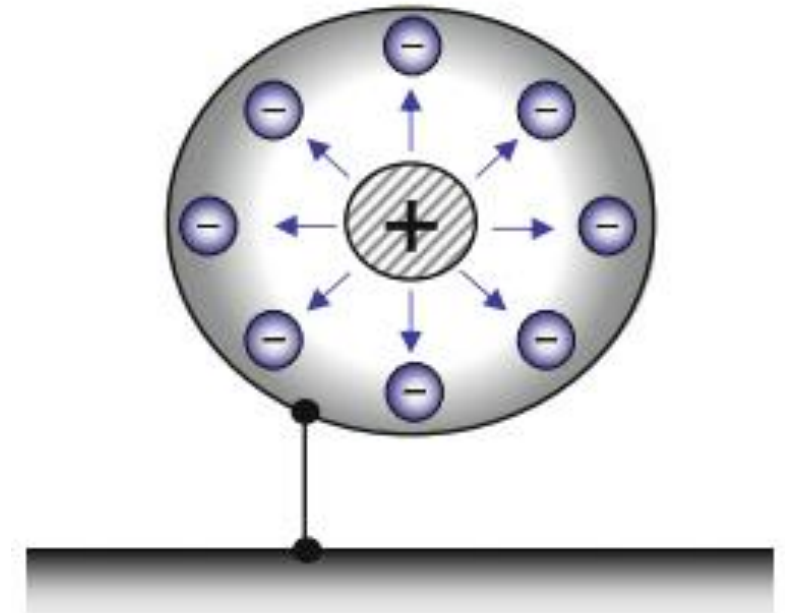
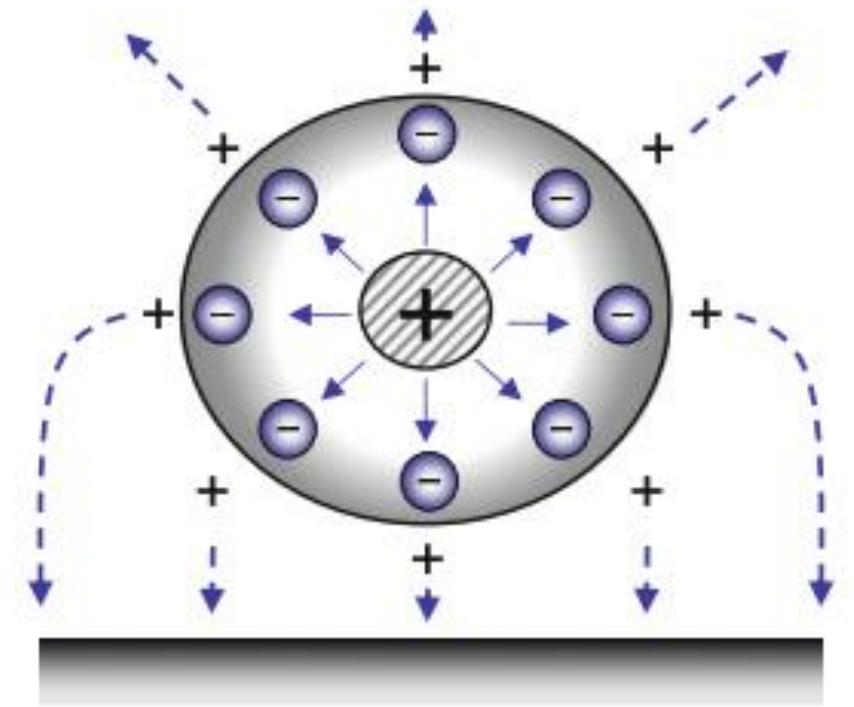


Electrostatic shielding-Faraday cage

- The charges are redistributed in the shield material by the forces $F = E_{\text{out}} Q$ acting on the mobile electric charges Q by the external field E_{out} .
- The charge redistribution is terminated when the field of the displaced charges and the external disturbance field cancel each other out in the shield interior.
- Thus, there are no potential differences on the inner surface of the shield, and $E_{\text{in}} = 0$ applies for internal field strength.
- Hence, the shielding effectiveness of a smooth electrically conductive shield tends to infinity ($SE_E = \infty$).
- The shield does not have to be made of a continuous material, but can also be a conductive metallic mesh-Faraday cage

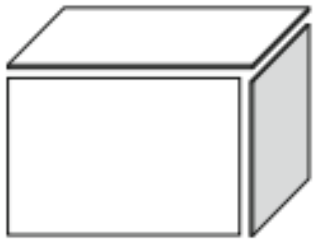
Grounded and ungrounded shell

- Ungrounded shell is not a shield
- A live positive conductor surrounded by a metallic shield has the field lines of the positive charge of the core conductor terminate at negative charges on the inner shield surface.
- New field lines emanate from the positive charges induced on the outer surface to the surroundings and end up at appropriate charges on the ground or the surrounding walls.
- Grounded shell is a shield
- If the shield is grounded, the field lines of the positive conductor also terminate at negative charges on the inner surface of the shield, while the external positive charges flow to ground.
- Conductor and inner surface of the shielding form a closed system from which no field lines emanate to the surroundings if the shield is a continuous entity.
- Coaxial cables are based on this design principle.

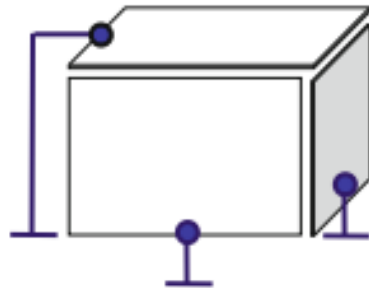


Ungrounded vs ungrounded shell with slow ac voltage

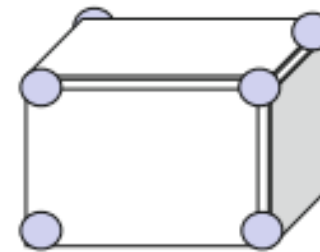
- For AC voltage, the internal field is continuously reversed
- **Grounded**: Induced charges on the outer shield surface are exchanged via the ground conductor.
- Impedance of shield and conductors should be minimized,
 - conductor cross sections should be large enough
 - line length to the reference conductors should be minimized
- **Ungrounded**: All side panels in an electrically conductive shielding enclosure must be joined at multiple points, particularly for the required equipotential points- to form a Faraday cage
- A minimum of three joints is needed.



Ineffective



Grounding



Connections for electrical
potential equalization

Dielectric shielding for electrostatic field

- A non-conductive (dielectric) object can also carry an electric flux W due to its “dielectric conductivity”, i.e., a relative permittivity $\epsilon_r \gg 1$.
- This bypass effect with respect to the surroundings with $\epsilon_r = 1$ yields $E_{in} < E_{out}$. Thick-walled masonry or plastics are thus suitable for shielding electrostatic fields as well.

•

Shielding Electroquasistatic Fields

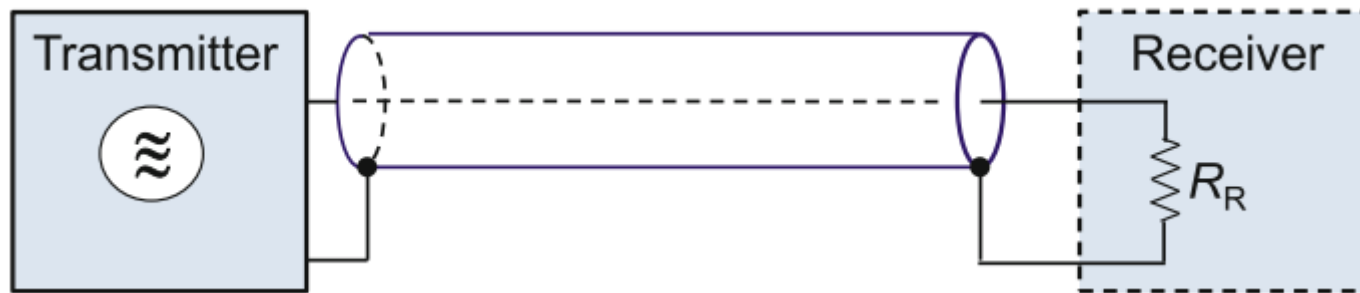
- electroquasistatic field - a slowly alternating electric field is dominant (approximately: $f < 100$ kHz)
- There is a phase shift in the charge distribution on the shield surface as the frequency increases, yielding a finite shielding effectiveness
- Shielding principles will be the same as electrostatic,
- but the shielding effectiveness will decrease from infinity available for a perfect 0 frequency.
- Electroquasistatic assumption is used when dimension meets $0 < L \leq \frac{\lambda}{10}$

Shielding electromagnetic fields

- When $L \geq \frac{\lambda}{10}$ use high frequency consideration with coupled capacitive and inductive effects. (λ is the shortest wavelength used)
- **Ex.** In digital electronics with an operating frequency of 100 MHz (the maximum disturbance frequency will be $f_s = 400$ MHz in this case), the conductor length $l_s = \frac{\lambda}{10} = 8$ cm for the start of EMI
- EMI spread by means of conductors (conductor-borne) and in free space (interference from radiation)

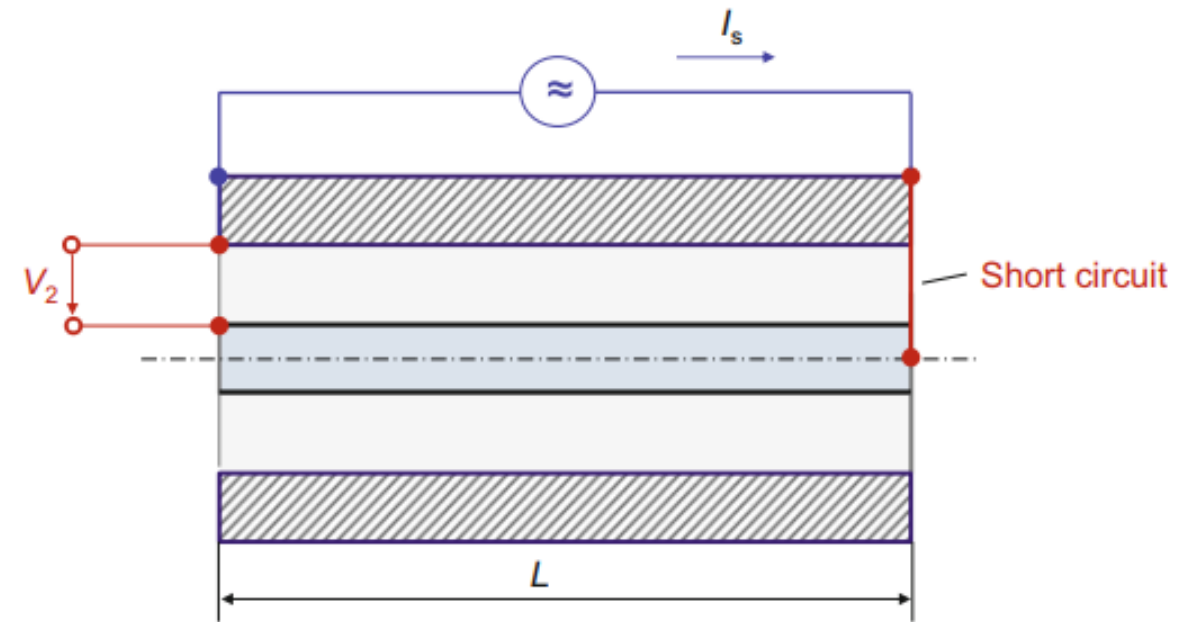
Shielding electromagnetic fields- conductor-borne

- **conductor-borne**: simultaneously acting capacitive and inductive interference between two conductors
- Mitigation of conductor-borne EMI:
 - Shields are used
 - Coaxial cables are a common solution: the core conductor carries the high-frequency signal and the exterior jacket carries the return signal.
 - Transmitter and receiver, in contrast to capacitive shielding, must connect their reference potential to the outer conductor of the coaxial cable “all-around.”
 - When the two opposite currents on the inner and outer conductors are balanced, the field emission from the coaxial cable is zero



Shielding electromagnetic fields- transfer impedance

- The transfer impedance Z_t is a measure of the shielding effect for conductor-borne EMI.
- Consider a coaxial cable of length $L < \frac{\lambda_s}{20}$ affected by a disturbance of wavelength λ_s
- If a parasitic current I_s flows through the (shorter) outer conductor, then, the voltage across the open-circuit input terminals of the conductor is V_2 when the conductor is shorted at the other end
- The transfer impedance per unit length Z_t is then
$$Z_t = \frac{V_2}{I_s \cdot L} \quad \text{m}\Omega/\text{m}$$
- Disturbance voltage is small if the transfer impedance is small

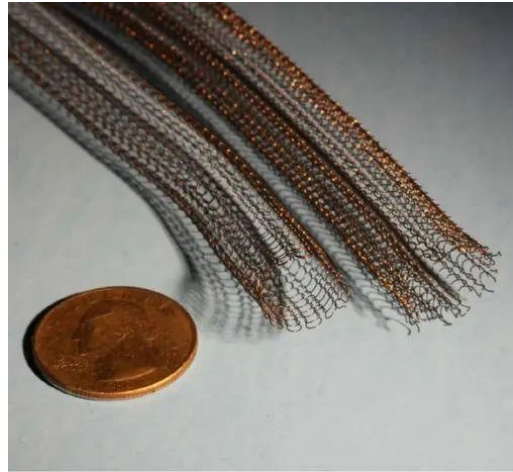


Shielding electromagnetic fields- transfer impedance

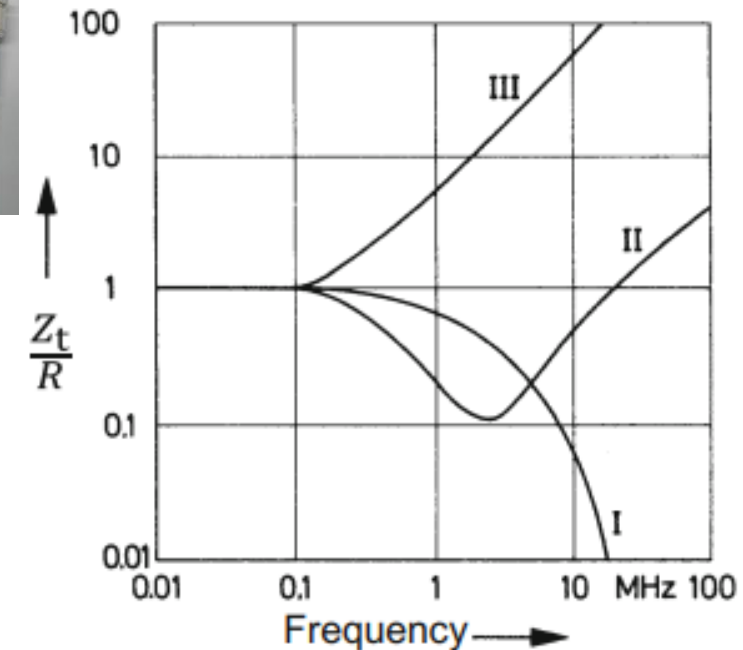
- The transfer impedance Z_t of the shield, normalized w.r.t. the DC resistance R , is different for different shielding types
- Case I homogeneous, RF-shielded conduit,

$$Z_t = \frac{V_2}{I_s \cdot L}$$

- Case II wire mesh,



- Case III helically wrapped tape or film)



<https://delikon.com/braid.html>
<https://www.indiamart.com/proddetail/rf-shielding-copper-mesh-24041349088.html>
<https://www.electriflex.com/products/type-ems/>

Shielding electromagnetic fields-Radiation-near vs far field

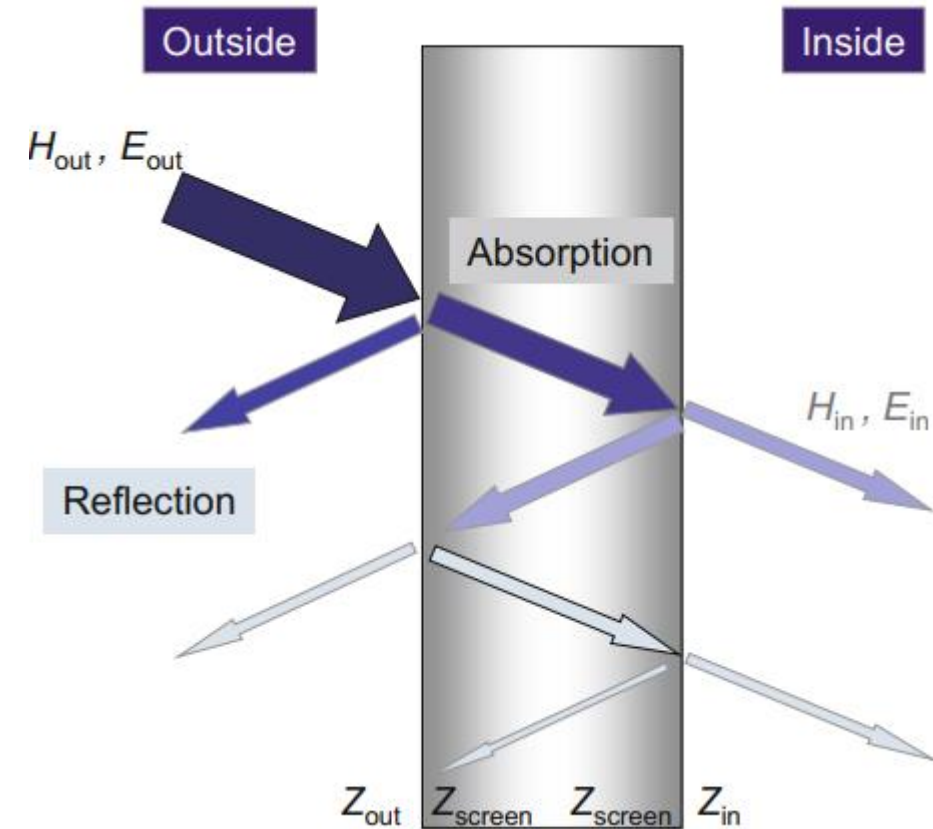
- In the case of interference from radiation, the culprit device generates electromagnetic waves
- The field and wave propagation characteristics depend on the distance r between observer and emitter.
- Near field, distance from emitter $r \ll \frac{\lambda}{2\pi}$,
 - The electric field is dominant for an open circuit.
 - The magnetic field dominates for a current loop
- Far field distance from emitter $r \gg \lambda/2\pi$
 - non-stationary field regardless of the type of radiating circuit
 - To calculate shield attenuations, the Maxwell equations must be solved
 - The Schelkunoff impedance concept is a much simpler approach.

Shielding electromagnetic fields- Schelkunoff impedance

- Three impedances play a role
 - Impedances in front of, behind, and in a shield.
- The shielding effect is based on
 - the **absorption loss A** due to absorption in the shield material,
 - the **reflection loss R** based on reflections at impedance boundaries Z_{out}/Z_{screen} and Z_{screen}/Z_{in} ,
 - A **re-flection coefficient B**, a reduction in attenuation (negative value) due to multiple shield penetrations caused by multiple internal reflections, ($B \approx 0$ for $A > 10$ dB)
- The shielding effectiveness SE can thus be expressed as

$$SE = A + R + B$$

- The **Schelkunoff impedance** concept and its application simplify the calculations
- The shielding effectiveness of electromagnetic radiation becomes a simple sum of absorption loss and reflection losses

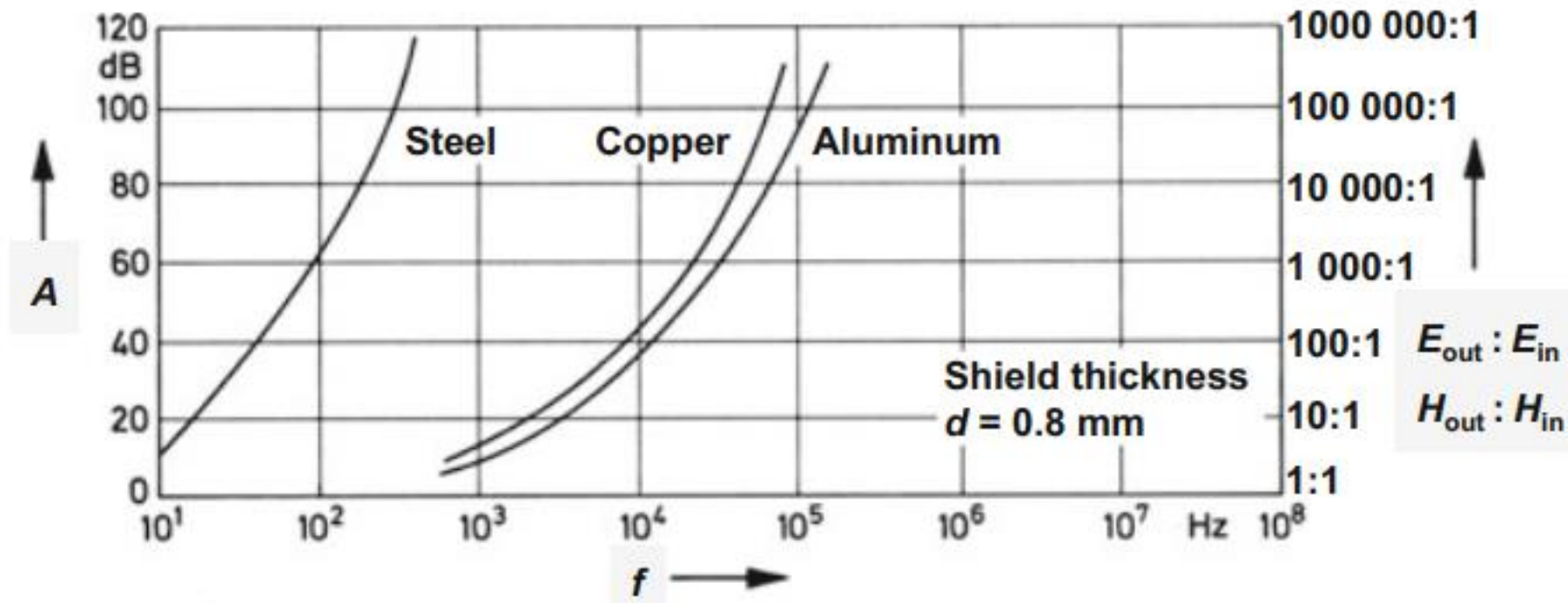


Shielding electromagnetic fields-Absorption loss

- The absorption loss A arises from the energy loss in the shield, resulting from the eddy currents generated by the radiation.
- The absorption loss A is directly proportional
 - to the shield **thickness**
 - to the **square root of the frequency**,
 - To the radiation barrier's **conductivity** and **permeability**.

$$A_E = 20 \cdot \log_{10} \frac{|E_{out}|}{|E_{in}|}$$

$$A_H = 20 \cdot \log_{10} \frac{|H_{out}|}{|H_{in}|}$$



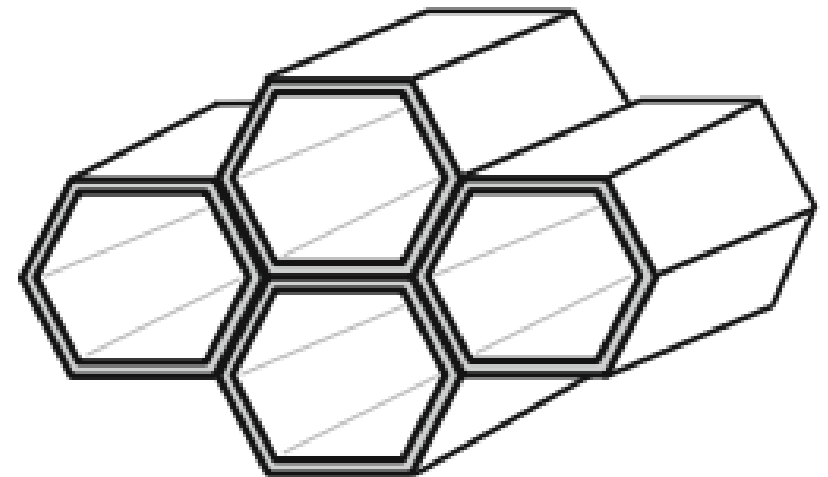
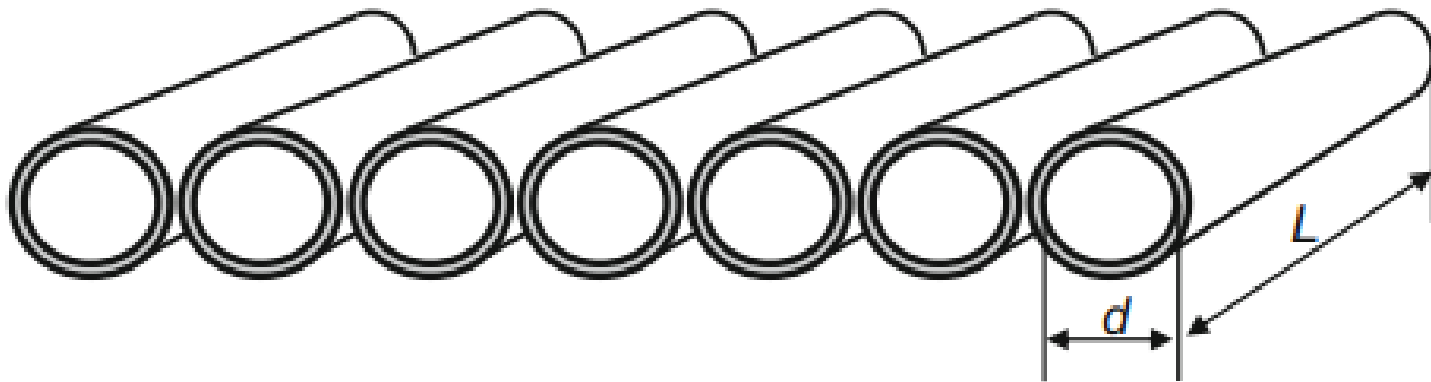
Shielding electromagnetic fields-Reflection loss

- The reflection loss R is caused by the partial reflection from the shield.
- It depends on the **conductivity** and **permeability** of the metal barrier, the **frequency**, and the **distance** of the shield from the culprit device.
- The enclosure should be completely sealed to allow **eddy currents** unrestricted passage.
- Joints should be **electrically bonded** along their entire length by means of soldering
- **Openings** significantly reduce the shielding effectiveness.
- **Many small** ($\ll \lambda$) openings are preferred to a few large ones
- **Best : Round holes | better : square | worst: rectangular ones.**
- Radiation emanating from a circular hole produces a **field strength proportional to the cube of the diameter** of the hole.
- Hence, the field strength increases by a factor of eight if the diameter is doubled.
- The **largest perforation should be $\ll 30 \text{ m} / f$ (in MHz) or $\ll \frac{\lambda}{10}$**
- At $30 \text{ m} / f$ (in MHz or $\frac{\lambda}{10}$) SE becomes $\sim 0 \text{ dB}$ (no shielding)

Shielding electromagnetic fields

- Deep honeycomb structures work like waveguides while allowing air path
- Waveguides act like high pass filters that will allow only high frequency ($f > f_c = c/2d$ for a hole of side d) to go through.
- The shielding effectiveness for an L m long d m wide honeycomb unit cell is

$$SE \approx 32 \times \frac{L}{w} \times \sqrt{1 - \left(\frac{f}{f_c} \right)^2}$$



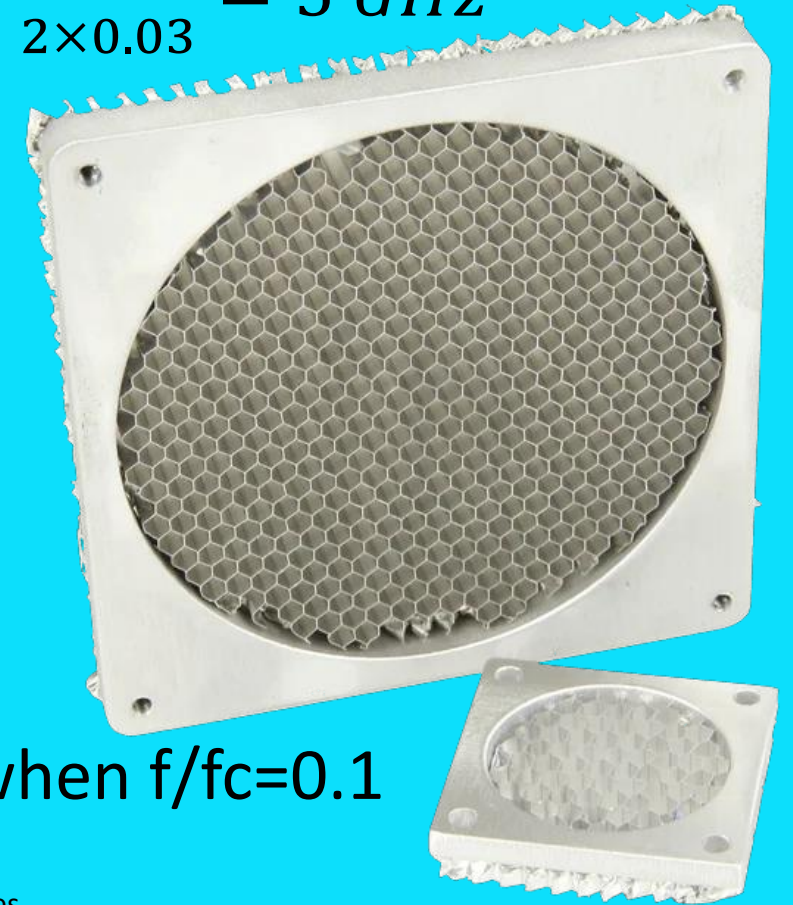
Q

- Calculate the shielding effectiveness of a honeycomb structured shield of width 3 cm and depth 12 cm at 500 MHz

- Cutoff frequency of the waveguides formed $f_c = \frac{c}{2d} = \frac{3 \times 10^8}{2 \times 0.03} = 5 \text{ GHz}$
- $f/f_c = 0.5/5 = 0.1$
- $L/d = 12/3 = 4$

$$SE \approx 32 \times \frac{L}{w} \times \sqrt{1 - \left(\frac{f}{f_c}\right)^2}$$

- $SE = 32 \times 4 \times \sqrt{(1 - 0.1^2)} = 127 \text{ dB}$
- So a depth/width ratio of 4 can give a SE of > 100 dB when $f/f_c = 0.1$

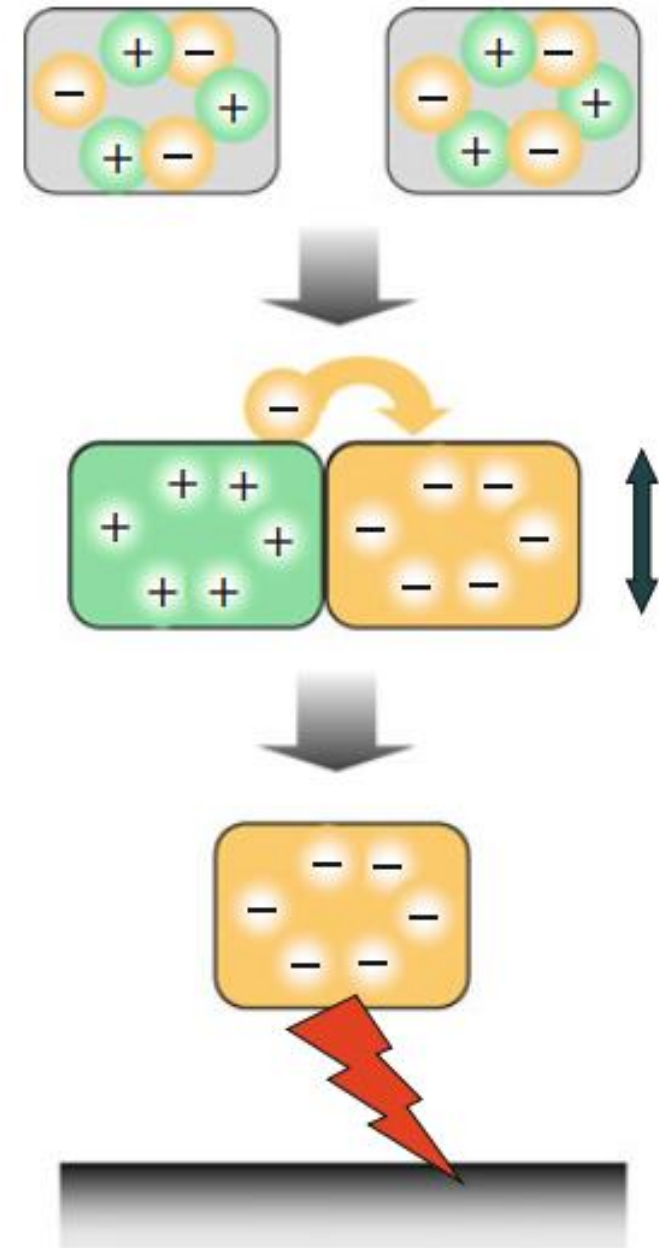


ESD

- Electrostatic discharge (ESD) is the sudden flow of electricity between two electrically charged objects.
- It is an electrical discharge from an electrically isolated material with a high potential difference (**typically 3 kV and higher**, which is much higher than the damage threshold of electronic devices) that causes a very short and high electrical current pulse.
- ESD must be considered in electronic system development to prevent system failures.
- Microelectronic devices are very vulnerable with **5-30 V cutoff voltages**.
- Components can be severely damaged or partially degraded by sparkovers
- Measures need to be put in place to prevent even small amounts of static build-up.

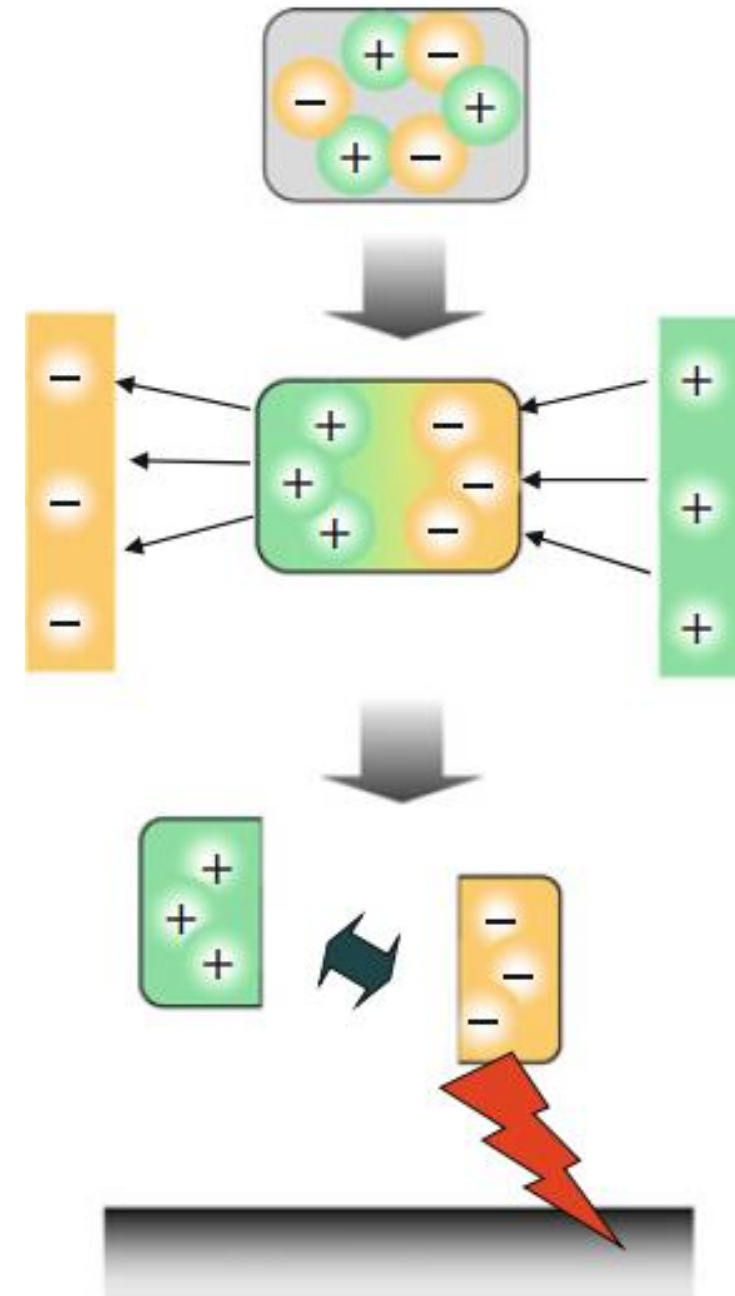
ESD-Causes-Triboelectricity

- Potential difference primarily caused by the **triboelectric effect** (also known as charging by friction).
- Two **materials with different Fermi levels are in frictional contact**, and the electrons tend to move from the material with a high Fermi level to the one with a lower.
- Hence, charges are transferred between the bodies while they are in contact; the electron charges are unbalanced when the bodies are separated again.
- Because the body's surface is now electrically charged, either negatively or positively, any contact with an uncharged object, such as a grounded body, may cause an electrical discharge of the built-up static electricity



ESD-Causes- Induction

- An ESD event can also be triggered by induction
 - Electronic components polarized due to the vicinity of an electric field.
- Such a field can come from equipment or feed belts that have been previously charged with an electrostatic charge.
- Even though the net electrostatic charge of the object has not changed, it now has regions of excess positive and negative charges.
- When components are separated from the object, for example, components from a feed belt, a potential difference is produced with the risk of ESD

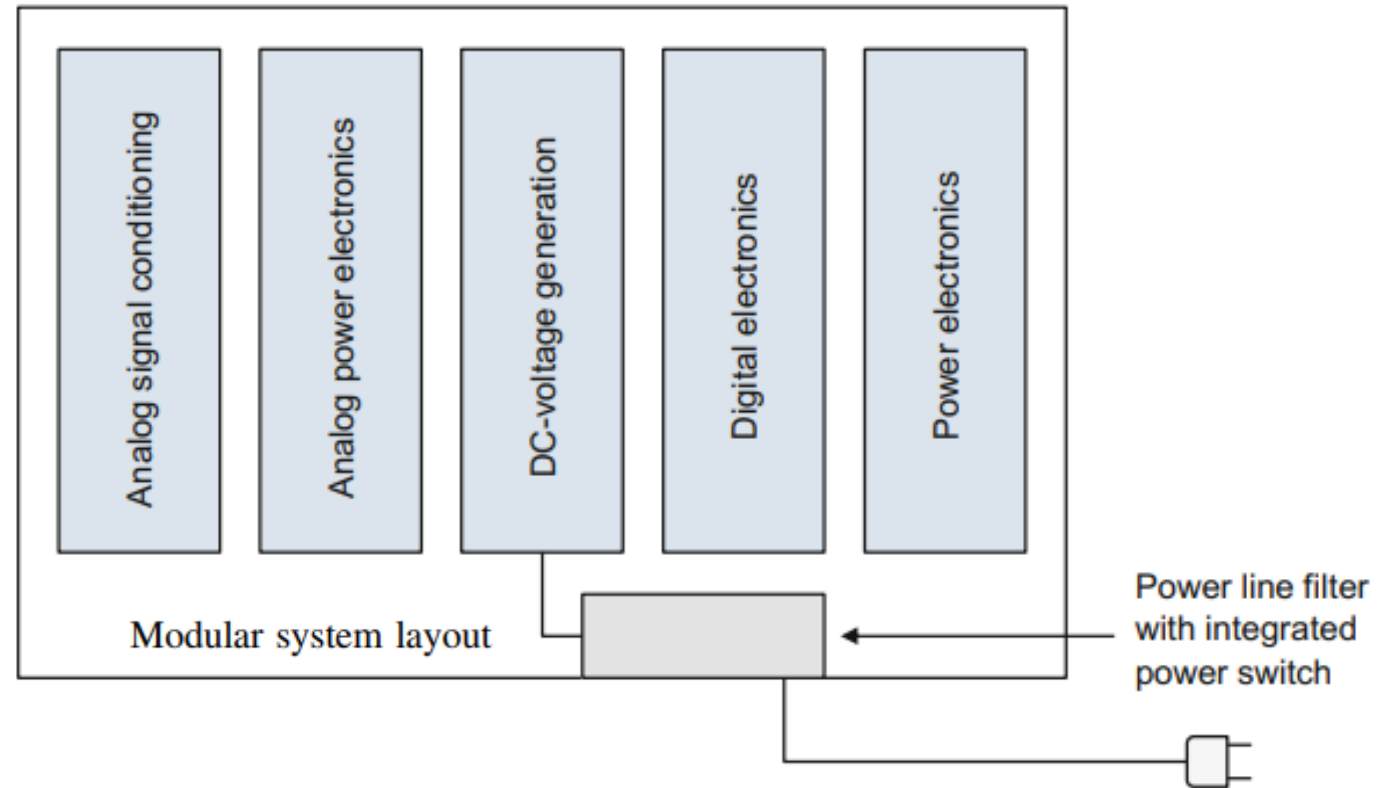
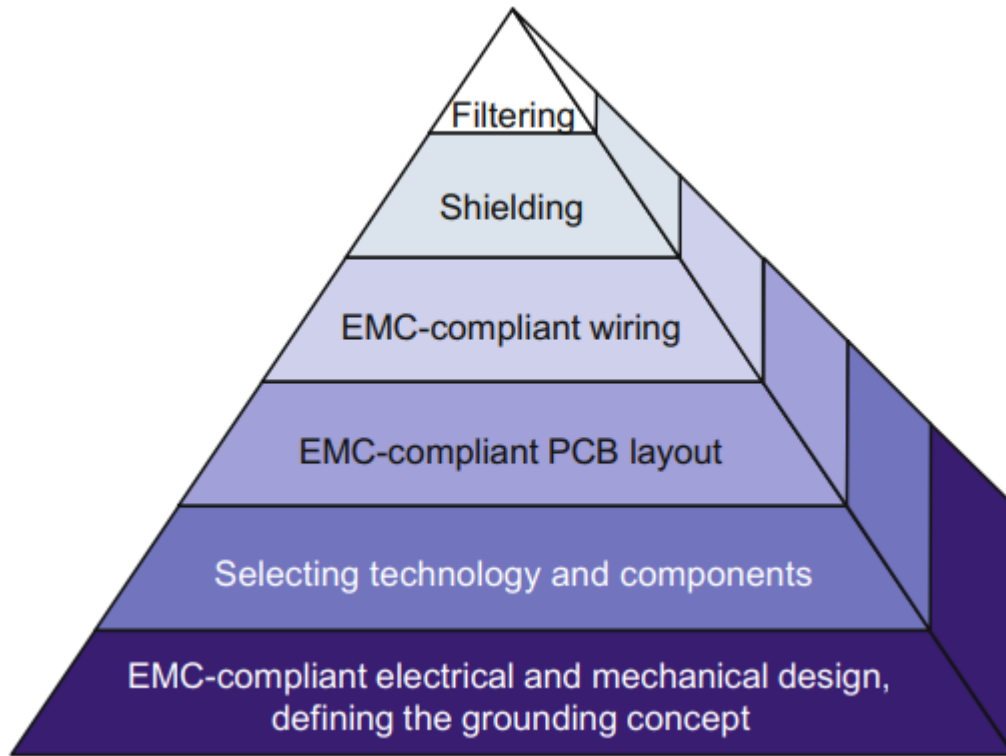


ESD-Suppression

- prevented by
 - safely **discharging** unavoidable parasitic static build-up and
 - **minimizing the build-up** of electrostatic charge in the vicinity of electronic devices
- integrated protection circuitry ensures that static build-up is discharged within electronic components
- **electrostatic discharge protected area (EPA)** comprises a range of protection measures for manufacturing and handling these components
- EPA-classified manufacturing areas
 - electric**conductive filaments on garments** ally **conductive and grounded workbenches**,
 - worn by assembly workers,
 - **conductive wrist and foot straps** and **floor covering**, **ionized air** and the **grounding** of all devices.
 - The **air humidity** should be at least 50%.
- **Packaging** for ESD sensitive components should be made of **electrically conductive and electrostatically dissipative plastics** – acts as a Faraday cage
- Conductive or metalized films, padding, and foams are typically used
- External pins for devices are connected with a **shorting jumper** for handling and transportation

- NPTEL course on EMI/EMC
- <http://www.digimat.in/nptel/courses/video/108105375/108105375.html>
- <https://nptel.ac.in/courses/108106138>
- Government EMI/EMC testing facilities
- <https://www.stqc.gov.in/en/emiemc-testing>
- <https://www.stqc.gov.in/ertlw-mumbai-0>
-

Summary-Designing for EMC



Rule 1: Physically group functional units together (and thus separate them from other groups) on a printed circuit board or in a system (Fig. 6.38).

Rule 2: Power supply lines should be filtered at the input (connector) on a printed circuit board.

Rule 3: Different types of circuits (analog, digital, power supply) should be grounded separately. Use multi-layer PCB with one or more ground planes for 10 MHz clocks or higher. It is wise to have ground tracks in place to the left and right of RF signal lines if you cannot use separate ground planes for cost reasons.

Rule 4: Keep the areas enclosed by loops at the power supply to a minimum. Small loops cap radiation and enhance immunity (Fig. 6.39).

Rule 5: Avoid current loops. Circuits require signal and return conductors; run signal and return conductors close to each other on a printed circuit board (Fig. 6.40). EMI and interference coupling are approximately proportional to the loop area.

Rule 6: Implement defined return paths for currents. Current always flows through the path of lowest impedance. A return current path always establishes itself in the proximity of the (forward) signal path in the case of RF signals. Defined return current paths are needed to mitigate radiation and immunity issues. The return signal automatically takes the path of lowest impedance on a printed circuit board with ground plane; coupling between two circuits via the impedance of the return current path is generally low (Fig. 6.41 left). Run the return line near the signal line

on PCB with no ground plane, ideally taking an identical route on a different layer. Implement the return line in the same way as the (forward) signal line when both lines cross (Fig. 6.41 on the right).

Rule 7: Place a bypass or energy-storage capacitor (ceramic, 0.001–1 μF) in the vicinity of every load to keep current spikes local and to prevent currents from flowing through long conductor loops (Rules 4 and 5). Choose the resonance frequency of the capacitor so that it is higher than the maximum signal frequency.

Rule 8: If there is a break in the ground plane in a printed circuit board, route the (forward) current trace around the break (Fig. 6.42).

Rule 9: If you want to provide capacitive decoupling between two signal traces on a board, introduce a grounded trace between the two traces (Fig. 6.43).

Rule 10: Route lines with rapidly switching signals, that is, with high current spikes (di/dt) or high voltage spikes (du/dt), away from “sensitive” lines, such as analog inputs.

Rule 11: We recommend keeping clock lines short, and laying them square to signal lines. If you want to take clock pulses from the printed circuit board, place the clock generator as near as possible to the board connector (plug).

Rule 12: Decouple aggressive circuit components with RC, LC or RLC filters to stop disturbances from impacting the rest of the circuit. Do not forget to consider the resonance frequency.

Rule 13: Use terminated transmission lines based on their characteristic impedance (using what is known as the microstrip technique) in order to connect components on PCB for RF applications (clock > 100 MHz). Make every effort to avoid reflection and transition points.

Rule 14: Lines longer than $\lambda/10$ (λ wavelength of clock frequency) that have not been matched should not be used.

Rule 15: Use small surface-mount devices (SMD) where possible, as they have a much better RF response. Keep line lengths associated with devices as short as possible to reduce series inductivity. Standard capacitors with interconnecting leads exhibit their first natural resonance at approximately 80 MHz.

Rule 16: Line drivers with symmetrical output (differential signaling; the signal difference is based on the 0 V potential) improve signal integrity, significantly cut interference emission and enhance immunity.

Rule 17: If several ICs are to be driven from one logic output, for example, routing a clock signal to a number of devices, split the trace near the devices (Fig. 6.44). The common line can be matched to its characteristic impedance at the distribution point, if needed.

Rule 18: Input and load capacitances should be as low as possible. This helps reduce charging currents for the state change, which in turn reduces magnetic-field radiation and ground return currents.

Rule 19: You need to prevent demodulation issues in analog circuits. Most EMC issues in analog semiconductor devices are caused by RF-signal demodulation. Analog circuits must be stable and continue working during a high-frequency disturbance event to prevent demodulation. This can only be achieved with an input filter or a feedback circuit.

Rule 20: Categorize lines on the printed circuit board and system wiring according to their signals in a similar manner as at the system level (e.g., clock, power, digital logic, or analog), and use a different wiring path for each category. Lay the different categories as far apart as possible, near the chassis ground, and successively from susceptible/non-disturbing to non-susceptible/highly disturbing.

Rule 21: A low-stray field arrangement can easily be achieved by using multi-wire flat ribbon cables. Strap them to the chassis ground. If too much cable is needed to run the cables to chassis ground, or if no chassis ground is available, running a metal foil underneath the ribbon cable will bring a significant improvement (imaging principle).

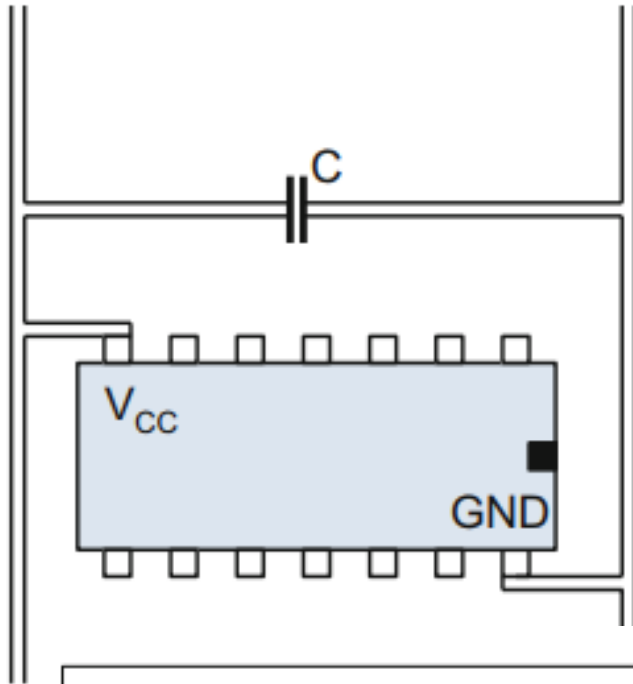
Rule 22: Eliminate all EMC passive conductors in an electronic system.

Rule 23: The power switch should, if possible, be integrated in the line filter. Use an LED on the low-voltage side to indicate power on.

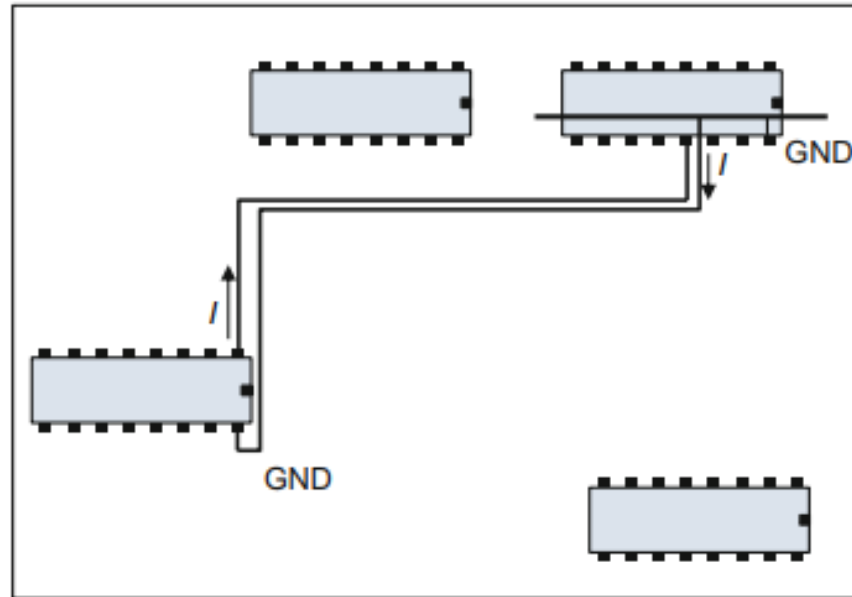
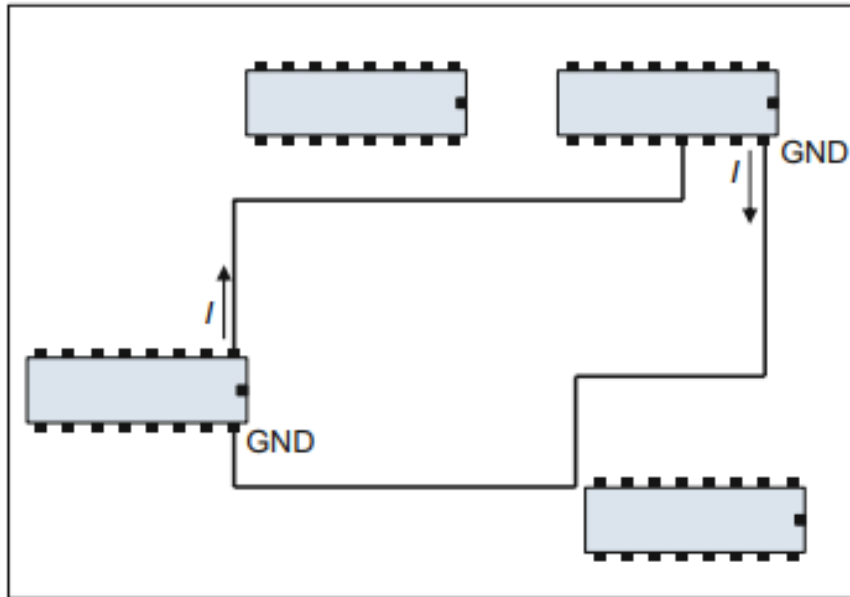
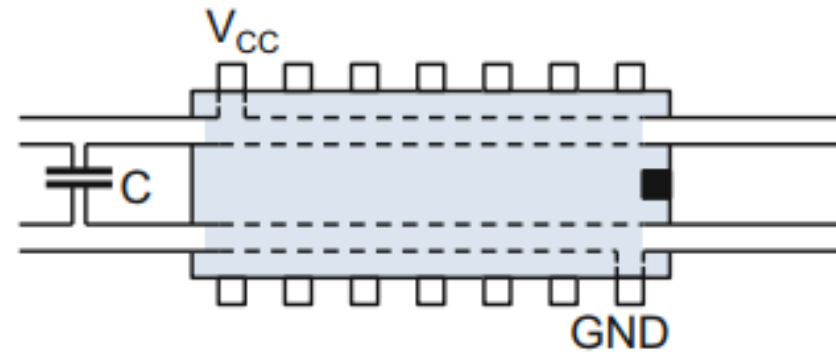
Rule 24: The following shielding rules should be observed when designing the shielding enclosure:

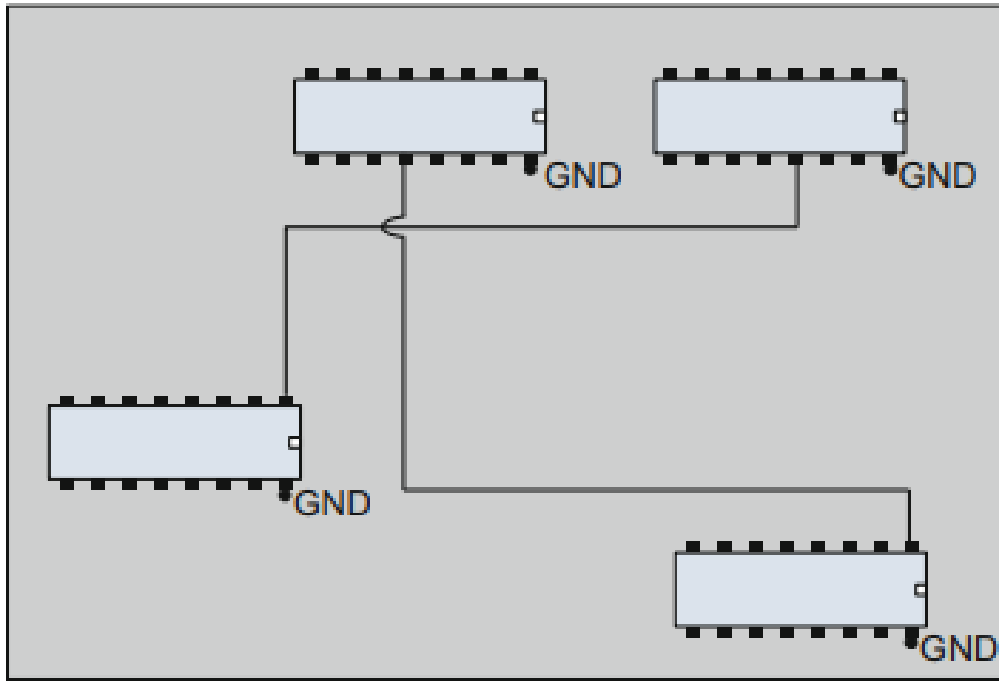
- Low-frequency electric fields (1 MHz max. at system level) can be shielded by a thin-walled metallic enclosure or a plastic enclosure with metal coating.
- Low-frequency magnetic fields (1 MHz max. at system level) require thick-walled metallic enclosures; highly permeable materials are needed for power frequency magnetic fields.
- The leaks (holes, slots) determine the shielding effectiveness more and more with increasing frequency. The perforation with the largest extension d determines the scale of the reduction in shielding effect. The shielding effectiveness reduces to 0 dB (i.e., no shielding) for a perforation with an extension d (in m) $\geq 30 \text{ m} / f$ (in MHz) (“ $\lambda/10$ rule”).
- Since the largest extension (in one direction) determines the shielding efficiency of shields with perforations, many small holes are better than a few large ones, and circular apertures are better than square ones.

Rule 25: The overall goal of all EMC measures for electronic systems should be to strengthen the electronics so that the emission limits and immunity requirements for the electromagnetic surroundings “budget” meet EMC standards without the need for an additional shielding enclosure.

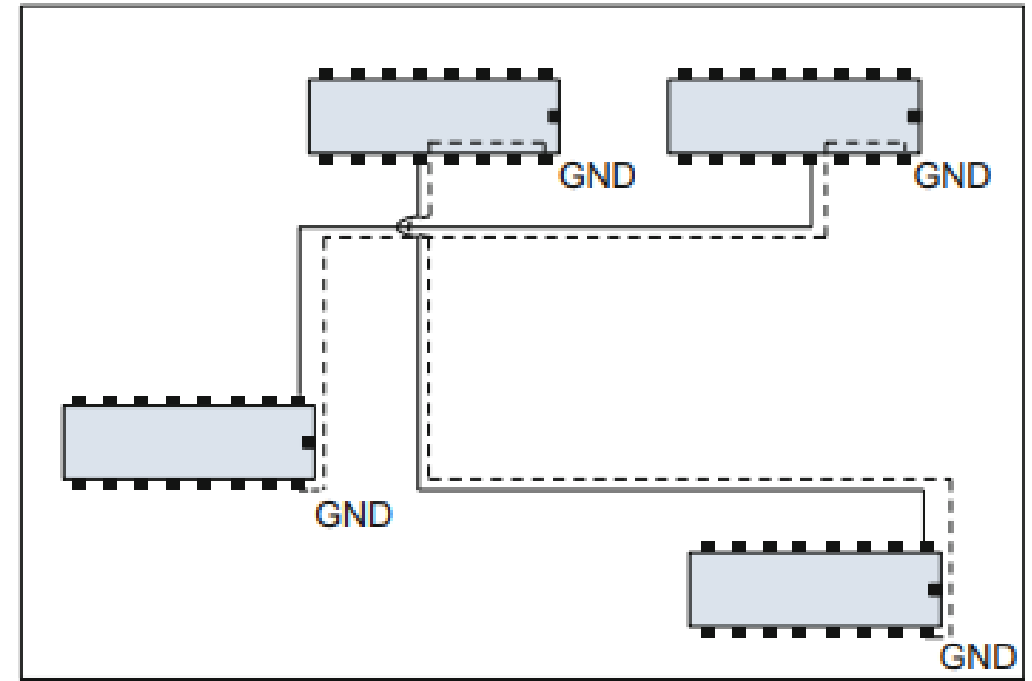


Reduce loop area



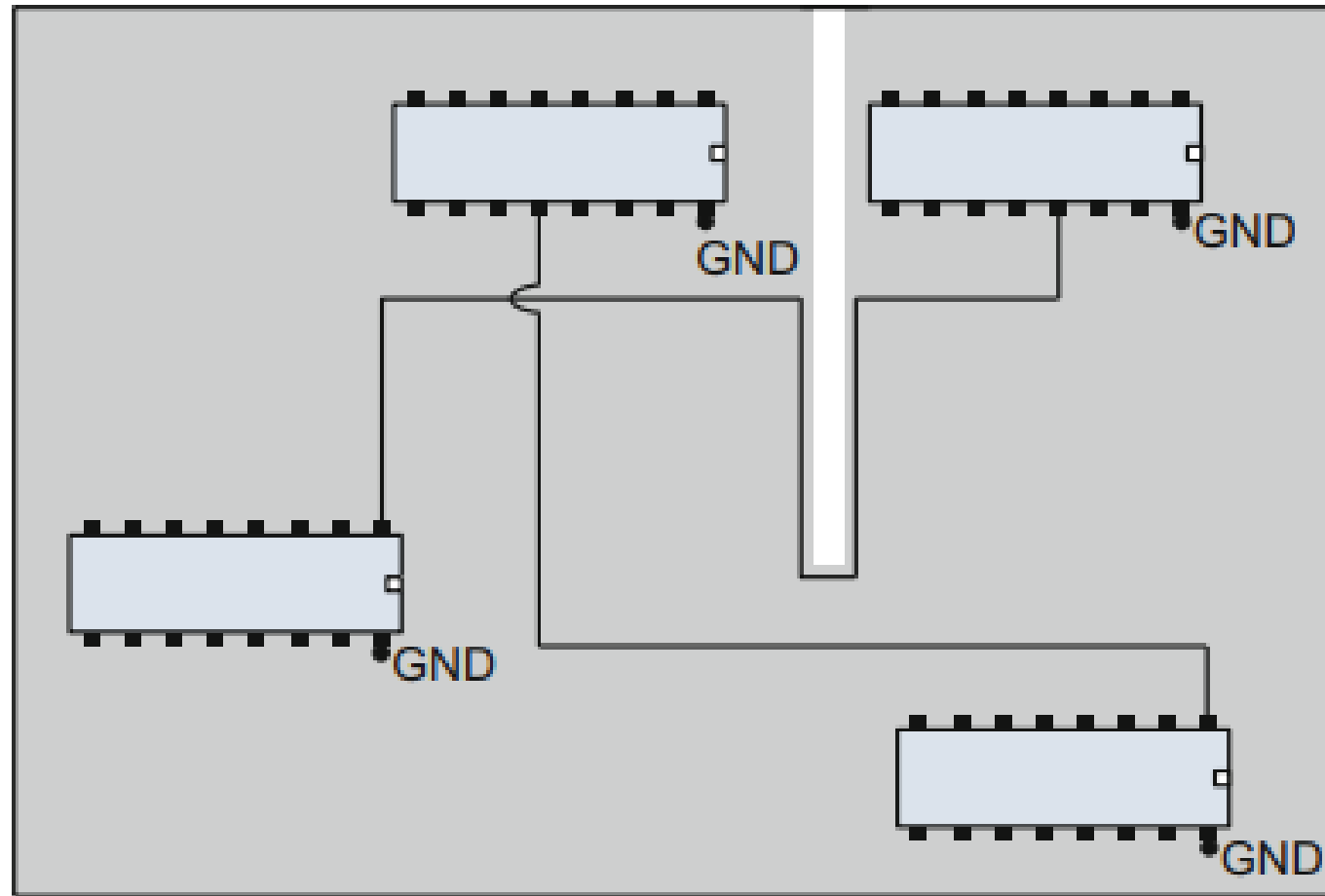


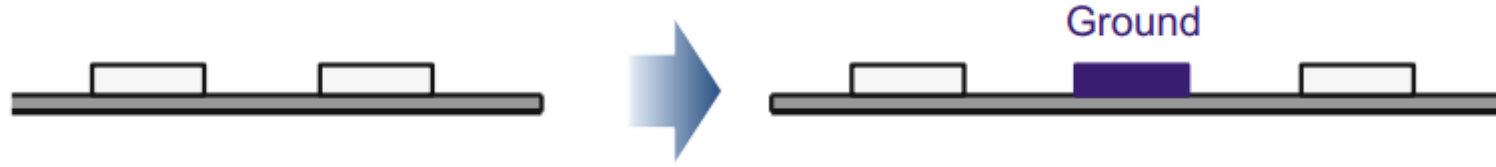
Have a ground plane for low ground impedance



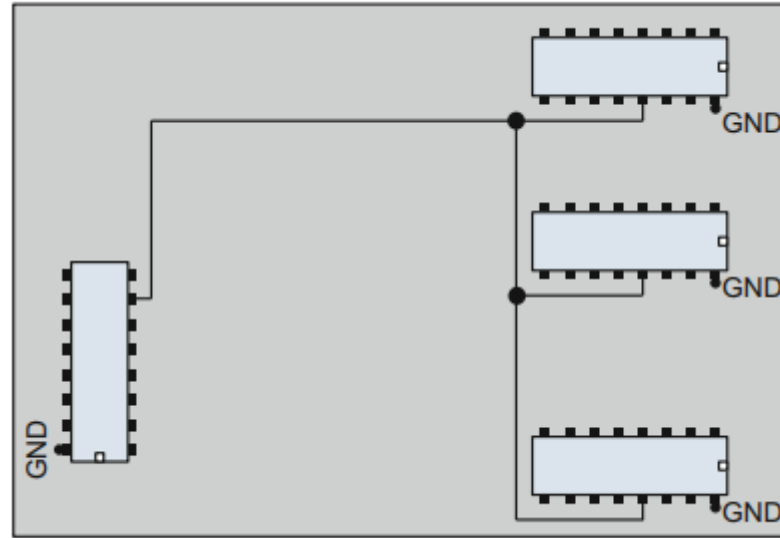
Have separate parallel ground paths for each signal line and maintain that even at crossings

- Route the signal lines around a break in the ground plane





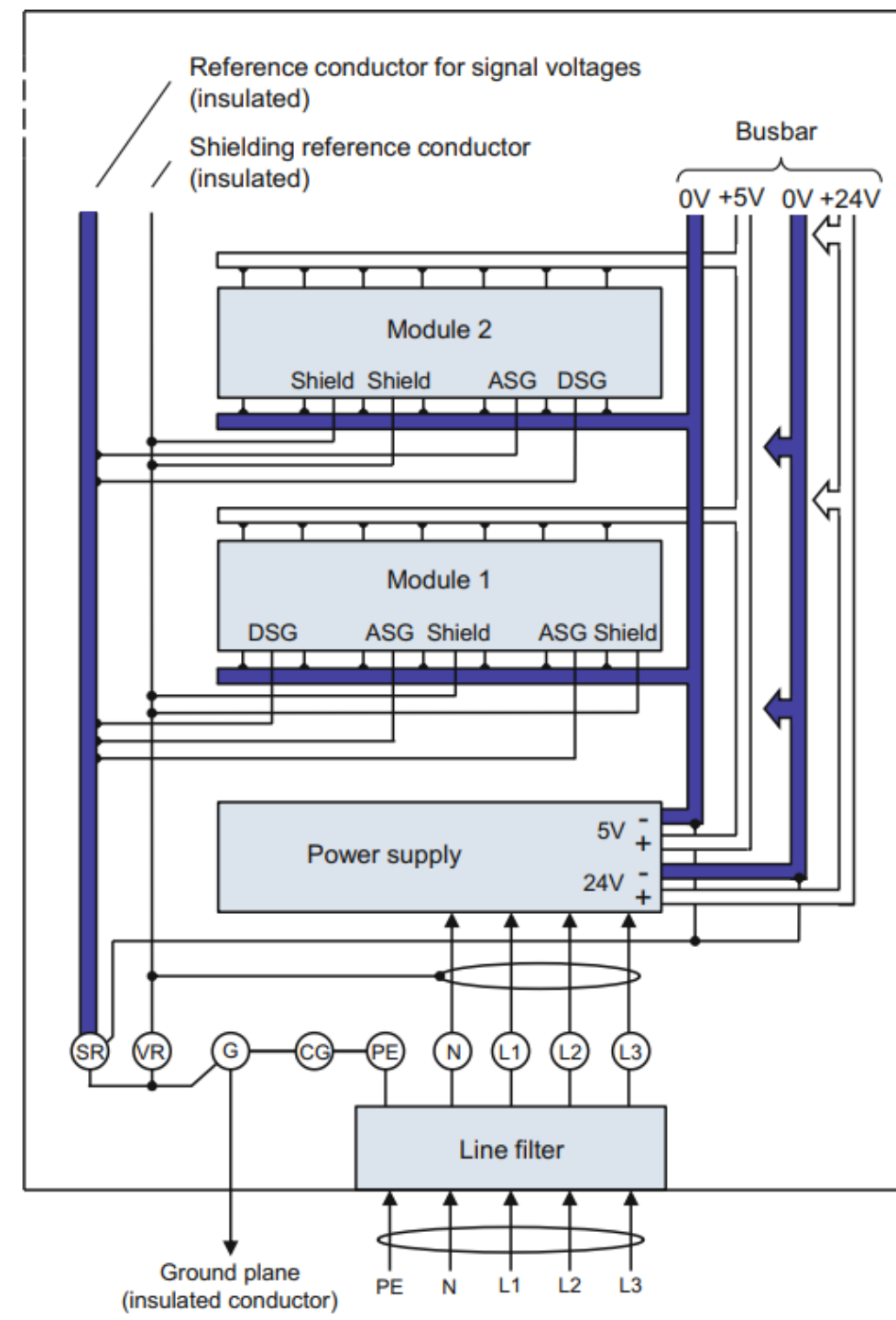
- Preventing capacitive coupling by inserting a ground trace between



- Split clock traces only once near the devices

Designing cabinets

- A modular system cabinet can reduce EMI
- Aluminum is generally unsuitable for the enclosure, as it provides no shielding against low-frequency magnetic fields
- Use round vents $d < 5 \text{ mm}$ in diameter instead of ventilation slots for effective shielding against high-frequency fields.
- Fit large apertures, for displays, for example, with metal screens or honeycomb windows
- Do not route the AC power supply through the cabinet, as it is difficult to filter out the low-frequency ripple voltages.
- Fit the line filter near the power cable entry point.
- Connect all metal parts, such as the cabinet and racks, with the protective conductor (PE, protective earth) as protection against contact voltages.
- No compensating currents should flow through these parts.



- The system ground G is the system reference potential.
- Avoid using the system protective conductor PE as system ground, as it could be the same as the neutral conductor N and can be highly “contaminated”.
- Instead, and if available, as well as the system protective conductor, use a system ground that is clear of disturbance voltages.
- Insulate the cable to the ground plane well.
- Reference conductors for signal voltages and shielding should be connected to the system ground.
- They should take the form of low inductance and low-resistance surfaces, a busbar, or cables. Insulate them from metal parts in the enclosure and in the assemblies to prevent the flow of compensating currents that would neutralize the shielding effect of the metal parts.
- Run the reference conductors for the shielding or ground potentials individually and insulated to the reference point for voltages in critical cases.
- Reference conductors should not be used as return conductors for the module power supplies.
- They should be shielded and twisted pairs or laminated busbars.
- You may need to filter out “stray” disturbance voltages from the power supplies in modules, such as PCB.

Connecting Peripherals

- The cables that provide the interconnections to external systems can also be very susceptible to disturbances.
- The propagation of wanted signals, the reduction of emissions, and also the couplings into the wanted signals must be considered while connecting cable shields to systems.
- In cable connectors, every ground conductor in an interconnect (cable) should have its own contact in the connector.
- If the distance between cable and circuit on the PCB is significant, implement traces on the board that mirror the cable design:
 - Lay a ground conductor parallel to every signal line on the PCB , and connect with the system ground only at one point.
 - Always twist the signal and associated return (ground) conductor in the cable.
 - If possible, fit all input and output lines with electrical isolators, like optocouplers or coupling transformers.
 - Conductors for ground and supply voltages and for shielding the two systems should not be connected together.

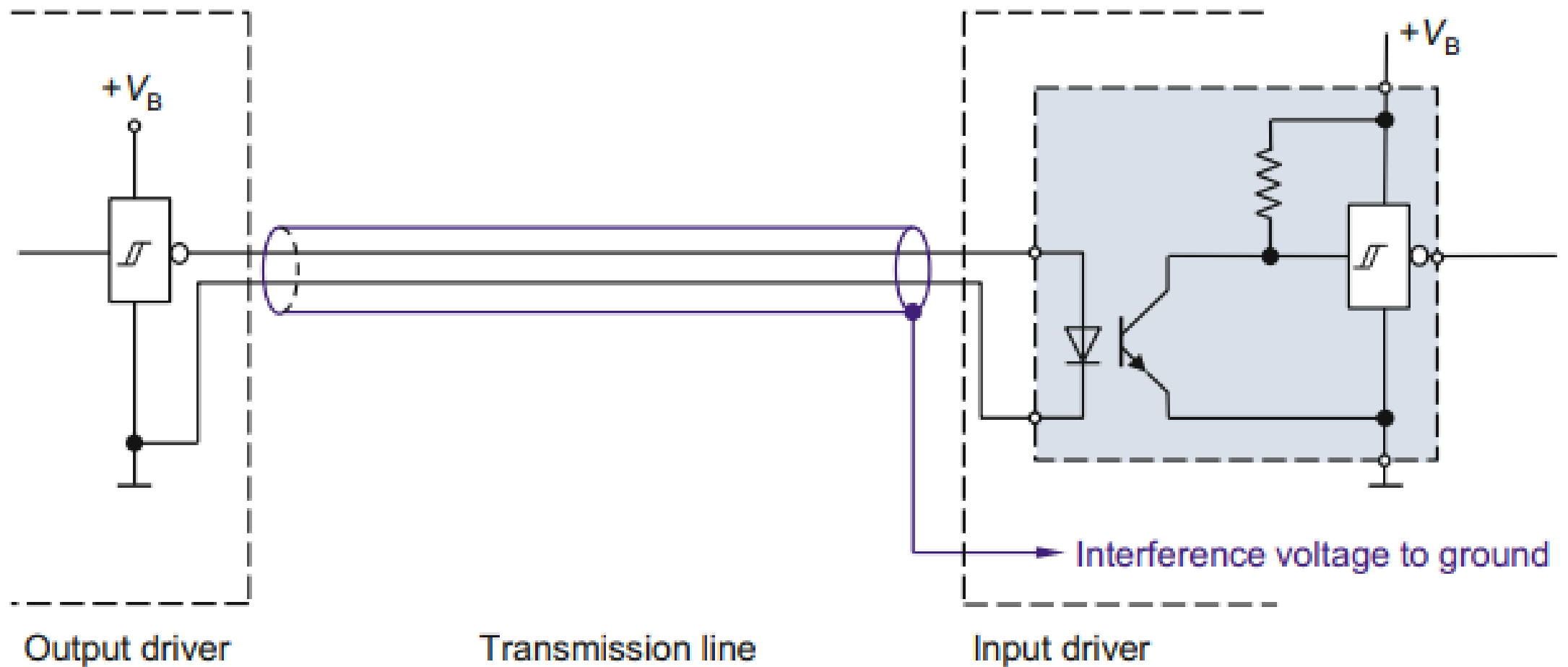
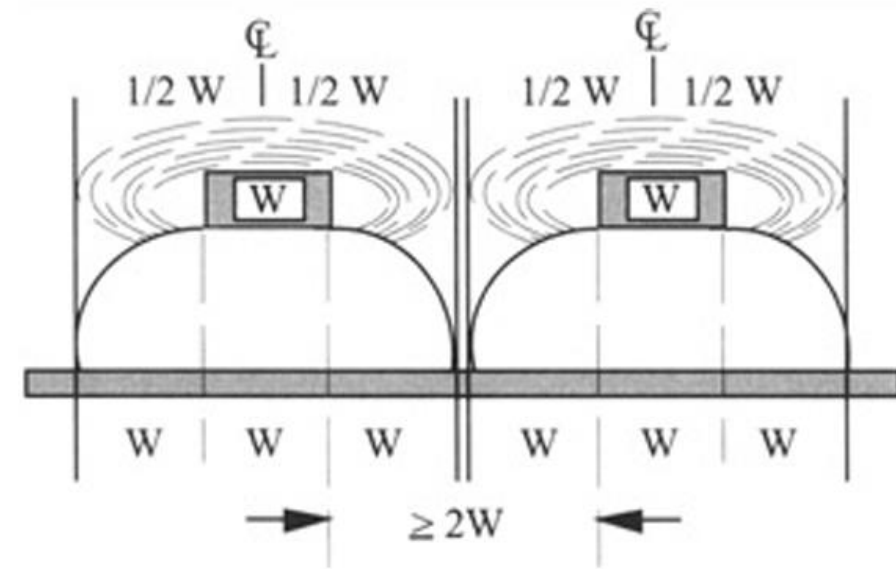


Fig. 6.46 Electrical isolation (galvanic separation) of two systems with an optocoupler

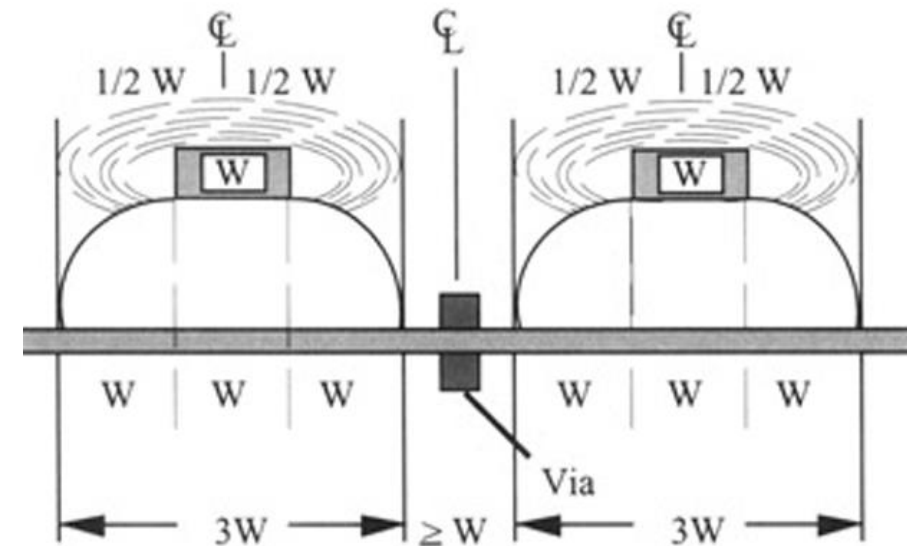
Additional reading

Trace separation and the 3-W rule

- The 3-W rule is used basically to minimize coupling between transmission lines or PCB traces.
- The rule states that the distance separation between traces must be three times the width of a single trace measured from centerline to centerline.
- Otherwise stated, the distance separation between two traces, edge-to-edge, must be greater than two times the width of a single trace.
- For example, a clock line is 6-mils wide.
- No other trace can be routed within a minimum of 12 mils ($2 * 6$ mils) of this trace, edge-to-edge.
- As observed, much real estate is lost in areas where trace isolation occurs.



The distance spacing between both traces must have a minimum overlap of $2W$.



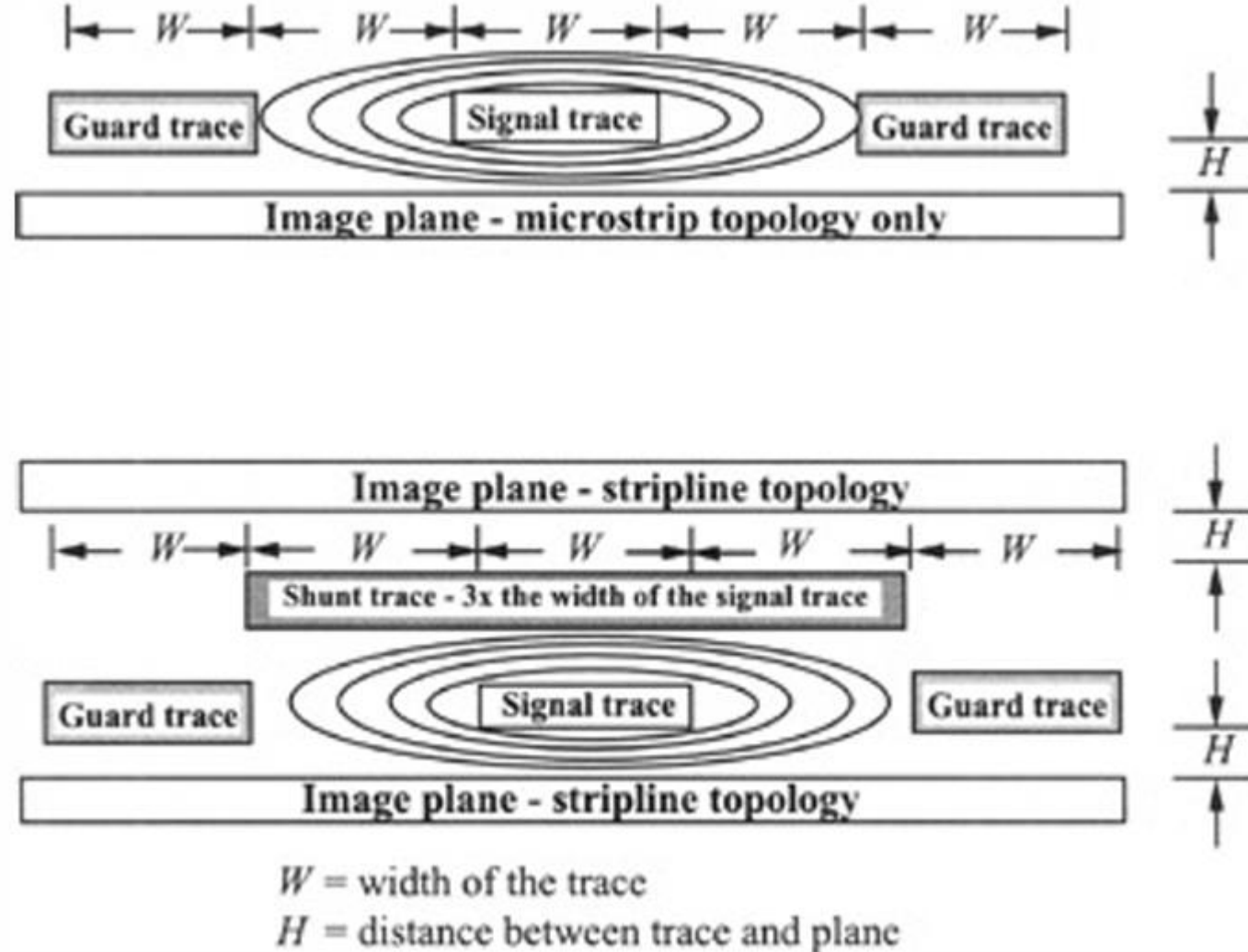
For the via, add annular keep-out diameter, which includes both the via and annular (anti-pad) clearance.

Rules for grounding

- The 3-W rule is affected by the presence of a reference plane and the distance of the signal trace above that plane.
- Prevention of magnetic flux coupling to adjacent traces occurs in the horizontal axis.
- If a reference (return) plane is physically closer to the signal trace than the trace-to-trace spacing, the plane will minimize magnetic flux within the transmission line, thus enhancing performance over that of the 3-W rule.
- Note that the 3-W rule represents the approximate 70% flux boundary at logic current levels.
- For the approximate 98% boundary, 10-W should be used.
- Use of the 3-W rule is mandatory for only high-threat signals, such as clock, differential pairs, video, audio, the reset line, or other system-critical nets.

Guard traces

- Guard traces are traces at 0V-potential that surround clocks, periodic signals, differential pairs, or system-critical (high-threat) nets from source to destination.
- Shunt traces are traces located directly above or below a high threat transmission line that parallels the trace along its entire route.
- Shunt traces are used when the sensitivity of a trace is critical (beyond achievable with 3W rule), related to crosstalk corruption, or when a known amount of excessive RF energy will be present within a particular transmission line or trace.
- Routing a trace at 0V-potential parallel, or adjacent, to the signal trace allows for enhanced flux cancellation in differential mode.
- The shunt trace acts as an image plane and must be three times the width of the signal trace to fully capture flux that surrounds the trace.
- If the trace has an image plane on one side, at 0V-potential, and a shunt trace on the other side, also at 0V-potential, a partial coaxial transmission line structure is developed.
- If guard traces are provided with a shunt trace, a true coaxial transmission line exists.



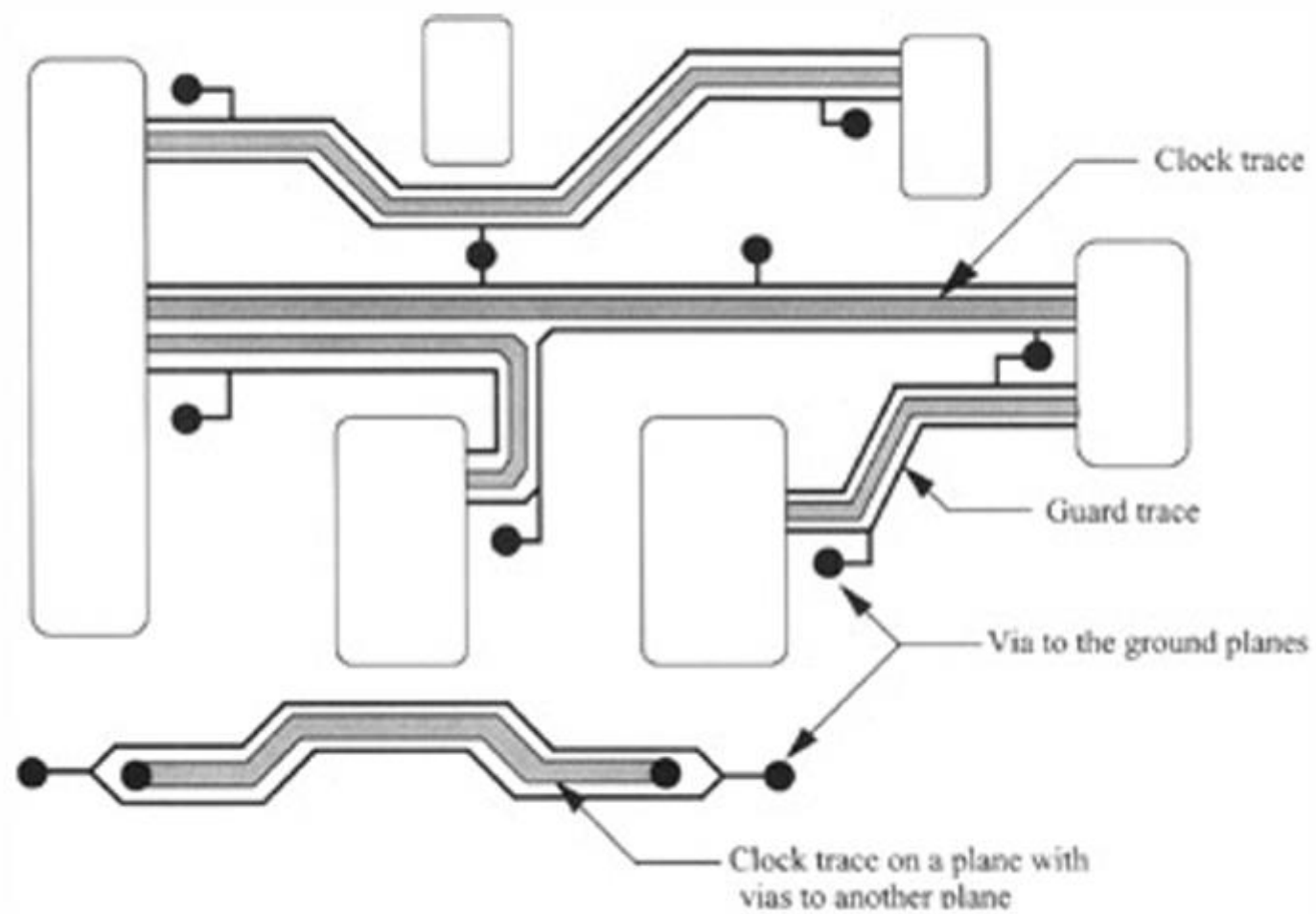


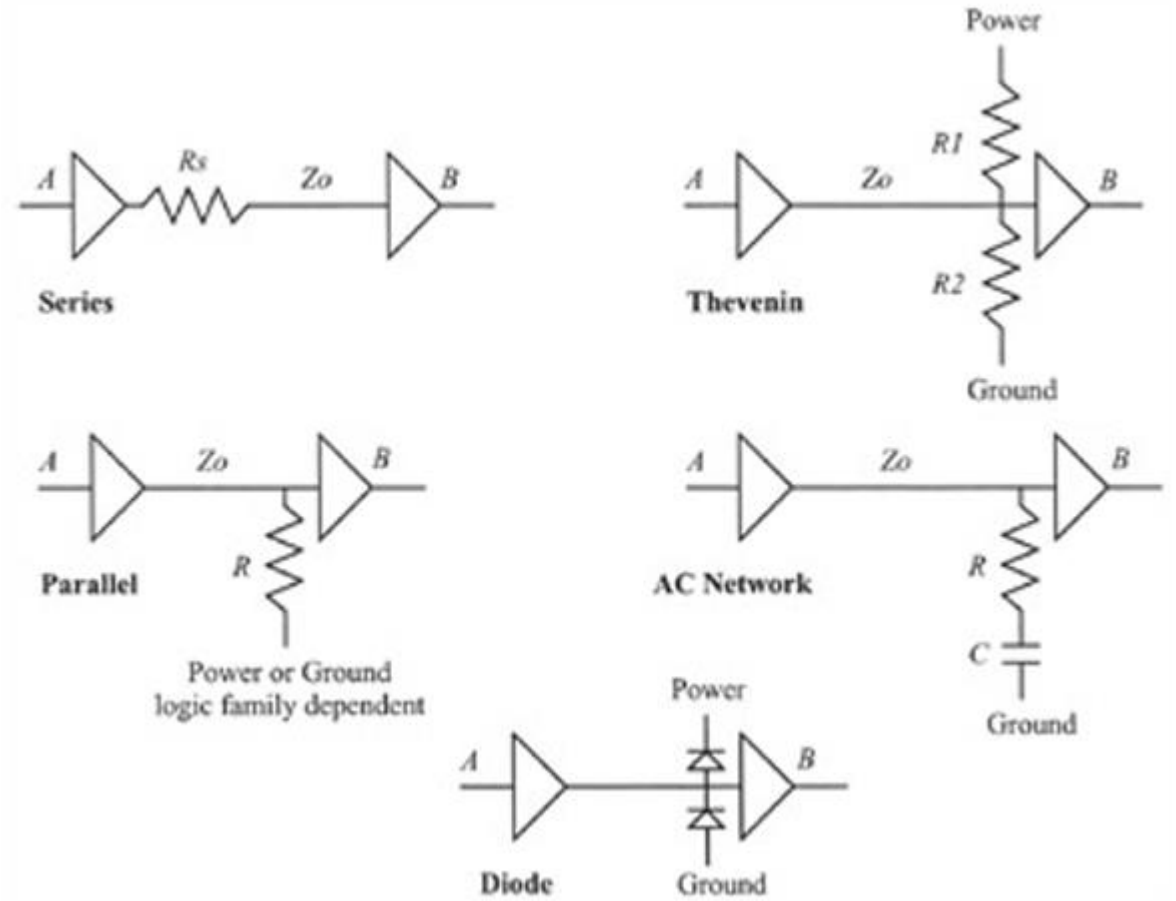
Figure 4.24: Guard trace implementation.

Trace termination

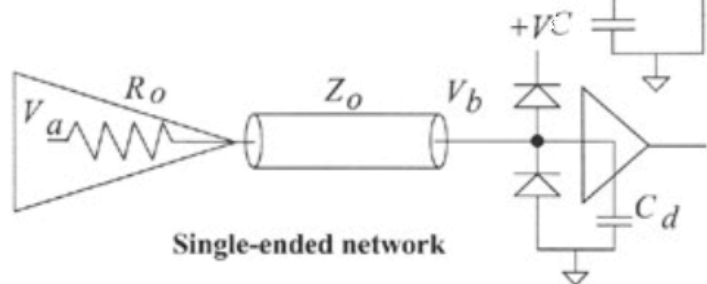
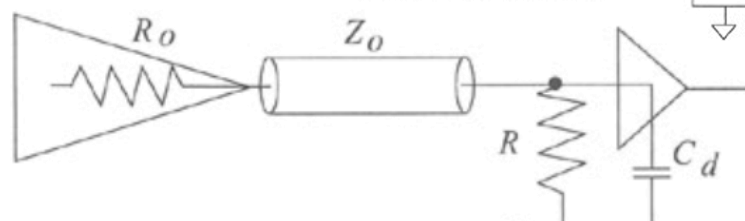
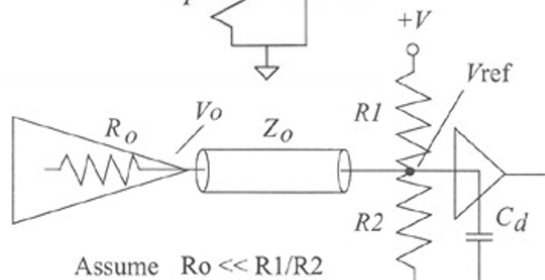
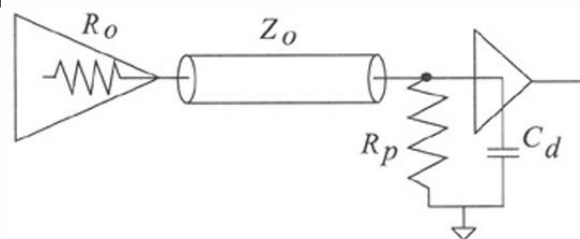
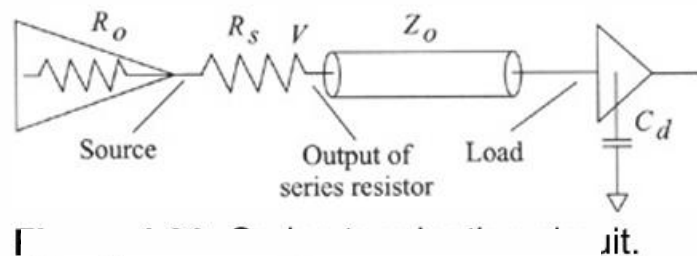
- Trace termination plays an important role in ensuring optimal signal integrity as well as minimizing the development of RF energy.
- To prevent impedance matching problems and provide higher quality signal transfer between circuits, termination may be required.
- Transmission line effects in high-speed circuits and traces must always be considered.
- If the clock speed is fast, for example, 100 MHz, and components are, for example, FCT series (2 ns edge rate being typical), reflections from a long trace route could cause the receiver to double clock on a single-edge transition.
- This is possible because it takes a finite time for the signal to propagate from source to load and return.
- If the return signal does not occur before the next edge transition event, functionality issues arise.
- Any signal that clocks a flip-flop is a possible candidate for causing transmission line effects, regardless of the actual frequency of operation.
- Termination absorbs excess or unwanted energy.

Trace termination

- Series termination affects this energy early in the transmission, while end termination absorbs it later.
- The topology chosen directs the energy to where it is needed, when it is needed.
- Excessive energy ends up going where it is not wanted, coming back later to cause functionality, EMI, and signal integrity problems.
- Each source driver must have its respective output properly terminated.

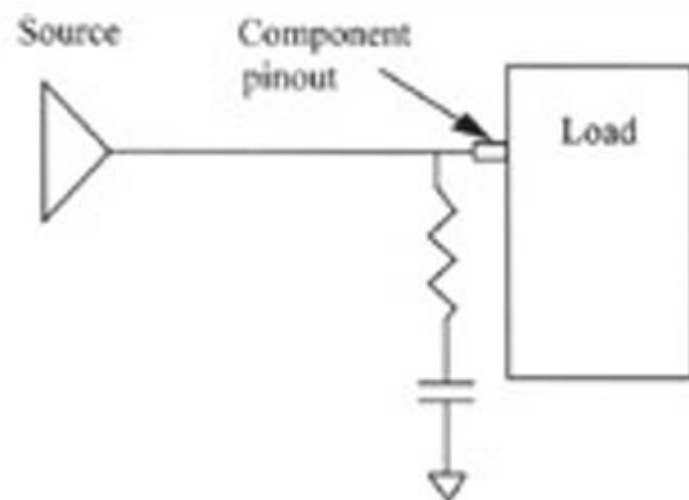


- The need to terminate is based on several design criteria, the most important of which is the existence of an electrically long trace.
- When a trace is electrically long, or when the length exceeds one-sixth of the electrical length of the edge transition time, the trace requires termination.
- Even if a trace is short, termination may still be required if the load is capacitive or highly inductive to prevent ringing.
- Termination not only matches trace impedance and removes (reduces) ringing and reflections, but it may also slow down the edge rate transition of the clock signal if incorrect values are applied.
- Inappropriate termination may degrade the signal amplitude integrity to the point of nonfunctionality.
- Reducing either dI/dt or dV/dt within the trace will reduce the creation of RF currents generated by high-amplitude voltage and current levels.
- The easiest way to terminate is to use resistive elements.

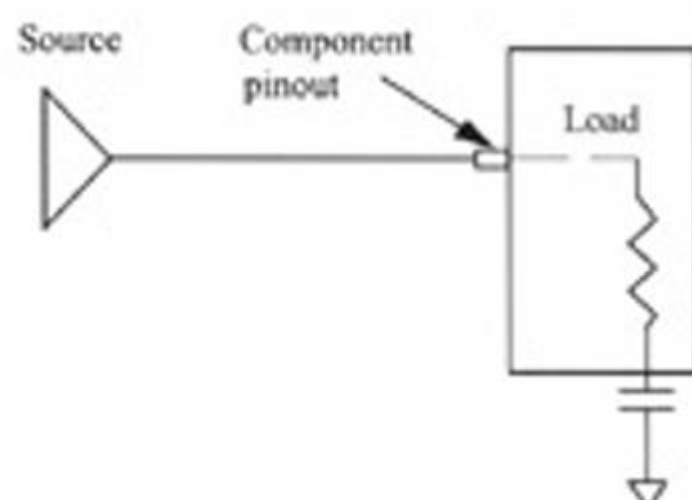


Single-ended network

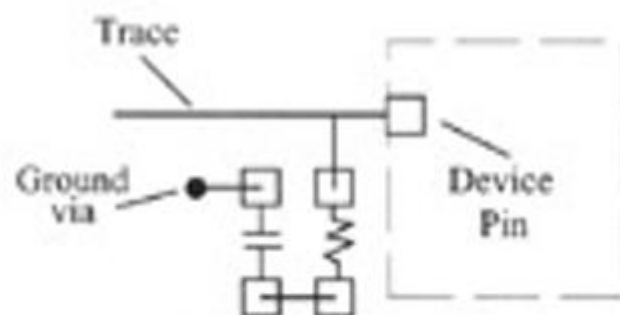
Termination Type	Added Parts	Delay Added	Power Required	Parts Values	Comments
Series	1	Yes	Low	$R_s = Z_o - R_o$	Good DC noise margin.
Parallel	1	Small	High	$R = Z_o$	Power consumption is a problem.
Thevenin	2	Small	High	$R = 2 * Z_o$	High power for CMOS. The ratio of R1/R2 determines the relative proportions of logic HI and LOW drive current.
AC (RC)	2	Small	Medium	$R = Z_o$ $C = 20-600 \text{ pF}$	Check bandwidth and added capacitance.
Diode	2	Small	Low	—	Limits overshoot; some ringing at diodes.



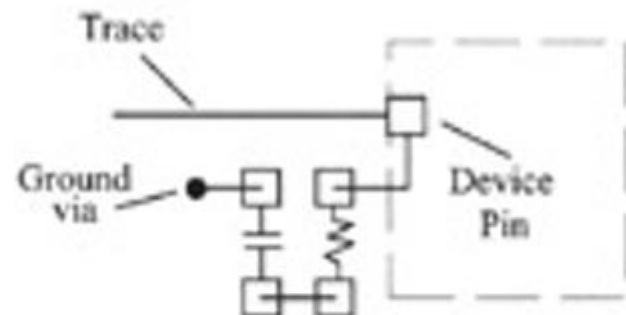
**Wrong implementation - end termination
(schematic representation)**



**Correct location of end termination
(schematic representation)**



**Wrong implementation - end termination
(actual layout representation)**



**Correct implementation - end termination
(actual layout representation)**